

Multiproduct Firms and Production Networks

Renata Abbott¹ and Marco Rojas²

¹UC San Diego

²Central Bank of Chile

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Abstract

Do product and customer margins interact in shaping firm size? Using the universe of firm-to-firm transactions in Chile, we show that they are interconnected. The largest growth episodes combine new buyers with new products, and within suppliers a 10% increase in buyers is associated with a 5.6% increase in product scope, much of it among products sold to incumbent buyers. We develop a model of multiproduct firms embedded in a production network in which the two margins are jointly determined through two channels: amortization, as a firm-level product-adoption cost is spread over more buyer relationships, and matching, as a broader buyer base gives each product more opportunities to find a profitable buyer. The model allows us to separate the two channels and assess their relative importance. Using only firm-level moments, amortization can explain up to about 40% of the co-movement, while adding within-link moments shifts most of the explanation toward matching. The model reproduces the buyer-product relationship and implies spillovers across margins. Frictions affecting buyer acquisition also shape firms' product scope, and those affecting product adoption also shape their buyer networks. For the firm-size distribution, the two margins are not symmetric: the buyer margin alone reproduces about 88% of the dispersion in firm sales, compared with 66% for the product margin, with product scope amplifying differences generated through the buyer network.

1 Introduction

How do firms’ product and customer scopes interact inside production networks, and how do these margins shape firm growth and the firm-size distribution? Using the universe of firm-to-firm transactions in Chile, we show that these two dimensions of firm scope are closely interconnected. We organize our empirical analysis around three model-facing facts. First, the largest growth episodes disproportionately combine new buyers with new products. Second, the composition of growth varies systematically across the initial firm-size distribution. Smaller suppliers grow relatively more through margins involving new products, often together with new buyers, while the buyer margin becomes increasingly important among larger suppliers, primarily through selling existing products to new customers. At the very top of the distribution, growth also relies heavily on existing buyer–product relationships. Third, buyer breadth and product scope co-move strongly within suppliers: conditional on supplier fixed effects, a 10 percent increase in the number of buyers is associated with approximately a 5.6 percent increase in product scope. About two thirds of this relationship remains when product scope is measured only among incumbent buyers, suggesting that the co-movement is not simply the mechanical consequence of new buyers creating additional product–buyer matches. Together, these facts motivate a model in which product adoption and customer acquisition are distinct but jointly determined dimensions of firm scope.

These two margins are quantitatively important, but are typically studied separately. Multi-product firms¹ account for a large share of production, trade, and employment (Arkolakis et al., 2021), and a large literature emphasizes the role of product introduction and product switching in firm performance and resource reallocation (Bernard et al., 2010, 2011; Mayer et al., 2014; Argente et al., 2018; Copeland and Shapiro, 2016; Hottman et al., 2016). Because these models focus primarily on firm and product dynamics, however, they generally abstract from heterogeneity across individual supplier–buyer relationships.

At the same time, a growing production-network literature shows that customer acquisition and buyer–supplier relationships are important determinants of firm growth and the firm-size distribution (Bernard et al., 2022; Eaton et al., 2026; Lim, 2018; Huneeus, 2018; Arkolakis et al., 2023; Afrouzi et al., 2023; Einav et al., 2021; Argente et al., 2021; Aekka and Khanna, 2024). This work has developed rich models of buyer–supplier matching and network formation (Bernard et al., 2018; Miyauchi, 2024; Demir et al., 2024; Dhyne et al., 2021); Bernard and Moxnes (2018) provide a survey of this literature. To the best of our knowledge, however, these models typically treat suppliers as single-product firms or subsume product expansion into match formation. When suppliers sell many products to many buyers, product expansion and customer acquisition are separate decisions. A firm can add products without gaining a new customer, gain a new customer without adding products, or expand along both margins at the same time. Our main point is that

¹Throughout the paper, whenever we refer to firms, we mean suppliers.

these decisions are still closely linked because expanding along one margin changes the incentives to expand along the other.

We study these margins using the universe of invoice-level firm-to-firm transactions in Chile (SII Factura Electrónica), matched to administrative firm characteristics, the same production network studied by [Huneus \(2018\)](#) and [Arkolakis et al. \(2023\)](#). We focus on manufacturing suppliers and define products at the 5-digit CPC level. Figure [1a](#) shows that larger suppliers operate along both extensive margins. Firms with higher sales serve more buyers and sell more products.

Figure [2](#) then examines how products are positioned across those buyer relationships. Larger suppliers sell each product to more buyers and, at the same time, sell more products within each buyer relationship. The first pattern is substantially stronger. As supplier size increases, customer reach expands faster than product scope. Figure [10](#) in Appendix [A](#) shows this asymmetry directly using the ratio of buyer breadth to product scope.

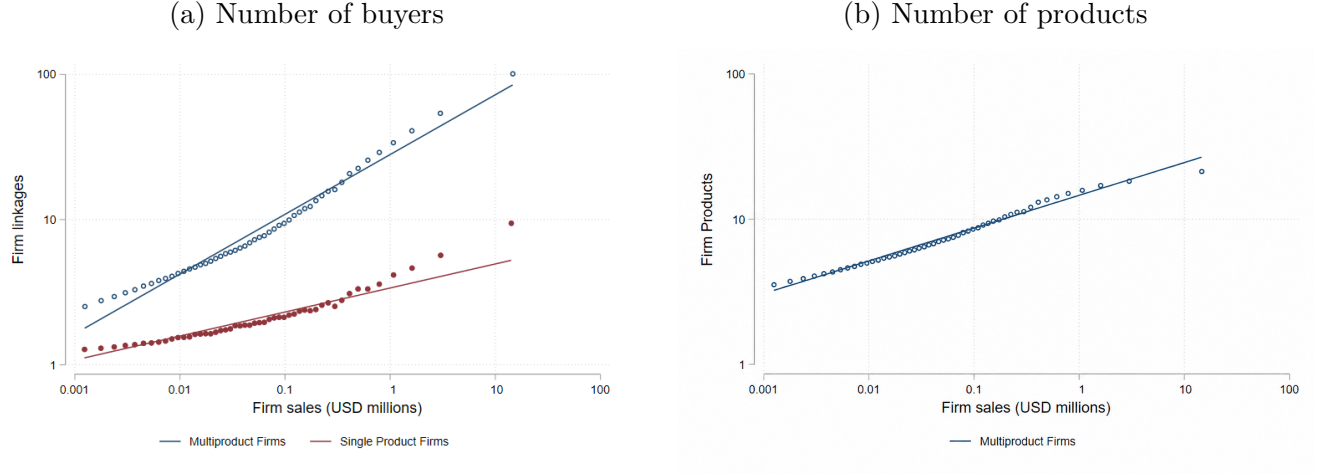
The buyer margin also plays a larger role in explaining differences in firm size across suppliers. In 2017, the number of buyers accounts for 42.9 percent of the variance in log supplier sales, while average sales per buyer account for the remaining 57.1 percent. On the product side, the number of products accounts for 27.3 percent of the variance, with 72.7 percent coming from average sales per product. Both margins matter, but differences in customer reach explain a larger share of firm-size dispersion than differences in product scope. Figure [11](#) in Appendix [A.1](#) reports these decompositions.

Together, Figures [2a](#) and [2b](#), along with the variance decompositions, show that firm-size differences are associated not only with broader customer and product scope, but also with how densely products are deployed across buyer relationships and how widely each product is sold across buyers. These cross-sectional patterns motivate the empirical analysis that follows, which studies how the buyer and product margins interact in firm growth and how their relative importance changes across the firm-size distribution.

We document three model-facing facts that characterize this interaction in more detail. First, the largest supplier-growth episodes disproportionately combine buyer acquisition and product expansion. Decomposing sales growth between 2017 and 2023 into old-link–old-product, new-link–old-product, old-link–new-product, and new-link–new-product components shows that simultaneous expansion along both margins becomes especially important among the fastest-growing suppliers. The largest expansions therefore cannot be understood as adjustments along either buyer or product scope in isolation.

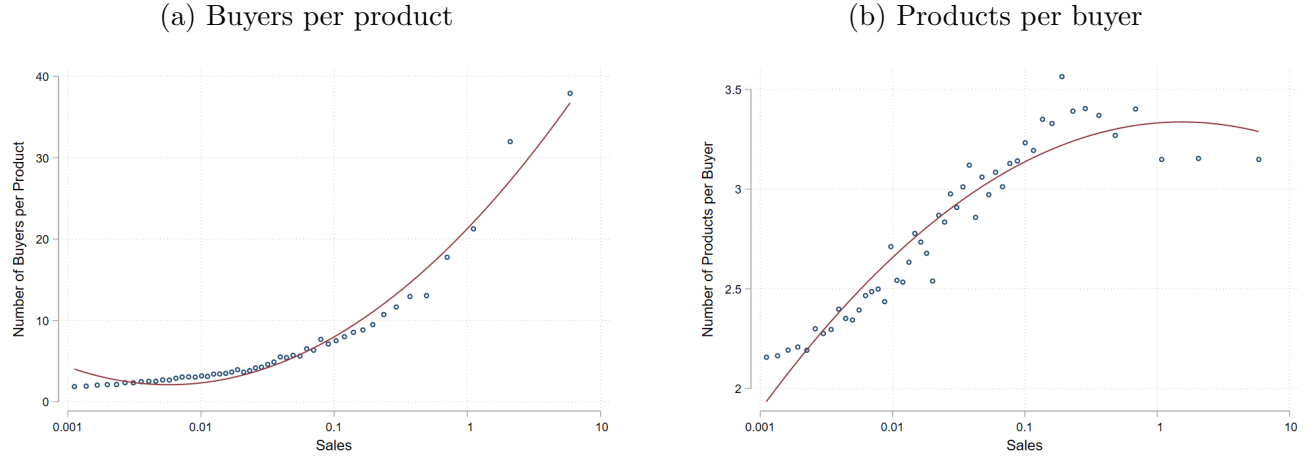
Second, the way firms expand varies systematically with their initial size. Smaller suppliers rely relatively more on margins involving new products, especially the joint addition of new buyers

Figure 1: Extensive margins and supplier size



Notes: The figure presents binned scatterplots with fitted regression lines relating supplier size to the number of buyers and products. Supplier size is measured by total sales at the tax-identifier level (million USD). Both axes are shown on a log scale. The sample excludes the top 0.1 percent of the sales distribution. Each dot represents a bin of suppliers based on total sales. Panel (a) distinguishes multiproduct and single-product firms, while Panel (b) reports multiproduct firms only, since single-product firms have exactly one product by definition.

Figure 2: Intensive margins and supplier size



Notes: The figure presents binned scatterplots relating supplier size to the density of buyer-product relationships. Panel (a) reports the average number of buyers per product, defined as the number of active buyer-product pairs divided by the number of products. Panel (b) reports the average number of products per buyer, defined as the number of active buyer-product pairs divided by the number of buyers. Supplier size is measured by observed firm-to-firm sales (million USD). The horizontal axis is shown on a log scale. Each dot represents one of 50 equally populated sales bins, and the solid line is a quadratic fit. The top 1 percent of the sales distribution is excluded.

and new products. As supplier size increases, buyer acquisition becomes more important, with larger suppliers increasingly expanding by selling products already in their portfolios to new customers. At the very top of the distribution, growth also relies heavily on existing buyer-product relationships. The relative importance of the two extensive margins therefore changes systematically with firm size. The composition of growth also varies with supplier age. Younger suppliers rely more heavily on margins involving new products and new buyers, whereas older suppliers rely relatively more on acquiring new buyers for products already in their portfolios. The initial-size patterns map directly into the model’s heterogeneous comparative statics, while the age patterns provide complementary descriptive evidence.

Third, buyer breadth and product scope co-move strongly within suppliers. In our preferred specification, a 10 percent increase in the number of buyers is associated with approximately a 5.6 percent increase in product scope. An alternative specification based on new product introductions yields the same qualitative result: a 10 percent increase in buyers is associated with approximately a 3.92 percent increase in product additions. Moreover, the relationship survives when product scope is measured only among incumbent buyers, with an elasticity of 0.38, about two thirds of the total. Because new buyers cannot change the products a supplier sells to buyers it already serves, this within-incumbent relationship suggests that a larger buyer base is associated with changes in the supplier’s product-development decisions. Importantly, this relationship is concentrated among peripheral rather than core products.² Around the acquisition of a large new buyer, sales of peripheral products rise sharply, while sales of core products change little. This pattern suggests that expanding the customer base disproportionately increases activity in products near the margin of adoption rather than simply increasing sales of products that were already central to the supplier.

We also ask whether this positive relationship simply reflects a reallocation of a fixed amount of activity across products or customers. We find little evidence of such cannibalization. When a supplier introduces a new product within an existing buyer relationship, sales of its existing products do not fall. Similarly, when a supplier acquires a large new buyer, sales to existing buyers do not systematically decline. This suggests that expansion along either margin increases the scale of supplier activity rather than simply redistributing a fixed amount of revenue across products or customers. Taken together, these exercises provide little evidence that the observed co-movement primarily reflects reallocation of a fixed amount of activity across products or buyers.

Guided by these facts, we build a static model that adapts the rank-based multi-product structure of [Arkolakis et al. \(2021\)](#)³ to a production network in which destinations are individual

²We define peripheral products as products that account for a relatively small share of the supplier’s revenue, while core products account for a relatively large share.

³In [Arkolakis et al. \(2021\)](#), a firm’s products are ordered by profitability, with a core product at the top of the ranking and progressively less profitable (peripheral) products at higher ranks. Firm product scope is therefore characterized by how far the firm extends down this common product ranking.

buyers. Suppliers differ in productivity and choose their firm-level product scope before deciding which buyer relationships to serve. The key mechanism comes from where the relevant fixed costs are paid. Developing a product requires a firm-level adoption cost, $F^P(G_s)$, that is paid once by the supplier, while serving an additional buyer requires a buyer-specific link-formation cost. Because the product-adoption cost is paid at the firm level while its returns accrue across customer relationships, a larger buyer base increases the return to developing an additional product. We call this the *amortization channel*. In the other direction, a broader developed product portfolio raises the expected surplus from a potential buyer relationship by creating more opportunities for profitable product–buyer matches, making it worthwhile to pay the link cost for buyers that would not be profitable with a narrower portfolio. Product development and buyer acquisition are therefore jointly determined rather than independent decisions.

The model also contains a second, conceptually distinct channel linking buyer breadth to observed product scope. Each developed product draws a buyer-specific match, and a product appears in the supplier’s observed portfolio only if it is sold to at least one buyer. As a result, a supplier with more buyers can sell a larger share of the products it has already developed even without changing its developed product portfolio. We refer to this as the *matching channel*. Both amortization and matching generate a positive relationship between buyer breadth and product scope, but they operate through different decisions. Amortization affects the incentive to develop new products, while matching affects whether developed products find a buyer and become part of observed product scope. This distinction also matters for policy because the same increase in observed product scope can reflect either additional product development or wider sales of products that already exist. Separating these two mechanisms is therefore a central objective of the paper.

The buyer-specific product shocks also allow the model to capture the fact that suppliers do not sell the same set of products to every customer. A supplier may develop many products, but each buyer purchases only the subset that is sufficiently profitable within that relationship. Products still have a systematic ranking at the firm level, with more peripheral products having lower expected profitability, but the idiosyncratic product–buyer component means that different buyers can purchase different, non-nested subsets of the same portfolio. This feature is disciplined directly by the data. A product’s importance in the supplier’s overall portfolio predicts its importance within individual buyer relationships, but the relationship is far from perfect. The baseline rank-alignment coefficient is approximately 0.27, and the supplier’s top product is not always the top product within every link. A common product ladder in which all buyers purchase products according to the same ranking would generate much stronger alignment than we observe.⁴

⁴This common-ladder structure follows the rank-based multiproduct framework of [Arkolakis et al. \(2021\)](#). Our contribution is to allow buyer-specific product heterogeneity, so different buyers can purchase different subsets of the supplier’s developed portfolio.

This flexibility is important both for reproducing the observed variation in product composition across buyer relationships and for distinguishing whether buyer–product co-movement comes from additional product development or from existing products reaching more buyers.

We calibrate the model by simulated method of moments (SMM) using the 2017 manufacturing cross-section sample of firms. We report two calibrations that differ in the information used to discipline the matching structure. The first targets firm-level moments as the distributions of product scope and buyer reach, buyers per product, sales concentration, product specialization, and the scope–buyer elasticity. The second adds within-link moments that describe how products are allocated across buyer relationships as the number of products per link, the share of single-product links, the decay of sales along the product ranking, the dispersion of sales across a supplier’s buyers, and portfolio coverage, the share of a supplier’s products sold to its broadest buyer.

The distinction between the two calibrations is important. Firm-level moments can discipline the overall relationship between buyer breadth and product scope, but they contain limited information about why that relationship arises. A model calibrated only to these moments can therefore reproduce the aggregate co-movement while assigning too much importance to product development and too little to buyer-specific matching. Within-link moments help separate these mechanisms because they reveal how a supplier’s products are actually distributed across its buyer relationships. Both calibrations reproduce the positive relationship between buyer breadth and product scope among suppliers with similar productivity and relationship costs: the model elasticity is 0.43 and 0.52, against 0.56 in the data.

The two calibrations differ sharply in why the relationship arises, which is the key mechanism result of the paper. We decompose the model elasticity with a relocated-cost experiment. The firm-level adoption cost is replaced by an equivalent cost paid on each buyer relationship on which the product is sold, rescaled to leave average product scope unchanged. This removes amortization while preserving matching. When the model is disciplined only by firm-level moments, product–buyer selection is weak and amortization accounts for up to about 40 percent of the elasticity. Once within-link moments are targeted, product–buyer selection becomes substantially stronger, with matching accounting for the majority of the elasticity, while amortization contributes only about 10 percent. Firm-level moments alone therefore cannot distinguish the two channels; the allocation of products across buyer relationships can. The evidence from incumbent buyers adds an important qualification. About two thirds of the empirical co-movement occurs within existing buyer relationships, where the mechanical effect of adding new buyers cannot explain the increase in product scope. This share is larger than what the static model generates under either calibration. We interpret this gap as evidence that the buyer base may have a stronger effect on product development than the static amortization mechanism captures, and as an important target for a

dynamic extension of the model.

We next use the calibrated model to quantify the importance of this interaction for the firm-size distribution and for responses to changes in fixed costs. The two margins are not equally important quantitatively. When we shut down the buyer margin by replacing each supplier’s buyer network with a single representative buyer, the standard deviation of log sales falls by 34 percent. Shutting down the product margin by restricting suppliers to a single product reduces sales dispersion by only 12 percent. Equivalently, the buyer margin alone reproduces approximately 88 percent of the sales dispersion generated by the full model, compared with approximately 66 percent for the product margin alone. The data point the same way. The number of buyers accounts for 42.9 percent of the variance of log supplier sales, against 27.3 percent for the number of products. Customer reach is therefore the quantitatively more important margin for firm-size dispersion, while product scope amplifies the heterogeneity generated through the buyer network. This parallels [Bernard et al. \(2022\)](#), who find that downstream factors, and the number of customers in particular, account for most firm-size heterogeneity in the Belgian production network; we show that the same ranking holds once suppliers are allowed to be multiproduct, with product scope amplifying rather than dampening this dispersion.

The two channels also determine how changes in one margin spill over to the other. Halving the cost of forming buyer relationships increases buyer reach by about 30 percent and product scope by about 9 percent. Halving the product-adoption cost increases product scope by about 12 percent and buyer reach by about 5 percent. In each case, the largest effect is on the margin directly affected by the cost change, but there are meaningful spillovers to the other margin. This distinction matters for policy. Reducing frictions to forming supplier–buyer relationships increases the number of products suppliers are observed to sell, but most of this increase comes from existing products reaching more buyers rather than from the development of new products. Because the model is partial equilibrium, these responses do not include adjustments in buyers’ price indices.

Finally, we connect the static model back to the growth decompositions that motivate the paper. We subject the calibrated economy to a common productivity shock, re-solve firms’ buyer and product decisions, and decompose the resulting change in sales into the same components used in the data. Although the model cannot reproduce an actual time path or relationship exit, it generates the same qualitative heterogeneity in the composition of expansion. Smaller and less-productive suppliers adjust relatively more through new buyers and products, whereas larger and more-productive suppliers rely more heavily on sales through existing relationships. We interpret this exercise as an untargeted validation of the model’s growth mechanics rather than as a structural reproduction of the 2017–2023 transition.

The distinction between buyer and product margins is also supported by their different lifecycle

patterns in the data. Sales in new supplier–buyer links peak earlier, decline faster, and exhibit lower survival than supplier–product relationships. These differences reinforce the treatment of buyer reach and product scope as distinct margins and are consistent with work viewing the customer base as a slowly accumulated stock (Gourio and Rudanko, 2014; Fitzgerald et al., 2024; Foster et al., 2016). Because the benchmark model is static, however, these lifecycle patterns serve as motivating evidence rather than quantitative targets.

The paper contributes to two related literatures by bringing them together around a common idea: product scope and customer scope are distinct but complementary endogenous dimensions of firm scope. First, relative to firm-to-firm matching models in which suppliers are effectively single-product or product expansion is subsumed into match formation (Bernard et al., 2018; Eaton et al., 2026; Bernard et al., 2022; Arkolakis et al., 2023), we keep firm-level product scope as a distinct choice and show that it interacts with buyer acquisition through two channels, cost amortization and product–buyer matching, which the model can separate. Second, we provide new empirical evidence on this interaction: large expansions disproportionately combine new buyers and products, the composition of expansion varies systematically with supplier size and age, and buyer breadth is positively associated with product scope within suppliers. Third, we develop a buyer-specific product-matching structure that allows firms to have a clear product ranking while still selling different sets of products to different buyers. This structure lets us separate the scope–buyer elasticity into the part coming from amortization and the part coming from matching. Importantly, distinguishing between these two channels requires information on how products are allocated across buyer relationships, which firm-level moments alone do not provide. Finally, our quantitative exercises show why this interaction matters: buyer reach is the dominant source of firm-size dispersion, and the relative weight of the two channels determines whether policies that ease buyer acquisition stimulate product development or mainly help existing products find buyers.

The remainder of the paper is organized as follows. Section 2 describes the data, sample, and key definitions. Section 3 presents the three model-facing empirical facts, with complementary lifecycle evidence reported in the appendix. Section 4 develops the model, and Section 5 describes the calibration strategy and empirical targets. Section 6 shows the quantitative behavior and fit of the calibrated model. Section 7 presents the mechanism decomposition, fixed-cost comparative statics, extensive-margin counterfactuals, and the comparative-static analog of supplier growth. Section 8 concludes.

2 Data Description

2.1 Administrative data sources

The main dataset combines three administrative sources. Our core data consist of domestic firm-to-firm transactions among Chilean firms, drawn from tax records collected by the Chilean tax authority, the Servicio de Impuestos Internos (SII). These records cover the universe of formal private-sector firms. The SII began collecting these data in 2014, and electronic invoicing became mandatory in 2017. Our sample contains invoices dated from January 2015 through December 2023 and is accessed through the Central Bank of Chile.⁵

Each invoice has two parts. The first contains transaction-level information: anonymized tax identifiers for the supplier and buyer, the value of the transaction, the municipality associated with each party (corresponding to the municipality of the firm’s registered headquarters), and payment terms (cash versus trade credit, including maturity). The second part contains line-item details on the goods and services traded, including a free-text product description, price and quantity (including discounts or surcharges when applicable), and a product identifier. Because the product identifier is assigned at the level of unique text descriptions, the same good is often split into multiple IDs due to minor differences in wording, and the IDs can be extremely granular.⁶ We therefore harmonize products using the Central Product Classification (CPC)⁷.

We merge the transaction records with two additional administrative datasets. The first contains firm balance-sheet and tax information from annual form F22 and monthly form F29, including total sales, intermediate-input expenditures, the wage bill, and fixed assets. The second is the employer–employee dataset from form DJ1887, which reports annual earnings paid by each firm to each worker and months of employment. Reported earnings include wages, salaries, bonuses, tips, and other taxable labor-income components. All datasets are anonymized.

2.2 Units of observation and network definitions

Throughout the paper, a firm is defined by a unique anonymized tax identifier. Unless stated otherwise, the term firm refers to a supplier. Buyers and suppliers can belong to any sector of the

⁵This paper was developed within the scope of the research agenda conducted by the Central Bank of Chile (CBC) in economic and financial affairs of its competence. The CBC has access to anonymized information from public and private entities through collaboration agreements. To protect the privacy of workers and firms, the CBC requires that the development, extraction, and publication of results not permit the direct or indirect identification of individuals or firms. CBC officials processed the disaggregated data. All analysis was implemented by the authors and does not involve or compromise the SII. Information in the SII databases is of a tax nature and originates in taxpayer self-declarations; its veracity is therefore not the responsibility of the SII.

⁶For example, “BOY SNEAKER” and “GIRL SNEAKER” are classified as different products.

⁷The CPC is an internationally harmonized classification system that organizes goods and services based on standardized concepts, definitions, and coding rules.

Chilean economy, although the main stylized facts focus on manufacturing suppliers. Suppliers include both upstream (primary) and downstream (secondary) firms in the network, i.e., any firm that sells a good or service to another firm. Transactions are strictly business-to-business because purchases by final consumers are not observed.

Let s denote a supplier, b a buyer, g a 5-digit CPC product, and t a year. Let x_{sbgt} denote sales from supplier s to buyer b of product g in year t . We define

$$\begin{aligned}\mathcal{B}_{st} &= \{b : \exists g \text{ such that } x_{sbgt} > 0\}, & \text{buyers of supplier } s, \\ \mathcal{G}_{st} &= \{g : \exists b \text{ such that } x_{sbgt} > 0\}, & \text{products sold by supplier } s, \\ \mathcal{R}_{st} &= \{(b, g) : x_{sbgt} > 0\}, & \text{active buyer-product relationships of supplier } s.\end{aligned}$$

Supplier sales observed in the transaction network are

$$S_{st}^{B2B} = \sum_b \sum_g x_{sbgt}.$$

When total supplier sales are available from tax filings, we denote them by S_{st}^{total} . The corresponding extensive-margin objects are $|\mathcal{B}_{st}|$, the number of buyers; $|\mathcal{G}_{st}|$, the number of products; and $|\mathcal{R}_{st}|$, the number of active buyer-product relationships. Therefore, average number of buyers per product is

$$\frac{|\mathcal{R}_{st}|}{|\mathcal{G}_{st}|},$$

while the average number of products per buyer is

$$\frac{|\mathcal{R}_{st}|}{|\mathcal{B}_{st}|}.$$

.

We also use the ratio

$$\frac{|\mathcal{B}_{st}|}{|\mathcal{G}_{st}|},$$

which measures buyer breadth relative to product scope.

Sales within a supplier-buyer link are

$$S_{sbt}^{\text{link}} = \sum_g x_{sbgt},$$

while sales associated with a supplier-product relationship are

$$S_{sgt}^{\text{prod}} = \sum_b x_{sbgt}.$$

2.3 Sample construction

We apply several cleaning and sample restrictions. First, we retain firms that report more than one paid worker. Second, we restrict attention to firms that file monthly tax form F29 in all sample years used in the relevant exercise and report positive sales. Following [Burstein et al. \(2024\)](#), we remove transactions that exceed USD 125 million, transactions with non-positive prices or quantities, and invoices in which the supplier and buyer are the same firm. Finally, we use product-level minimum and maximum price filters from the Chilean National Statistics Institute (INE) to remove line items with implausible unit prices. The exact estimation sample varies across exercises because different results require different levels of aggregation and support.

2.4 Descriptive statistics

Table 1 reports descriptive statistics for manufacturing suppliers in 2017 and 2023, separately for multi-product and single-product firms. Multi-product production is pervasive in the Chilean manufacturing sector. Multi-product suppliers account for 86.2% of suppliers in 2017 and 83.8% in 2023, and they operate at substantially greater scale than single-product suppliers. In 2017, the average multi-product supplier has total sales of 2.35millionandserves68.9buyers, comparedwith0.97 million in sales and 7.6 buyers for the average single-product supplier. These differences remain large in 2023. This suggests that the interaction between product and customer scope is relevant for most manufacturing suppliers and is especially important among larger firms.

Multi-product suppliers also exhibit substantial heterogeneity in both dimensions of scope. In 2017, they sell an average of 16.1 products and serve 68.9 buyers, while the median supplier sells only nine products and serves nine buyers. Similar gaps between means and medians remain in 2023 and are especially large for the number of buyers. Sales are also highly right-skewed. This means that a relatively small group of suppliers operate with very broad product portfolios and customer networks. Capturing this heterogeneity across firms is an important goal of the quantitative model.

The comparison across years reveals a different pattern for the two margins. Product scope is relatively stable: among multi-product suppliers, the mean number of products changes only from 16.1 in 2017 to 15.8 in 2023, while the median declines from nine to eight. Buyer reach contracts more noticeably, with the average number of buyers falling from 68.9 to 59.9 and the median from nine to eight. Consistent with this decline, average buyers per product falls from 3.75 to 3.22. At the same time, buyer concentration rises modestly, from 0.51 to 0.54, whereas product specialization changes little, from 0.54 to 0.55. Thus, over this period, suppliers' product portfolios remain comparatively stable while their customer networks become somewhat narrower and more concentrated. We interpret these changes descriptively rather than as evidence about the mechanisms studied below. The empirical analysis instead exploits the detailed buyer–product structure of

supplier growth and the within-supplier relationship between buyer breadth and product scope.

Table 1: Descriptive Statistics for Chilean Manufacturing Suppliers

	2017			2023		
	Mean	Median	Std. Dev.	Mean	Median	Std. Dev.
<i>Panel A: Multiproduct firms (suppliers)</i>						
Total sales (million USD, firm level)	2.35	0.08	43.95	4.22	0.10	84.29
B2B intermediate sales (million USD)	0.96	0.02	29.53	1.67	0.02	53.283
Number of buyers	68.904	9.000	784.554	59.914	8.000	667.048
Number of products	16.061	9.000	20.955	15.764	8.000	21.583
Number of buyers per product	3.749	1.000	22.768	3.220	1.000	20.357
Product specialization ratio	0.541	0.512	0.243	0.553	0.528	0.245
Buyer concentration ratio	0.514	0.475	0.287	0.544	0.514	0.286
Number of firms	37,179	–	–	28,938	–	–
Share of all suppliers (%)	86.2	–	–	83.8	–	–
<i>Panel B: Single-product firms (suppliers)</i>						
Total sales (million USD, firm level)	0.971	0.022	25.378	1.166	0.023	27.427
B2B intermediate sales (million USD)	0.268	0.002	10.445	0.272	0.002	6.572
Number of buyers	7.638	1.000	32.583	4.767	1.000	22.324
Number of products	1.000	1.000	0.000	1.000	1.000	0.000
Number of buyers per product	7.638	1.000	32.583	4.767	1.000	22.324
Product specialization ratio	1.000	1.000	0.000	1.000	1.000	0.000
Buyer concentration ratio	0.801	1.000	0.295	0.845	1.000	0.259
Number of firms	5,935	–	–	5,585	–	–
Share of all suppliers (%)	13.8	–	–	16.2	–	–

Notes: Sales are in million USD. Shares are computed within year among single-product and multiproduct suppliers. A supplier is classified as multi-product in a year if it sells at least two distinct 5-digit CPC products in that year. The product-specialization ratio is the share of supplier sales accounted for by its largest product. The buyer-concentration ratio is defined analogously using the supplier’s largest buyer. Buyers per product are calculated at the supplier level before averaging across suppliers.

3 Empirical Facts

This section documents three model-facing facts about how buyer reach and product scope interact within suppliers. First, the largest supplier-growth episodes disproportionately combine the addition of new buyers and new products. Second, the composition of growth varies systematically with suppliers’ initial size: smaller suppliers rely relatively more on margins involving new products, while larger suppliers increasingly expand by selling established products to new buyers and through existing buyer–product relationships. Third, buyer breadth and product scope co-move strongly within suppliers, providing direct evidence on the interaction between the two margins.

The first two facts characterize how suppliers move between an initial and a final allocation and therefore have a natural comparative-static counterpart in the model. The third fact disciplines the model’s central mechanism linking buyer breadth and product scope.

Appendix B provides complementary evidence that supplier–buyer links and supplier–product

relationships are empirically distinct objects. Link sales peak earlier and decline faster, cohort profiles show a more systematic deterioration in sales with link age, and survival rates are consistently lower for supplier–buyer links than for supplier–product relationships. Because our model is static, these patterns are not quantitative targets of the model. Rather, they motivate treating buyer reach and product scope as separate margins of adjustment.

3.1 Fact 1: Supplier growth combines buyer and product expansion

We begin by studying how suppliers move between an initial and a final allocation. Although this decomposition is measured over time, its model counterpart is a comparative-static exercise: the static model can compare a baseline equilibrium with a counterfactual equilibrium and decompose the resulting change across buyer and product margins.

Let $t \in \{2017, 2023\}$. Define the sets of buyers and products of supplier s in the baseline year:

$$\mathcal{B}_{s0} = \{b : \exists g \text{ such that } x_{sbg,2017} > 0\}, \quad \mathcal{G}_{s0} = \{g : \exists b \text{ such that } x_{sbg,2017} > 0\}.$$

We classify each unit of 2023 sales according to whether its buyer and product were present in 2017:

$$\begin{aligned} S_{s,2023}^{\text{OLOP}} &= \sum_{b \in \mathcal{B}_{s0}} \sum_{g \in \mathcal{G}_{s0}} x_{sbg,2023}, & \text{old link, old product,} \\ S_{s,2023}^{\text{NLOP}} &= \sum_{b \notin \mathcal{B}_{s0}} \sum_{g \in \mathcal{G}_{s0}} x_{sbg,2023}, & \text{new link, old product,} \\ S_{s,2023}^{\text{OLNP}} &= \sum_{b \in \mathcal{B}_{s0}} \sum_{g \notin \mathcal{G}_{s0}} x_{sbg,2023}, & \text{old link, new product,} \\ S_{s,2023}^{\text{NLNP}} &= \sum_{b \notin \mathcal{B}_{s0}} \sum_{g \notin \mathcal{G}_{s0}} x_{sbg,2023}, & \text{new link, new product.} \end{aligned}$$

Because all 2017 sales occur with buyers and products that are old relative to the 2017 baseline,

$$S_{s,2017} = S_{s,2017}^{\text{OLOP}}.$$

The four components of supplier sales growth are therefore

$$\begin{aligned} \Delta S_s^{\text{OLOP}} &= S_{s,2023}^{\text{OLOP}} - S_{s,2017}, \\ \Delta S_s^{\text{NLOP}} &= S_{s,2023}^{\text{NLOP}}, \\ \Delta S_s^{\text{OLNP}} &= S_{s,2023}^{\text{OLNP}}, \\ \Delta S_s^{\text{NLNP}} &= S_{s,2023}^{\text{NLNP}}. \end{aligned}$$

They satisfy the accounting identity

$$\Delta S_s = S_{s,2023} - S_{s,2017} = \Delta S_s^{\text{OLOP}} + \Delta S_s^{\text{NLOP}} + \Delta S_s^{\text{OLNP}} + \Delta S_s^{\text{NLNP}}. \quad (1)$$

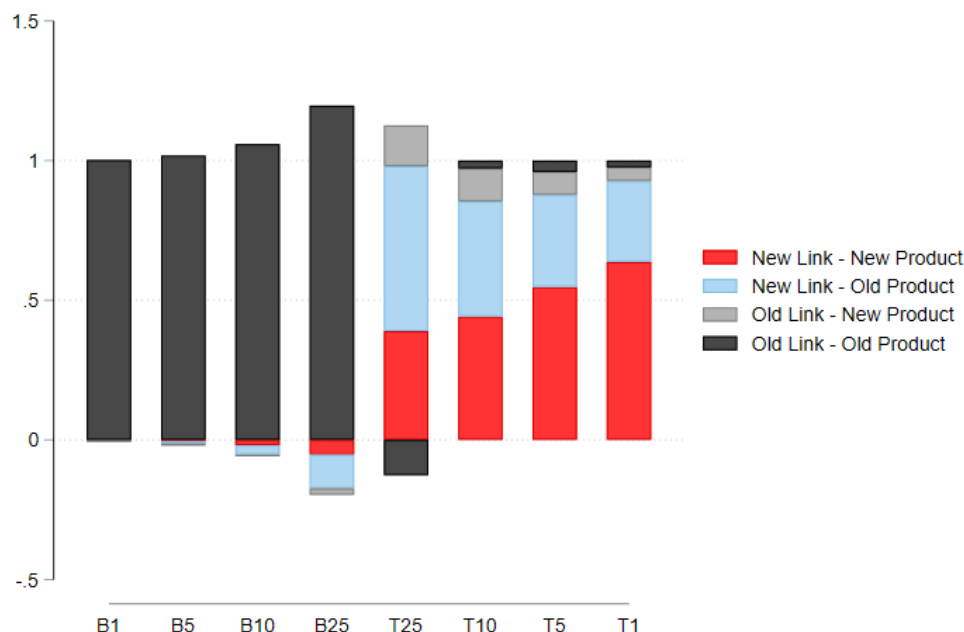
The OLOP component is a net intensive-and-exit term. It includes changes in sales among continuing old-buyer–old-product relationships and losses from old relationships that are no longer active in 2023. The other three components involve at least one extensive-margin change relative to the baseline.

Decomposition by realized growth. We first group suppliers by their realized 2017–2023 sales growth and compute the average contribution of each component to ΔS_s . Figure 3 asks an ex post accounting question: among suppliers that subsequently shrink or grow by different amounts, which margins account for their realized change in sales?

The lower tail displays little offsetting expansion through new buyers or new products. Contraction is concentrated in OLOP, reflecting lower sales in the continuing portion of the supplier’s baseline buyer–product space and losses from exiting relationships. Because each component is normalized by the supplier’s total change in sales, the OLOP share appears positive for shrinking suppliers: both the OLOP contribution and total sales growth are negative, so their ratio is positive. By contrast, any positive contribution from new buyers or new products is divided by a negative total change and therefore appears as a negative share. In the upper tail, extensive margins account for a much larger share of growth. The NLNP component shows that the fastest-growing suppliers disproportionately combine new buyers and new products rather than expanding along only one margin.

The increasing importance of NLNP among the fastest-growing suppliers is also visible when we isolate the three margins involving a new buyer or a new product. Appendix C.1 and C.2 provide two additional decompositions that reinforce these patterns. The first one isolates the three margins involving a new buyer or a new product by excluding OLOP and renormalizing NLOP, OLNP, and NLNP relative to their net extensive-margin adjustment. The composition shifts strongly toward NLNP as realized supplier growth increases, with the joint new-buyer–new-product margin accounting for the largest share among the fastest-growing suppliers. Second, appendix C.2 separates OLOP into changes among baseline buyer–product relationships that remain active through 2023 and losses from relationships that disappear. Relationship destruction accounts for most of the contraction in the lower tail, whereas the largest expansions rely increasingly on the formation of new buyer–product combinations. Together, these additional decompositions reinforce the main result that large supplier expansions disproportionately involve simultaneous growth along the buyer and product margins.

Figure 3: Decomposition of supplier sales growth, 2017–2023



Notes: The figure decomposes supplier sales growth between 2017 and 2023 across intervals of the realized-growth distribution. Each bar reports the average contribution of new link–new product (NLNP), new link–old product (NLOP), old link–new product (OLNP), and old link–old product (OLOP). B denotes intervals in the bottom half and T intervals in the top half. Each category corresponds to the range between adjacent reported cutoffs. For example, B5 denotes the interval from the 1st to the 5th percentile, while T90 denotes the interval from the 90th to the 95th percentile.

3.2 Fact 2: As suppliers grow larger and older, expansion shifts from new products and buyers toward existing products and new buyers

The previous figures classify suppliers according to their realized change in sales. The sorting variable is therefore the outcome being decomposed. We now ask a different question: do suppliers that begin from different initial conditions expand through different combinations of buyer acquisition, product introduction, and sales within continuing relationships? We examine two dimensions of initial heterogeneity: suppliers’ position in the 2017 sales distribution and their stage in the firm life cycle.

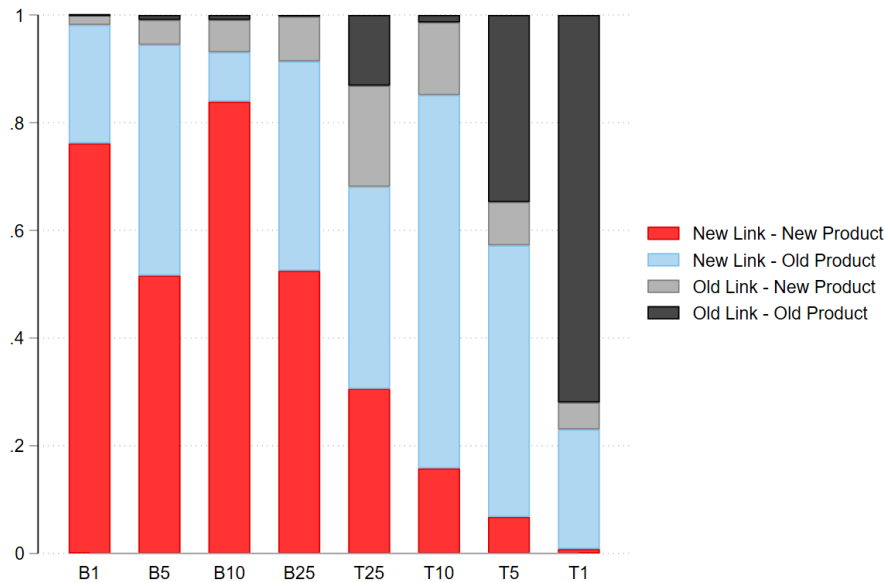
Heterogeneity by initial supplier scale. We first assign suppliers to groups according to their total sales in 2017 and then decompose their sales growth between 2017 and 2023. The current version of this exercise restricts attention to suppliers with positive growth. Accordingly, the figure describes the composition of growth among expanding suppliers; it does not establish which initial-size groups grow faster or are more likely to grow.

Figure 4 reveals a clear pattern across the initial-size distribution. Among suppliers with

positive growth, initially smaller suppliers rely relatively more on margins involving new products, particularly the joint addition of new buyers and new products (NLNP). As initial supplier size increases, growth shifts toward existing products: larger suppliers rely relatively more on sales through existing buyer–product relationships (OLOP) and on forming new buyer links for products already in their portfolios (NLOP). Thus, product expansion is relatively more important for initially smaller suppliers, while larger suppliers expand relatively more by deepening existing relationships and extending established products to new buyers.

This increasing importance of the buyer margin among larger suppliers is consistent with its broader role in accounting for cross-sectional differences in firm size. In 2017, a decomposition of the variance of log supplier sales into the number of buyers and average sales per buyer attributes 42.9 percent to the buyer extensive margin and 57.1 percent to sales per buyer. The analogous product-side decomposition attributes only 27.3 percent to the number of products, with the remaining 72.7 percent accounted for by sales per product. Thus, although both dimensions of scope are quantitatively important, differences in customer reach account for a larger share of supplier-size dispersion than differences in product scope. Appendix Figure 11 reports both decompositions.

Figure 4: Composition of positive sales growth by initial supplier size



Notes: Suppliers are grouped according to their position in the 2017 total-sales distribution. Within each initial-size group, the figure decomposes positive supplier sales growth between 2017 and 2023 into new link–new product (NLNP), new link–old product (NLOP), old link–new product (OLNP), and old link–old product (OLOP). Because the sample is restricted to suppliers with positive growth, the figure describes how expanding suppliers grow rather than differences in average growth rates or growth probabilities across initial-size groups. Each category corresponds to the interval between adjacent reported percentile cutoffs.

The distinction between Figures 3 and 4 is important. The first is an ex post accounting

exercise: it asks which margins characterize different realized growth outcomes. The second is an ex ante heterogeneity exercise that asks whether the composition of subsequent expansion varies with a supplier characteristic measured before that expansion occurs. The latter is therefore especially useful for disciplining heterogeneous comparative-static responses in the model.

This heterogeneity has a direct model counterpart. Initial supplier sales can reflect baseline differences in productivity, buyer reach, or product scope. The model can therefore apply a common change in the economic environment and ask whether initially small and large suppliers respond through different combinations of buyer acquisition, product adoption, and changes within continuing relationships. The cross-sectional variance decomposition provides a complementary benchmark for assessing the relative quantitative importance of the buyer and product margins for the firm-size distribution.

Appendix C.3 provides an additional view of the initial-size gradient by grouping suppliers into broad low-, middle-, and high-sales categories. It also reports the decomposition after excluding OLOP, which focuses on growth coming from the extensive margins. The same pattern remains. The contribution of NLNP declines with initial supplier size, while NLOP becomes increasingly important. Even among transactions involving a new buyer or a new product, smaller suppliers are more likely to expand both margins together, whereas larger suppliers rely relatively more on selling products already in their portfolios to new customers.

Appendix C.4 examines the second dimension of initial heterogeneity, supplier age, while conditioning on sales in 2017. The life-cycle decomposition shows a similar pattern. Younger suppliers rely relatively more on NLNP, while NLOP becomes more important as firms age. More mature suppliers increasingly expand by selling products already in their portfolios to new buyers. Because age is not an explicit state variable in our static model, we treat this evidence as complementary to the initial-size results rather than as a separate quantitative target of the model.

3.3 Fact 3: Buyer breadth and product scope co-move within suppliers

The previous facts establish that buyer relationships and product scope are distinct margins. We now ask whether these two margins are systematically related within suppliers. A priori, their relationship could reflect either a trade-off, whereby expanding along one margin comes at the expense of the other, or a complementarity, whereby a larger buyer base raises the return to maintaining a broader product portfolio. And broader product portfolio, in turn, raises the value of forming additional buyer relationships because it gives the supplier more products through which to generate surplus with a potential buyer, making the fixed cost of forming the relationship easier to cover.

This question is closely related to recent work on the interaction between demand-expansion investments, such as advertising, and innovation. For example, [Cavenaile and Roldan-Blanco](#)

(2021) provide evidence of substitution between R&D and advertising among U.S. firms, whereas recent models often assume complementarity between customer expansion and innovation, as in Einav et al. (2021) and Afrouzi et al. (2023). These studies focus primarily on final-consumer settings. Our setting instead allows us to study product and customer scope directly within a production network.

We begin by relating a supplier’s product scope to its buyer breadth. Let Buyers_{st} denote the number of distinct buyers served by supplier s in year t , and let Products_{st} denote the number of distinct products (CPCs) it sells. We estimate:

$$\log(\text{Products}_{st}) = \beta_1 \log(\text{Buyers}_{st}) + \theta \log(\text{Sales}_{st}) + \gamma_t + \delta_s + \varepsilon_{st}, \quad (2)$$

where γ_t are year fixed effects and δ_s are supplier fixed effects, respectively. The supplier fixed effects are particularly important for our purposes because they absorb persistent differences across firms that may jointly shape buyer breadth and product scope. In the model below, these persistent firm characteristics are summarized by supplier productivity. Thus, the preferred specification compares the same supplier over time, holding fixed its underlying productivity type. Contemporaneous sales are included as an additional control for the supplier’s current scale. The coefficient β therefore captures whether, conditional on persistent supplier heterogeneity and current firm size, years in which a supplier serves more buyers are also years in which it operates a broader product portfolio. We interpret this relationship as a conditional association rather than a causal effect of buyer acquisition on product scope.

Table 2 reports the results. The relationship is strong even within suppliers. With only year and supplier fixed effects, the elasticity of product scope with respect to buyer breadth is 0.555, which is the empirical moment that maps most closely into the model because the model is static and therefore cannot reproduce within-supplier changes over time literally. Controlling additionally for supplier sales reduces the coefficient only modestly, to 0.486. Because sales are themselves an outcome of the two margins, we treat the specification without the sales control (column 5) as the benchmark target for the model.

Appendix D decomposes this relationship. When product scope is measured using only products sold to incumbent buyers, the elasticity is 0.382, about 69 percent of the total (59 percent when controlling for sales). This decomposition is informative about mechanisms. A new buyer can add products to a supplier’s measured scope simply by purchasing products the supplier already had but was not selling; it cannot add products to relationships that already existed. The incumbent component therefore rules out the mechanical new-buyer matching channel, while the remainder combines development with products that new buyers reveal. Section 7.1 compares this empirical decomposition with its model counterpart.

The positive relationship is not specific to the stock of products. Appendix E considers product additions, defined as products sold by the supplier in year t that were not sold in $t - 1$. With year and supplier fixed effects, a 10 percent increase in buyers is associated with approximately 3.92% additional product introductions. Thus, within suppliers, years with more buyers are also associated with more product introductions. The magnitude is economically meaningful.

Taken together, these results establish a robust positive relationship between the two dimensions of firm scope. Firms with broader customer bases also operate broader product portfolios, and years in which a supplier serves more buyers are associated with more product additions. These patterns do not by themselves identify a structural complementarity, since unobserved firm-level shocks may affect both margins. Instead, we treat their systematic co-movement as a model-facing fact. The evidence therefore motivates a framework in which buyer acquisition and product adoption are jointly determined rather than independent choices. In the model, the two margins are linked through two channels. The two margins are linked through the organization of fixed costs (amortization) and buyer-specific product matching. A broader buyer base raises the return to developing products and gives existing products more chances to find a buyer, while a broader product portfolio increases the value of forming buyer relationships. We then ask whether these mechanisms can account for the empirical co-movement between buyer breadth and product scope, together with some of the other facts documented above.

Table 2: Buyer Breadth and Product Scope

	<i>Dependent variable: Log Number of Products</i>					
Log Number of Buyers	0.498*** (0.00157)	0.464*** (0.00183)	0.518*** (0.00147)	0.458*** (0.00179)	0.555*** (0.00324)	0.486*** (0.00352)
Log Sales		0.0487*** (0.00135)		0.0877*** (0.00139)		0.127*** (0.00258)
Observations	138,784	138,534	138,784	138,534	138,219	137,966
Firms	23,382	23,380	23,382	23,380	22,817	22,812
R^2	0.434	0.439	0.565	0.579	0.886	0.890
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Sector FE	No	No	Yes	Yes	No	No
Supplier FE	No	No	No	No	Yes	Yes

Notes: The dependent variable is the log number of distinct products (CPCs) sold by supplier s in year t . The main explanatory variable is the log number of distinct buyers served by the supplier in the same year. Columns (1)–(2) include year fixed effects; columns (3)–(4) additionally include supplier-sector fixed effects; and columns (5)–(6) include supplier fixed effects instead. Even-numbered columns additionally control for contemporaneous log supplier sales. Column (5) is the model’s calibration target; column (6) is reported for robustness. Heteroskedasticity-robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

4 Model

The empirical analysis shows that firm size in production networks reflects two distinct extensive margins: product scope and buyer relationships. These margins are strongly interconnected within firms, and their interaction matters for the firm-size distribution because differences in scope can amplify underlying firm heterogeneity and generate substantial sales dispersion. Since both margins are endogenous and respond to common profitability forces, reduced-form correlations alone do not identify the mechanism linking them. We therefore develop a tractable static model in which suppliers jointly choose product scope and buyer relationships. The model serves three purposes. First, it isolates the mechanism generating complementarity between the two margins. Second, it quantifies how shocks to product-adoption or buyer-linkage costs propagate across both margins and reshape the firm-size distribution. Third, it assesses their contribution to firm-size dispersion while accounting for this complementarity.

Our model builds on the heterogeneous-firm logic of [Hopenhayn \(1992\)](#) and the rank-based multiproduct structure of [Arkolakis et al. \(2021\)](#), adapting them to a production-network environment in which destinations are individual buyers. The link-formation margin follows the literature modeling network formation through fixed costs per relationship ([Lim, 2018](#); [Huneus, 2018](#); [Bernard et al., 2022](#)), rather than through search ([Miyauchi, 2024](#); [Arkolakis et al., 2023](#)). Suppliers are heterogeneous in productivity and choose how many products to develop and which buyers to serve. Product-adoption costs are paid once at the firm level and can therefore be amortized across buyer relationships, whereas link formation requires buyer-specific fixed costs. This distinction generates the model’s first source of complementarity, amortization. A larger buyer base increases the return to developing additional products, while a broader product portfolio makes forming new buyer relationships more valuable. Buyer-specific product matching creates a second source of complementarity, which we describe below.

Motivated by the absence of systematic cannibalization across products and buyer relationships documented in [Appendix F](#), the benchmark abstracts from displacement of existing sales as suppliers expand product or buyer scope. We consider within-link cannibalization separately in the nested-CES extension. A central modeling choice is how a supplier’s products map to individual buyers. Our baseline allows this mapping to be buyer-specific. Firms develop products at the supplier level, but their relevance for a particular buyer depends on an idiosyncratic product–buyer match. This choice is disciplined by the data. Firm-level and within-link product rankings are positively but imperfectly aligned, a pattern difficult to reconcile with a rigid common product ladder.⁸ The matching structure also connects our framework to models of two-sided heterogeneity in firm-to-firm trade ([Bernard et al., 2018](#); [Eaton et al., 2026](#); [Demir et al., 2024](#); [Miyauchi, 2024](#)),

⁸A common product ladder, as in [Arkolakis et al. \(2021\)](#), imposes a shared ranking of products across buyers, so that buyers differ in how far down the same product ranking they purchase.

in which trading relationships and their intensity are endogenous.

4.1 Setting

Suppliers are indexed by $s \in \mathcal{S}$ and buyers by $b \in \mathcal{B}$.⁹ Supplier s has productivity $\phi_s > 0$ and chooses a firm-level portfolio of G_s products. Supplier productivity follows a Pareto distribution,

$$\phi_s \sim \text{Pareto}(\theta_\phi, \phi_{\min}), \quad \phi_s \geq \phi_{\min} > 0. \quad (3)$$

Products are ordered from core to peripheral by rank $g \in 1, \dots, G_s$, with product-level productivity

$$\phi_{s,g} = \frac{\phi_s}{h(g)}, \quad h(g) = g^\alpha, \quad \alpha \geq 0. \quad (4)$$

Products farther from the core are therefore less productive on average.

For each developed product g and potential buyer b , supplier s draws an idiosyncratic product–buyer match

$$\zeta_{sb} \sim \text{Pareto}(\theta_\zeta, \zeta_{\min}), \quad \theta_\zeta > 1, \quad \zeta_{\min} = \frac{\theta_\zeta - 1}{\theta_\zeta}, \quad (5)$$

so that $\mathbb{E}[\zeta_{sb}] = 1$. Product rank captures systematic differences in profitability across the supplier’s portfolio, while ζ_{sb} allows product relevance to vary across buyers. Conditional on an active relationship, the set of products sold to buyer b is therefore buyer-specific,

$$\mathcal{G}_{sb} \subseteq 1, \dots, G_s, \quad G_{sb} = |\mathcal{G}_{sb}| \leq G_s, \quad (6)$$

and need not form a common prefix of the supplier’s product ranking.

Fixed costs and the amortization mechanism. The supplier faces three fixed costs that operate at different levels of its problem and govern distinct margins of adjustment. First, developing product g requires the firm-level cost

$$f^P(g) = f^P(1)g^\delta, \quad \delta \geq 0, \quad (7)$$

so that a supplier developing G_s products pays total product-development costs

$$F^P(G_s) \equiv \sum_{g=1}^{G_s} f^P(g) = f^P(1) \sum_{g=1}^{G_s} g^\delta. \quad (8)$$

⁹We interpret suppliers as atomless from the perspective of each buyer’s input bundle. The finite supplier population used in the quantitative implementation approximates this underlying environment.

Crucially, this cost is paid once at the firm level. Once product g has been developed, it can potentially generate profits across all buyers served by the supplier.

Second, forming relationship (s, b) requires paying the link-level cost f_{sb}^L . Third, conditional on having developed product g and formed link (s, b) , selling that product to buyer b requires the product–buyer activation cost

$$f_{sb}^A(g) = f_{sb,0}^A g^\mu. \quad (9)$$

If $\mathcal{G}_{sb} \subseteq \{1, \dots, G_s\}$ denotes the set of developed products that supplier s activates for buyer b , total activation costs on the link are

$$F_{sb}^A(\mathcal{G}_{sb}) \equiv \sum_{g \in \mathcal{G}_{sb}} f_{sb}^A(g) = f_{sb,0}^A \sum_{g \in \mathcal{G}_{sb}} g^\mu. \quad (10)$$

Thus, if $\mathcal{B}_s$ denotes the set of buyers served by supplier s , its total fixed costs are

$$F_s^{\text{fixed}} = F^P(G_s) + \sum_{b \in \mathcal{B}_s} [f_{sb}^L + F_{sb}^A(\mathcal{G}_{sb})]. \quad (11)$$

This distinction across levels is central to the model’s complementarity between buyer breadth and product scope. Product-development costs are incurred once by the supplier and can therefore be amortized across buyer relationships. As the supplier serves more buyers, a newly developed product has more potential relationships over which to generate surplus, increasing the return to product adoption. Link and activation costs, by contrast, are not amortized across buyers: acquiring an additional buyer requires paying a new f_{sb}^L , and selling a particular product on that relationship requires paying the corresponding $f_{sb}^A(g)$. The firm-level location of $F^P(G_s)$ is therefore essential for the model’s amortization force.

The complementarity also operates in the opposite direction. A supplier with a broader developed product portfolio can generate surplus from more potential product–buyer matches, raising the value of forming an additional buyer relationship. Buyer breadth and product scope are consequently jointly determined. Additional buyers increase the return to product development, while additional products increase the return to link formation. This extends the standard product-ladder structure of [Arkolakis et al. \(2021\)](#) to a production-network environment in which individual buyer relationships constitute the relevant destinations.

In the quantitative model, heterogeneity in link and activation costs is parameterized as

$$f_{sb}^L = \bar{f}_b^L \eta_{sb} c_s, \quad f_{sb,0}^A = \bar{f}_b^A \eta_{sb} c_s, \quad (12)$$

where $\eta_{sb} > 0$ captures pair-specific cost heterogeneity and $c_s > 0$ captures persistent supplier heterogeneity in relationship costs. We normalize $\mathbb{E}[\eta_{sb}] = \mathbb{E}[c_s] = 1$. The supplier component c_s is common across all relationships of supplier s , while η_{sb} varies across supplier–buyer pairs.

The pair-level cost shock η_{sb} and the product-level match shock ζ_{sb} capture distinct sources of heterogeneity. η_{sb} makes some supplier–buyer relationships systematically easier to establish and operate, while ζ_{sb} determines the suitability of an individual developed product for a particular buyer.

Demand and operating profits. We work in partial equilibrium. Buyer b has CES demand across available supplier–product varieties with elasticity $\sigma > 1$. Let E_b denote buyer expenditure and P_b its CES price index. Under monopolistic competition, product-level revenue is

$$r_{sb} = D_b \phi_s^{\sigma-1} g^{-\alpha(\sigma-1)} \zeta_{sb}, \quad (13)$$

where

$$D_b \equiv E_b \left(\frac{P_b}{[\sigma/(\sigma-1)]w} \right)^{\sigma-1} \quad (14)$$

summarizes buyer profitability. Variable operating profits are a constant share of revenue:

$$\pi_{sb} = \frac{1}{\sigma} r_{sb}. \quad (15)$$

The same match shock governs both selection into product–buyer trade and realized sales conditional on activation. Thus, unlike the deterministic-ladder benchmark, the matching specification does not require a separate ex-post product appeal shock to generate residual within-link sales dispersion. Appendix I derives these expressions from the buyer’s CES problem.

4.2 Supplier’s two-stage problem

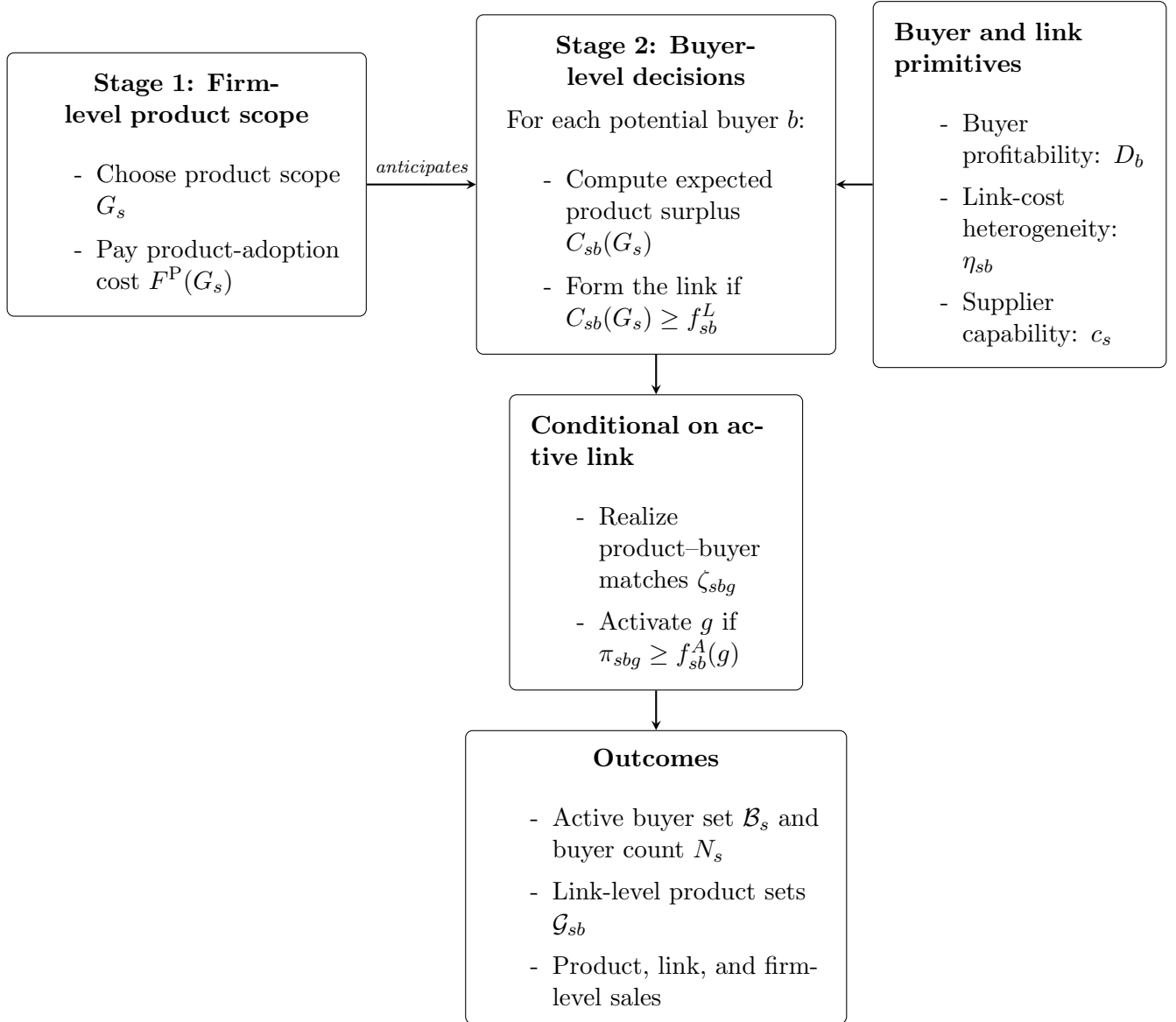
The supplier’s problem has a recursive two-stage structure, which we solve by backward induction in expected value. In Stage 1, supplier s chooses its firm-level product scope G_s and pays the corresponding product-development costs. This determines the set of products that the supplier is capable of selling, but not the subset ultimately sold to each buyer.

In Stage 2, conditional on G_s , the supplier evaluates each potential buyer relationship. Link formation is an ex-ante decision. The supplier compares the expected surplus generated by its G_s developed products on relationship (s, b) with the buyer-specific link cost f_{sb}^L . If the relationship is formed, the product–buyer match shocks ζ_{sb} are realized. Each developed product is then activated for buyer b whenever its realized operating profit covers its product–buyer activation cost $f_{sb}^A(g)$. Thus, firm-level scope G_s places an upper bound on link-level product scope, while the realized matches determine which subset of the developed portfolio is actually traded with each buyer.

Anticipating these Stage 2 decisions, the supplier chooses G_s in Stage 1 to maximize expected

profits across all potential buyer relationships net of firm-level development costs. Figure 5 summarizes the sequence of decisions and uncertainty.

Figure 5: Timing of the supplier's two-stage problem



Notes: In Stage 1, the supplier chooses firm-level product scope G_s and pays $F^P(G_s)$, anticipating the buyer-level problem. In Stage 2, buyer profitability D_b and pair-specific cost heterogeneity η_{sb} determine the expected value of each potential relationship. The supplier forms the link when expected product surplus covers f_{sb}^L . Conditional on an active link, product-buyer match shocks ζ_{sbg} determine which developed products are activated and sold to that buyer.

4.3 Optimality conditions

Fix firm-level scope G_s . Define the supplier–buyer profitability shifter

$$A_{sb} \equiv \frac{D_b}{\sigma} \phi_s^{\sigma-1}, \quad (16)$$

and let

$$\kappa \equiv \mu + \alpha(\sigma - 1). \quad (17)$$

Developed product g is profitable on link (s, b) whenever its realized match quality is sufficiently high:

$$\zeta_{sb} \geq z_{sb}(g) \equiv \frac{f_{sb,0}^A}{A_{sb}} g^\kappa. \quad (18)$$

The threshold $z_{sb}(g)$ is the minimum match quality required to activate product g for buyer b . It is lower for more productive suppliers and more profitable buyers, making products easier to activate on those relationships. When $\kappa > 0$, the threshold also rises with product rank, so more peripheral products require stronger product–buyer matches to be traded.

Given the Pareto distribution of match quality, the corresponding probability that developed product g is activated for buyer b is

$$p_{sb}(g) \equiv \Pr(\zeta_{sb} \geq z_{sb}(g)) = \begin{cases} 1, & \text{if } z_{sb}(g) \leq \zeta_{\min}, \\ \left(\frac{\zeta_{\min}}{z_{sb}(g)}\right)^{\theta_\zeta}, & \text{if } z_{sb}(g) > \zeta_{\min}. \end{cases} \quad (19)$$

Because $\zeta_{\min}$ is the lowest possible realization of match quality, a product whose activation threshold lies below this bound is activated with probability one. Otherwise, activation occurs only when the realized match exceeds the threshold, with probability given by the Pareto tail.

The parameter θ_ζ governs the dispersion of product–buyer match quality and therefore how the supplier’s firm-level portfolio maps into buyer-specific product sets. Conditional on firm-level scope G_s , expected product scope on link (s, b) is

$$\mathbb{E}[G_{sb} \mid G_s] = \sum_{g=1}^{G_s} p_{sb}(g). \quad (20)$$

Thus, G_s determines the set of products that the supplier can potentially sell, while product–buyer matches determine which subset of that portfolio is actually activated for each buyer.

Let expected net surplus from developed product g for buyer b be

$$\Psi_{sb}(g) \equiv \mathbb{E}_\zeta \left[\left(A_{sb} g^{-\alpha(\sigma-1)} \zeta - f_{sb,0}^A g^\mu \right)^+ \right]. \quad (21)$$

Appendix J reports the corresponding closed-form Pareto expression. Expected gross surplus from potential relationship (s, b) is

$$C_{sb}(G_s) \equiv \sum_{g=1}^{G_s} \Psi_{sb}(g). \quad (22)$$

The supplier forms the relationship whenever this surplus covers the link cost:

$$I_{sb}^*(G_s) = \mathbf{1} \ C_{sb}(G_s) \geq f_{sb}^L. \quad (23)$$

The corresponding active buyer set is

$$\mathcal{B}_s(G_s) = \{b \in \mathcal{B} : C_{sb}(G_s) \geq f_{sb}^L\}, \quad N_s(G_s) = |\mathcal{B}_s(G_s)|. \quad (24)$$

Anticipating these buyer-level decisions, the supplier chooses firm-level scope to solve

$$G_s^* \in \arg \max_{G_s \in \{0, \dots, |\Omega|\}} \sum_{b \in \mathcal{B}} [C_{sb}(G_s) - f_{sb}^L]^+ - F^P(G_s), \quad (25)$$

where

$$F^P(G_s) = \sum_{g=1}^{G_s} f^P(g). \quad (26)$$

Conditional on an active relationship and realized matches, the set of products sold to buyer b is

$$\mathcal{G}_{sb} = \{g \leq G_s : \zeta_{sb}g \geq z_{sb}(g)\}, \quad G_{sb} = |\mathcal{G}_{sb}|. \quad (27)$$

Product sets can therefore differ across buyers and need not be nested.

4.4 Equilibrium definition

Because the benchmark is partial equilibrium, buyer demand shifters $D_{bb \in \mathcal{B}}$ and the distribution of potential buyers are taken as given. Therefore, given model primitives and buyer demand shifters $D_{bb \in \mathcal{B}}$, an equilibrium is a collection

$$\{G_s^*, \mathcal{B}_s^*, \mathcal{G}_{sb}^*, r_{sb}^*\}_{s,b}$$

such that: (i) G_s^* solves (25); (ii) buyer relationships satisfy the link-formation rule in (23); (iii) products within active links satisfy the activation rule in (27); and (iv) product-level sales satisfy (13). For each candidate G_s , the link-formation rule is already embedded in the objective in (25). The equilibrium can therefore be computed directly by evaluating expected profits over the finite set of possible product scopes and selecting the maximizing value. Appendix K provides additional details.

4.5 Characterization

The model's central implication follows directly from the different levels at which the fixed costs are paid. Product scope and buyer breadth reinforce one another in both directions.

More products imply weakly more buyers. Because $\Psi_{sb}(g) \geq 0$,

$$C_{sb}(G_s + 1) = C_{sb}(G_s) + \Psi_{sb}(G_s + 1) \geq C_{sb}(G_s). \quad (28)$$

Hence,

$$\mathcal{B}_s(G_s) \subseteq \mathcal{B}_s(G_s + 1). \quad (29)$$

A broader product portfolio therefore weakly enlarges the set of profitable buyer relationships.

A broader buyer base raises the return to product adoption. Define the marginal value of adding product g as

$$\Delta\Pi_s(g) \equiv \Pi_s(g) - \Pi_s(g - 1).$$

It can be decomposed as

$$\Delta\Pi_s(g) = \underbrace{\sum_{b \in \mathcal{B}_s(g-1)} \Psi_{sb}(g)}_{\text{surplus on existing links}} + \underbrace{\sum_{b \in \mathcal{B}_s(g) \setminus \mathcal{B}_s(g-1)} [C_{sb}(g) - f_{sb}^L]}_{\text{newly profitable buyer links}} - f^P(g). \quad (30)$$

The first term makes the amortization mechanism transparent. Since $f^P(g)$ is paid once at the firm level, every additional active buyer adds another nonnegative expected-surplus term to the return from developing product g . The second term reinforces this effect because the new product may also make additional buyer relationships profitable. A simple analytical approximation makes the first channel especially clear. In the interior Pareto region,

$$\Psi_{sb}(g) = \psi_{sb} g^{\mu - \kappa \theta_\zeta}, \quad (31)$$

where

$$\psi_{sb} = \frac{\zeta_{\min}^{\theta_\zeta}}{\theta_\zeta - 1} A_{sb}^{\theta_\zeta} (f_{sb,0}^A)^{1 - \theta_\zeta}. \quad (32)$$

Holding the active buyer set locally fixed and abstracting from the discrete gain from newly

activated links, optimal product scope is approximately

$$G_s^* \approx \left[\frac{\Lambda_s^\psi}{f^P(1)} \right]^{\frac{1}{\delta + \kappa \theta_\zeta - \mu}}, \quad \Lambda_s^\psi \equiv \sum_{b \in \mathcal{B}_s} \psi_{sb} = N_s \bar{\psi}_s, \quad \bar{\psi}_s \equiv \frac{1}{N_s} \sum_{b \in \mathcal{B}_s} \psi_{sb}. \quad (33)$$

This expression separates buyer breadth from average buyer value. Holding $\bar{\psi}_s$ fixed, a larger buyer base N_s increases the return to the marginal product and therefore raises optimal product scope.

Equivalently, defining

$$\Lambda_s^M \equiv \sum_{b \in \mathcal{B}_s} D_b^{\theta_\zeta} (f_{sb,0}^A)^{1-\theta_\zeta}, \quad (34)$$

we obtain

$$G_s^* \propto \left[\frac{\phi_s^{\theta_\zeta(\sigma-1)} \Lambda_s^M}{f^P(1)} \right]^{\frac{1}{\delta + \kappa \theta_\zeta - \mu}}. \quad (35)$$

Thus, conditional on supplier productivity, a more valuable buyer base raises optimal product scope. Together with the monotonicity of buyer-link formation in G_s established above, this delivers the model's two-way interaction between buyer breadth and product scope. The closed-form expression is used for economic interpretation; the quantitative model is solved using the exact discrete problem in equation (25).

A second channel: matching. Amortization operates on developed scope G_s . What the data record, however, is observed scope: the number of developed products sold to at least one buyer. Given G_s and the active buyer set, expected observed scope is

$$\mathbb{E}[\widehat{G}_s \mid G_s, \mathcal{B}_s] = \sum_{g=1}^{G_s} \left[1 - \prod_{b \in \mathcal{B}_s} (1 - p_{sb}(g)) \right], \quad (36)$$

which is increasing in the buyer set even when G_s is held fixed. A supplier with more buyers gives each developed product more independent chances to clear its activation threshold, so a larger share of its portfolio is observed in trade. This matching channel works entirely through selection. It does not require firms to develop additional products or pay any additional firm-level cost. Its strength is governed by how tightly selection binds, which depends on the activation cost $\bar{f}_0^A$ and the dispersion of match quality θ_ζ . When activation almost never binds, $p_{sb}(g) \approx 1$ and observed scope equals developed scope, so all co-movement comes from amortization. When selection binds, matching can dominate. Section 7.1 shows how to separate the two channels quantitatively and that the within-link moments, which discipline $\bar{f}_0^A$ and θ_ζ , determine their relative importance.

The same decisions map directly into firm size. Conditional on realized matches, link sales and supplier sales are

$$t_{sb} = \sum_{g \in \mathcal{G}_{sb}} r_{sbg}, \quad T_s = \sum_{b \in \mathcal{B}_s} t_{sb}. \quad (37)$$

Heterogeneity in productivity, buyer composition, and the two endogenous scope margins therefore jointly shapes the firm-size distribution.

4.6 Estimation strategy

We discipline the quantitative model using the 2017 cross-section of Chilean manufacturing suppliers. The benchmark matching specification is calibrated by simulated method of moments (SMM). For a candidate parameter vector, we solve the supplier problem exactly, simulate the corresponding product and buyer networks, and compare model moments with their empirical counterparts.

The targeted moments are chosen to separately discipline the product and buyer margins as well as their interaction. They include moments of firm-level product scope and buyer breadth, buyers per product, product specialization and buyer concentration, the alignment between firm-level and within-link product importance, and the relationship between buyer breadth and product scope conditional on supplier productivity. The rank-decay and rank-alignment moments are particularly informative about the systematic product ranking and the dispersion of product–buyer match quality, while the levels and dispersion of product and buyer scope discipline the corresponding fixed-cost parameters and cost heterogeneity.

We report two calibrations. Calibration A uses 11 firm-level moments to discipline 8 free parameters. Calibration B adds within-link moments and the upper percentiles of product scope and buyer reach, for 20 moments and 10 free parameters. Parameters that are fixed externally or normalized are held constant throughout the estimation. We minimize equally weighted squared relative deviations between simulated and empirical moments. Moments not included in the SMM objective, including additional distributional statistics and the single-product share, are used as validation exercises.

This strategy exploits information not only on firms’ total product and buyer scope, but also on how products are allocated across individual buyer relationships. The latter moments discipline the matching structure separately from the overall size of the two extensive margins. After estimation, we evaluate model fit and use the calibrated model to quantify the contribution of product scope, buyer breadth, and their complementarity to the firm-size distribution.

5 Model Calibration

5.1 Calibration strategy and empirical targets

We calibrate the model to the cross-sectional distribution of manufacturing suppliers in 2017. We use 2017 as the baseline year because electronic invoicing was mandatory by that year and because it provides a common cross section in which buyer relationships, product scope, and supplier sales can all be measured consistently. Using 2017 also allows us to avoid contamination from large aggregate shocks that affected the Chilean economy in subsequent years, including the social unrest that began in late 2019 and the COVID-19 pandemic. These events generated unusually large disruptions to firm sales, buyer–supplier relationships, and production decisions that are outside the mechanisms captured by our static model.

The calibration is designed to discipline the joint distribution of the two extensive margins at the center of the model, buyer base and product scope, as well as the allocation of products and sales across those margins.

The quantitative analysis focuses on the matching model developed in Section 4. We report two calibrations that differ in the information used to discipline product–buyer matching. Calibration A targets firm-level moments only, as is standard in the literature. Calibration B adds within-link moments that describe how a supplier’s portfolio is allocated across its buyers. Comparing the two is informative in its own right, the two calibrations fit the scope–buyer elasticity similarly well but imply very different contributions of amortization and matching (Section 7.1). Calibration B is our preferred specification, and all counterfactuals below use it unless noted.

The deterministic common-ladder benchmark of Appendix M, in which all buyers purchase prefixes of a common product ranking, serves as a diagnostic. Calibrated to the same within-link moments, it fits worse than the matching model (criterion 1.54 against 1.18 on the ladder’s own 18 targets) and, by construction, cannot generate portfolio coverage below one or non-nested product sets, against 0.61 and 0.70 in the data.

Calibration method. Let Θ denote the vector of parameters calibrated in the model, and let $\mathcal{M}$ denote the corresponding set of targeted moments. We estimate Θ by simulated method of moments (SMM), minimizing an equally weighted sum of squared relative deviations between simulated and empirical moments:

$$\hat{\Theta} = \arg \min_{\Theta} \sum_{j \in \mathcal{M}} \left(\frac{m_j^{\text{model}}(\Theta) - m_j^{\text{data}}}{m_j^{\text{data}}} \right)^2. \quad (38)$$

Using relative deviations places moments measured in products, buyers, slopes, and shares on a comparable scale. Appendix K.1 describes the distributions of all primitives and the simulation

design.

Table 4 summarizes the empirical moments and their model counterparts. It reports the moments used in the SMM objective, additional untargeted moments used to evaluate model fit, and the model predictions for objects for which we do not yet have a direct empirical counterpart. Calibration A targets eleven moments and calibration B twenty. Moments that are not used in the objective function are reported as validation rather than being mechanically fit.

The simulated moments are constructed from realized trade using definitions that mirror those in the data. In particular, a developed product counts toward observed firm scope only if it is sold to at least one buyer, and a buyer counts toward observed buyer reach only if positive trade occurs on the supplier–buyer relationship. This distinction matters in the matching model because a developed product may fail to match with any buyer, and different buyers may trade different subsets of the supplier’s portfolio.

Common empirical targets. The first group of moments in Table 4 disciplines the distribution of product scope and buyer reach. We target the median, mean, and standard deviation of the number of products, $\widehat{G}_s$, together with the median and mean number of buyers, N_s . In the 2017 manufacturing sample, multiproduct suppliers sell a median of 9 and an average of 16.1 products, while serving a median of 9 and an average of 68.9 buyers. The large gaps between the means and medians are important because they require the model to reproduce not only the typical supplier but also the strong right skewness in both extensive margins.

We additionally target the median and mean number of buyers per product, defined as the number of realized buyer–product relationships divided by observed product scope. These moments discipline the joint behavior of buyer reach and product scope rather than either marginal distribution separately. A model could reproduce the average number of buyers and products while generating the wrong way in which the two margins are combined within firms. Buyers per product helps rule out this possibility.

Sales concentration provides a further restriction on the composition of the buyer network. We target the mean buyer-concentration ratio, defined as the share of supplier sales accounted for by its largest buyer. This moment is informative about buyer-quality heterogeneity, relationship costs, and selection into the supplier’s active buyer set.

Finally, we target the conditional relationship between buyer breadth and product scope. Its empirical counterpart is the supplier-fixed-effect specification in Table 2, column (5), in which the coefficient on $\log N_{st}$ is 0.555. Because the model is static, it has no within-supplier time variation and therefore cannot reproduce the empirical fixed-effect regression literally. We construct its

closest model counterpart by estimating

$$\log \hat{G}_s = \beta^{\text{model}} \log N_s + \Gamma(\phi_s) + \varepsilon_s. \quad (39)$$

where $\Gamma(\phi_s)$ denotes twenty productivity-bin fixed effects, and we additionally control for $\log c_s$. Supplier fixed effects in the data absorb all persistent supplier characteristics; in the model these are productivity and the relationship-cost shifter, so the like-for-like counterpart controls for both. The resulting coefficient captures the relationship between buyer reach and product scope among suppliers with similar productivity and relationship costs. We use this moment to discipline the strength of the buyer–product co-movement, whichever channel generates it.

Moments specific to product–buyer matching. The matching model uses these moments to discipline how the supplier’s firm-level product hierarchy maps into individual buyer relationships. The first is the mean product-specialization ratio, defined as the share of supplier sales accounted for by its largest product. The second is the within-link rank-alignment coefficient documented in Table 11. In the data, the relationship between a product’s importance in the supplier’s overall portfolio and its importance within an individual supplier–buyer link is positive but far from one. Calibration A targets the rank-score coefficient of 0.27; calibration B targets the share-on-share coefficient of 0.82, which is the statistic the model computes.

Within-link moments (calibration B). Calibration B adds five moments that capture how products are distributed across a supplier’s buyer relationships. These are the mean number of products per link (3.24), the share of single-product links (0.42), the decay of within-link sales along the supplier’s lagged product ranking (0.85), the within-supplier dispersion of log link sales (1.52), and portfolio coverage, defined as the share of a supplier’s products sold to its broadest buyer (0.61). It also targets the 75th and 90th percentiles of product scope and buyer reach. The within-link moments are what separate the two channels. Products per link and single-product links pin down how often activation binds, and coverage pins down how differently buyers select from the portfolio. Together they determine how much of the observed scope–buyer co-movement is produced by selection rather than by development.

This moment is particularly informative about the dispersion of $\zeta_{sb\gamma}$. A larger θ_ζ implies less dispersed product–buyer matches and therefore a closer correspondence between the firm-level and within-link product hierarchies. A smaller θ_ζ generates greater buyer-specific variation and weaker rank alignment. Product specialization also helps distinguish systematic rank decay, governed by α , from idiosyncratic product–buyer matching, governed by θ_ζ .

Buyer heterogeneity and the calibration of D_b . The model allows buyers to differ in profitability through D_b . In the data, we construct measures of buyer scale independently of the focal

supplier, including the buyer’s total purchases and total sales. These variables provide empirical proxies for the expenditure and market-size components summarized by D_b . Figure 6 documents substantial heterogeneity in these measures even among buyers served by the same supplier, motivating the inclusion of buyer-level profitability heterogeneity in the quantitative model.

We also examine the relationship between these buyer characteristics and supplier–buyer link sales to distinguish persistent buyer-level profitability from pair-specific compatibility, which is captured separately by η_{sb} . In the quantitative implementation, however, we do not map the observed buyer-size measures one-for-one into D_b . Instead, D_b is represented by a reduced-form distribution whose dispersion, ω_B , is estimated jointly with the remaining parameters using the firm-level moments in Table 4. The buyer-size evidence in Figure 6 therefore provides empirical motivation and discipline for buyer-level heterogeneity, while the exact dispersion of D_b is determined jointly in the SMM calibration. The calibrated dispersion of buyer profitability, $\omega_B = 2.30$ in calibration B, is of the same order as the within-supplier dispersion of log buyer size in Figure 6 (medians of about 2.2 to 2.7), although the two objects are not identical: D_b combines buyer expenditure with its price index, and the figure measures dispersion among buyers a supplier actually serves.

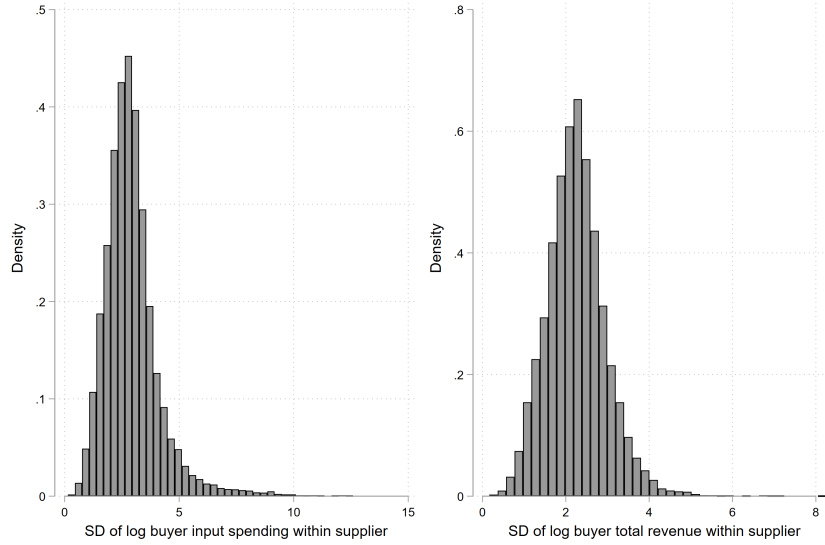
Consistent with the atomless-supplier assumption in the theoretical model, the buyer-level price index embedded in D_b is not recomputed in response to the realized shocks of individual suppliers in the finite simulation. The S simulated suppliers should instead be interpreted as a Monte Carlo approximation to the underlying supplier distribution. Accordingly, we do not renormalize simulated supplier–buyer sales so that they sum exactly to E_b for each buyer, and buyer-level expenditure exhaustion is not a targeted moment of the current calibration.

5.2 Parameter discipline

Table 3 reports the parameter values, their sources, and the moments that provide the main discipline for each free parameter. Table 4 reports how well the matching model fits both the targeted and untargeted moments. The parameters are jointly identified, since changing any one cost or heterogeneity parameter typically affects several moments at the same time. The moment assignments in Table 3 should therefore be interpreted as the primary sources of discipline rather than as one-to-one identification statements.

The level of the buyer-specific product-activation cost, $\bar{f}_0^A$, governs how easily developed products are activated across buyer relationships and is therefore primarily disciplined by product scope, buyers per product, and sales concentration. The link-cost level, $\bar{f}^L$, acts directly on the extensive buyer margin and is primarily disciplined by the buyer-count and buyers-per-product moments. The level of the firm-level product-adoption cost, f_0^P , determines the overall cost of expanding the supplier’s global portfolio and therefore affects both product scope and, through

Figure 6: Within-supplier dispersion in buyer size, 2017



Notes: The figure reports the distribution across manufacturing suppliers of within-supplier heterogeneity in buyer size in 2017. For each supplier, we calculate the standard deviation across its active buyers of the logarithm of buyer input spending (left panel) and the logarithm of buyer total revenue (right panel). Buyer input spending is measured excluding purchases from the focal supplier, so that the measure does not mechanically reflect the size of the supplier–buyer relationship. Buyer total revenue is obtained from firms’ administrative tax filings. The sample is restricted to suppliers with at least five buyers and nonmissing information for the corresponding buyer-size measure. Each observation in the histograms is one supplier, and the vertical axes report density.

amortization, the buyer base that can support that scope.

The parameter δ governs how the incremental firm-level adoption cost changes as the supplier moves toward more peripheral products. It is therefore particularly important for the dispersion and upper tail of the product-scope distribution. Supplier-level cost heterogeneity, summarized by ω_c , generates common variation in relationship costs across all buyers of a supplier. It therefore creates variation in buyer reach that survives aggregation across links and, through the amortization mechanism, generates co-movement between buyer reach and product scope conditional on productivity. For this reason, the within- ϕ slope β in equation (39) is an informative moment for ω_c . Buyer-quality dispersion, ω_B , instead governs heterogeneity in the profitability of potential buyers and contributes to dispersion in both buyer reach and product scope.

In the model, θ_ζ governs product–buyer match dispersion, while α governs systematic sales decay along the supplier’s product ranking. The two parameters are estimated jointly because both affect the relationship between firm-level and link-level product importance. Within-link rank alignment is particularly informative about θ_ζ , while product specialization provides additional discipline on α and helps separate systematic rank decay from idiosyncratic matching.

The activation-cost rank parameter μ is fixed in calibration A, where the firm-level moments carry little information about it, and estimated in calibration B, where products per link and

the share of single-product links discipline it directly. The currently available aggregate moments provide little independent information about μ once the remaining rank and cost parameters are free to adjust. A direct empirical measure of the distribution of product scope within supplier–buyer links would provide substantially sharper discipline for this parameter.

The CES elasticity, the parameters of the supplier-productivity distribution, and the remaining idiosyncratic cost shocks are fixed externally or normalized rather than estimated from the same SMM moment set. Table 3 reports these values and their sources.

Targeted versus validation moments. Not every moment reported in Table 4 enters the SMM objective. We report the standard deviation of buyer counts, the standard deviation of buyers per product, and the median concentration measures as over-identification checks. These moments are strongly influenced by the upper tail of the supplier distribution and are not assigned to a separate structural parameter in the current calibration.

We also report the share of single-product suppliers as a validation moment. Although the model has no force designed to generate a mass at one product, calibration B produces a single-product share of 7.9 percent (13.8 in the data), because once selection binds some suppliers find only one product worth developing. Calibration A produces essentially none. We also report the non-prefix share of links and the frequency with which a supplier’s top product is missing from a link as untargeted diagnostics of the matching structure. The comparison of targeted moments, untargeted moments, and model-only predictions in Table 4 therefore provides a transparent assessment of both model fit and the dimensions along which additional structure or empirical discipline may be useful.

Because developed scope is bounded by $G_{\max}$, the expected-surplus aggregator is finite for every supplier. For sufficiently productive suppliers every product lies in the region where activation is certain, and expected surplus is linear in $\phi_s^{\sigma-1}$. The population moments therefore exist whenever $\theta_\phi > \sigma - 1 = 3$, which the value $\theta_\phi = 4.5$ satisfies. The stronger condition $\theta_\phi > \theta_\zeta(\sigma - 1)$, which would be required if all products lay in the interior of the Pareto support, does not bind at the calibrated parameters.

Calibration B fits the product side of the data closely. The distribution of product scope, specialization, the scope–buyer elasticity, and the within-link allocation of products. Its main misses are on the buyer side and in the heterogeneity of buyers’ product sets. The model’s buyer distribution is thinner-tailed than in the data (mean buyers 35 against 69), coverage is too high (0.85 against 0.61), and the supplier’s top product is almost always sold to every buyer. Richer buyer-side heterogeneity is a natural extension.

Table 3: Parameters of the matching model: values, sources, and main sources of discipline

Parameter	Calib. A	Calib. B	Source	Main sources of discipline
<i>Fixed or normalized</i>				
σ (CES elasticity)	4	4	Literature	Standard value.
θ_ϕ (productivity tail)	4.5	4.5	Literature	Standard value.
$\phi_{\min}$ (productivity floor)	1	1	Normalization	Units of productivity.
ω_{fL}, ω_{fA} (buyer-cost dispersion)	0.3	0.6	A: assumption; B: data	B: residual SD of log buyer degree (0.607) and of buyer mean link scope (0.598).
<i>Calibrated (SMM)</i>				
$\bar{f}_0^A$ (activation cost)	0.944	272	SMM	Products per link, one-product links, mean $\hat{G}_s$, buyers per product.
$\bar{f}^L$ (link cost)	947	261	SMM	Median and mean N , percentiles of N , buyers per product.
f_0^P (adoption cost)	21.5	1.01	SMM	Mean $\hat{G}_s$, mean N , buyers per product.
δ (adoption convexity)	1.58	0.219	SMM	SD and percentiles of $\hat{G}_s$; β .
μ (activation convexity)	1	0.141	A: fixed; B: SMM	B: products per link and one-product share.
θ_ζ (match dispersion)	1.53	1.76	SMM	Within-link alignment, specialization, coverage.
α (rank decay)	0.0487	0.404	SMM	Specialization, rank decay.
ω_η (pair-cost dispersion)	0.7	1.01	A: fixed; B: SMM	B: within-supplier SD of log link sales.
ω_c (supplier cost dispersion)	0.0806	1.31	SMM	β , dispersion of N .
ω_B (buyer-quality dispersion)	1.57	2.3	SMM	Dispersion of $\hat{G}_s$ and N , buyer concentration.

Notes: Calibration A targets the firm-level moments of Table 4 (11 moments, 8 parameters); calibration B adds the within-link moments and the percentiles of $\hat{G}_s$ and N (20 moments, 10 parameters). Identification is joint; the last column lists the most informative moments. Values from a single simulation seed with $S = 2,000$ suppliers and $B = 8,000$ potential buyers.

Table 4: Identifying moments: data and model

Moment	Data	Calibration A (firm-level moments)	Calibration B (adds within-link moments)
<i>Product scope</i>			
Median $\hat{G}$	9	8 [†]	9 [†]
Mean $\hat{G}$	16.1	14.3 [†]	15.9 [†]
SD of $\hat{G}$	21.0	18.6 [†]	23.0 [†]
75th / 90th percentile of $\hat{G}$	19 / 38	16 / 30	18 / 34 [†]
<i>Buyer base</i>			
Median N	9	7 [†]	11 [†]
Mean N	68.9	79.7 [†]	35.1 [†]
75th / 90th percentile of N	30 / 100	44 / 158	31 / 83 [†]
<i>Link breadth</i>			
Buyers per product (median)	1.000	1.000 [†]	1.333 [†]
Buyers per product (mean)	3.749	2.159 [†]	1.573 [†]
<i>Sales composition and product hierarchy</i>			
Product specialization ratio (mean)	0.541	0.300 [†]	0.540 [†]
Buyer concentration ratio (mean)	0.514	0.435 [†]	0.498 [†]
Within-link alignment, link share on firm share ^a	0.820	0.240 [†]	0.549 [†]
<i>Within-link structure</i>			
Products per link (mean)	3.244	37.1	3.845 [†]
Share of one-product links	0.423	0.000	0.335 [†]
Rank decay (log sales on log leave-out rank)	0.852	-0.070	0.595 [†]
Within-supplier SD of log link sales (median)	1.523	0.971	1.526 [†]
Portfolio coverage $\max_b G_{sb} / \hat{G}_s$ (mean)	0.612	1.000	0.854 [†]
<i>Complementarity</i>			
Scope-buyer elasticity β (Table 2, col. 5)	0.555	0.427 [†]	0.521 [†]
<i>Untargeted</i>			
SD of N	784.6	237.4	87.1
Buyers per product (SD)	22.8	2.709	0.972
Product specialization ratio (median)	0.512	0.264	0.505
Buyer concentration ratio (median)	0.475	0.386	0.453
Single-product share	0.138	0.004	0.079
Products per link (SD)	4.176	22.7	7.308
Non-prefix share of links	0.701	0.695	0.420
Top product missing from link (leave-one-out)	0.363	0.178	0.043
Residual SD, rank-decay regression	1.227	0.704	0.653
Sales tail index, top 5%	0.841	0.861	1.255
β also controlling for log sales (col. 6)	0.486	0.307	0.337
Targeted moments		11	20
Free parameters		8	10
SMM criterion		0.575	1.228

Notes: [†] targeted in that calibration. Data moments refer to multiproduct manufacturing suppliers in 2017, except the within-link matching statistics (coverage, non-prefix share, top product missing, alignment), which pool 2015–2023. Model moments are computed from realized trade on the calibrated economy ($S = 2,000$ suppliers, $B = 8,000$ potential buyers, one seed), for multiproduct suppliers except β and the single-product share. β : $\log \hat{G}_s$ on $\log N_s$ with twenty $\log \phi_s$ -bin fixed effects and $\log c_s$.

^a Calibration A targets this statistic at 0.266, the rank-score coefficient of Table 11; the share-on-share coefficient in the data is 0.820, which calibration B targets. The SMM criterion sums squared relative deviations over each calibration’s own targets and is not comparable across columns.

6 Quantitative Results

This section presents the quantitative behavior of the calibrated matching model (calibration B of Section 5). The calibrated values are reported in Table 3. Selection binds in this calibration. The activation cost is high, links carry on average 3.8 products, and portfolio coverage is below one. Every firm-level moment is computed from realized transactions. A supplier's products are the distinct codes it actually sells, and its buyers are those that actually receive at least one product.

The section is organized around the model's central testable prediction: conditional on productivity, firms with more buyers should have more products, and then inspects the model's other objects to assess where the calibration succeeds and where it is strained.

6.1 The central prediction: within-productivity co-movement of scope and buyer base

Figure 7 plots the relationship between observed product scope $\hat{G}_s$ and buyer base n_s in two ways.

Panel (a) plots $\log \hat{G}_s$ against $\log n_s$ separately by productivity quintile. Firms are first divided into five groups based on ϕ_s , and within each group average product scope is plotted across bins of buyer count. The relationship is positive in every productivity quintile. Even among firms with similar productivity, suppliers that reach more buyers also sell more products.

Panel (b) residualizes both variables on productivity bins and on $\log c_s$, the model counterpart of a supplier fixed effect; the residual relationship is clean and positive with slope $\beta = 0.521$. Table 5 traces the same object. The unconditional elasticity is 0.690 (column 1); conditioning on productivity alone barely changes it (0.672, column 4); controlling also for the relationship-cost shifter lowers it to 0.521 (column 6), against 0.555 in the data. The drop reflects that suppliers with high relationship costs have both fewer buyers and fewer products, a common driver that supplier fixed effects absorb in the data.

The variance decomposition in Appendix L Table 13 sharpens the point. Productivity alone explains only 20% of the variance of $\log \hat{G}_s$, and of the remaining within- ϕ variation, buyer count absorbs about 80%. Table 14 shows the within-quintile correlation between $\log \hat{G}_s$ and $\log n_s$ rising from 0.87 in the lowest productivity quintile to 0.93 in the highest.

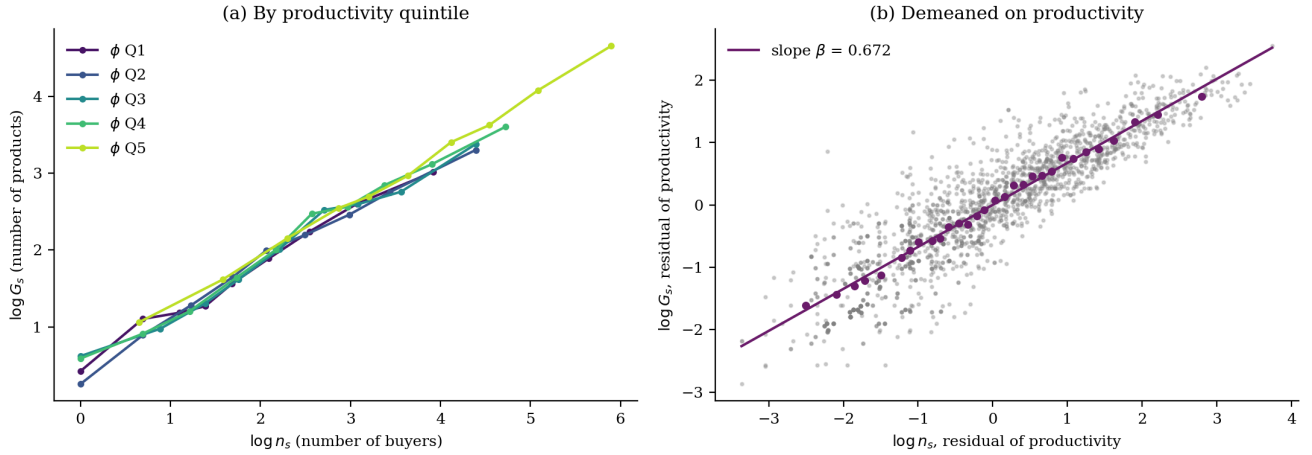
The within-productivity slope is the model's central prediction, but by itself it does not reveal which channel generates it. A firm with more buyers may develop more products because the adoption cost is spread over more relationships (amortization), or may sell more of an unchanged portfolio because each product has more chances to find a buyer (matching). Both produce the

same slope. Section 7.1 separates them. The unconditional slope alone would be uninformative, because more productive firms mechanically have both more buyers and more products; the conditional slope instead isolates buyer–product co-movement from productivity and relationship-cost sorting. By itself, however, it does not distinguish amortization from matching.

Match with the data. The model’s 0.521 compares with an empirical within-supplier elasticity of 0.555 (Table 2, column 5). Controlling in addition for log sales, the model gives 0.337 against 0.486 in the data.¹⁰

Appendix L.1 shows that the return to product adoption is concentrated in a few high-value buyer relationships. This means that even suppliers with hundreds of buyers have about four effective buyers in the adoption aggregator. This concentration already suggests that amortization is quantitatively limited in this calibration, because an additional marginal buyer adds little to the return to developing a product.

Figure 7: Within-productivity co-movement of product scope and buyer breadth



Notes: Calibrated matching model. Panel (a) plots $\log \hat{G}_s$ against $\log n_s$ within productivity quintiles. Panel (b) residualizes both variables on twenty $\log \phi_s$ bins and $\log c_s$; the resulting slope is $\beta = 0.521$.

6.2 Marginal distributions and calibration fit

Figure 8 reports the simulated distributions of firm-level product scope $\hat{G}_s$, buyer count N_s , and total sales T_s . Product scope and buyer breadth are both strongly right-skewed, with long upper tails, while the distribution of log sales is approximately bell-shaped.

The model reproduces the central tendency of the two extensive margins well. Among multi-product suppliers, the median simulated supplier sells 9 products and serves 11 buyers, compared

¹⁰Across simulation seeds, the calibrated-counterpart elasticity ranges from approximately 0.38 to 0.56, with a mean near 0.50.

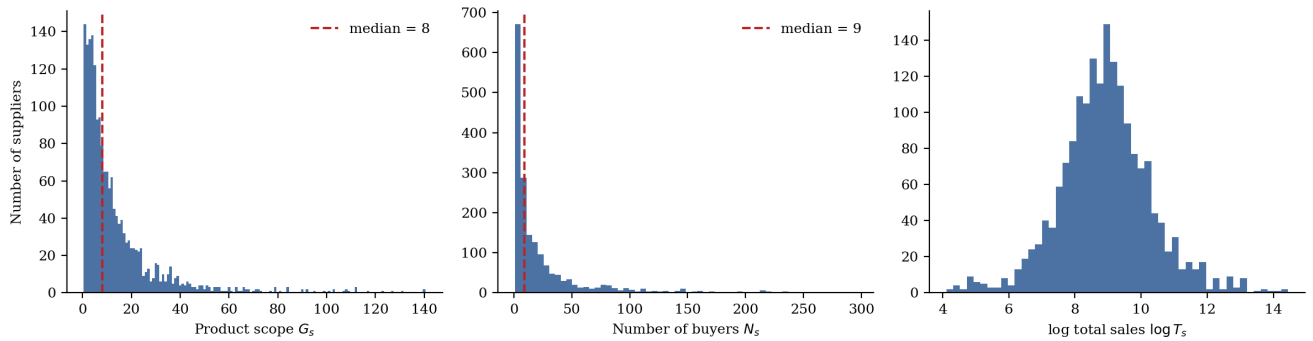
Table 5: Regressions of $\log \hat{G}_s$ on $\log n_s$, $\log \phi_s$, and $\log c_s$ (calibrated model)

	(1)	(2)	(3)	(4)	(5)	(6)
$\log n_s$	0.690 (0.007)		0.671 (0.008)	0.672 (0.008)	0.474 (0.030)	0.521 (0.028)
$\log \phi_s$		2.124 (0.099)	0.263 (0.050)		0.909 (0.108)	
$\log c_s$					-0.233 (0.035)	-0.180 (0.031)
ϕ -bin FE	No	No	No	Yes	No	Yes
R^2	0.836	0.202	0.839	0.840	0.843	0.843

Notes: Dependent variable $\log \hat{G}_s$ (products sold to at least one buyer). ϕ -bin FE are twenty $\log \phi_s$ bins. Column (6) is the calibrated counterpart of Table 2, column (5). Standard errors in parentheses.

with 9 products and 9 buyers in the 2017 data. Thus, the model exactly matches median product scope and slightly overstates median buyer reach in the calibration sample. The dispersion and upper percentiles of product scope are also close to the data (standard deviation 23.0 against 21.0; 90th percentile 34 against 38), and the model produces a single-product share of 7.9 percent against 13.8 percent. The buyer distribution is thinner-tailed than in the data. We view these discrepancies as limitations of the quantitative fit rather than as objects central to the buyer–product complementarity mechanism.

Figure 8: Marginal distributions of firm-level outcomes in the calibrated model



Notes: The figure reports simulated distributions of firm-level product scope $\hat{G}_s$ (left), buyer count N_s (middle), and log total sales $\log T_s$ (right) in the calibrated model (calibration B), for all active suppliers, including single-product suppliers. Red dashed lines indicate medians. Table 4 reports moments for multiproduct suppliers only, to match the empirical sample, which is why its medians are higher. The top percentile of the buyer-count distribution is trimmed for display.

Appendix L.2 shows that simulated sales co-move positively with buyer breadth, product scope, and supplier productivity, consistent with the corresponding cross-sectional patterns in the data.

7 Counterfactuals

7.1 Decomposing the scope–buyer elasticity: amortization versus matching

The within-productivity co-movement of product scope and buyer breadth can arise through two channels (Section 4.5). Under amortization, a firm with more buyers develops more products because the firm-level adoption cost is spread over more relationships. Under matching, a firm with more buyers sells more of an unchanged portfolio because each product has more chances to find a buyer. Both deliver the same regression coefficient. The model, however, lets us separate them.

The relocated-cost experiment. We remove the firm-level adoption cost and charge the same cost on each buyer relationship on which the product is sold. The activation cost of product g on link (s, b) becomes $f_{sb}^A(g) + \lambda f^P(g)$, with λ chosen so that average log product scope is unchanged. In this economy the decision to sell a product to one buyer is independent of the supplier’s other relationships, so there is nothing to amortize. Matching, however, is untouched, meaning more buyers still give each product more chances to be sold. The elasticity in the relocated-cost economy therefore measures the matching channel, and the difference from the benchmark measures amortization.

The decomposition depends on what disciplines the model. Table 6 reports the result for both calibrations. In calibration A, disciplined only by firm-level moments, activation rarely binds, so observed scope is close to developed scope and selection is weak. Relocating the cost lowers the elasticity from 0.43 to about 0.25, so amortization accounts for about 40 percent of the co-movement and matching for the remaining 60 percent. In calibration B, which also targets the within-link allocation of products, selection binds. Links carry about four products and suppliers’ broadest buyer takes only part of the portfolio. Relocating the cost now leaves the elasticity essentially unchanged (0.521 against 0.470), so matching accounts for about 90 percent of the co-movement.

This contrast is the central mechanism result of the paper. Firm-level moments, including the scope–buyer elasticity itself, are consistent with a wide range of mechanisms. The two calibrations fit that elasticity comparably well but attribute it to very different channels. What identifies the decomposition is how products are allocated across buyer relationships. Products per link, the share of single-product links, and portfolio coverage reveal how tightly selection binds, and hence how much of the observed co-movement a static model must attribute to matching.

Evidence from incumbent buyers. The data offer a direct check. Products sold to a supplier’s incumbent buyers cannot be added by matching with new buyers, so their response to buyer breadth isolates development. In the data, the incumbent elasticity is 0.38, about 69 percent of the total (Appendix D). Development therefore plays a larger role in the data than the static model generates when disciplined by within-link moments. We view this gap as informative rather than as a failure of fit. A static model in which products are developed once and matched to buyers ex post can reproduce the within-link allocation of products only by loading the co-movement onto selection. The incumbent-buyer evidence suggests instead that suppliers respond to a larger buyer base by developing and extending products within existing relationships over time, a force that a dynamic model of product adoption would capture. Reconciling the within-link moments with the incumbent-buyer evidence is the main direction for future work.

Table 6: Sources of the scope–buyer elasticity: amortization versus matching

	β	Amortization share	Matching share
<i>Model: relocated-cost decomposition</i>			
Calibration A (firm-level moments only)	0.427	$\approx 40\%$	$\approx 60\%$
Calibration B (adds within-link moments)	0.521	$\approx 10\%$	$\approx 90\%$
<i>Data: incumbent-buyer decomposition</i>			
Products sold to incumbent buyers	0.382	development share $\approx 69\%$	

Notes: The model rows decompose the scope–buyer elasticity by comparing the benchmark model with a counterfactual economy in which the product-adoption cost is paid separately on each buyer link rather than once at the firm level. The link-level cost is rescaled so that average product scope is unchanged. The resulting reduction in β is attributed to the amortization channel, with the remaining share attributed to matching.

The data row provides a complementary empirical decomposition. The within-supplier elasticity of product scope sold to incumbent buyers with respect to buyer breadth is 0.382 (Table 9, column 5), compared with a total product-scope elasticity of 0.555 (Table 2, column 5). Their ratio, $0.382/0.555 \simeq 0.69$, implies that approximately 69 percent of the total co-movement occurs through products sold to incumbent buyers. With sales controls, the corresponding share is 59 percent. We interpret this “development share” as an upper bound on the importance of amortization: product scope among incumbent buyers can respond to buyer breadth through forces other than the amortization of firm-level product-development costs.

Removing the adoption cost altogether. Setting $F^P = 0$ does not provide a useful test of the mechanism because, without the firm-level development cost, firms expand product scope up to the model’s imposed upper bound. In calibration B, however, observed scope stays bounded (median 9) because selection determines which developed products are sold, and the elasticity remains positive (0.551). This is the matching channel at work. Even when all products have already been developed, suppliers with more buyers end up selling a larger share of them. The deterministic ladder behaves similarly (0.490 to 0.486), through the deepest-buyer order statistic discussed in Appendix M.

7.2 Comparative statics in the fixed costs

Having decomposed the channels that tie the two margins together, we now ask how the calibrated economy responds to reductions in the fixed costs governing each margin.

Table 7 reports the response to halving, in turn, the firm-level product-adoption cost F^P and the buyer-relationship cost $\bar{f}^L$, holding all other parameters fixed.

First, each cost moves mainly its own margin. Halving the buyer-relationship cost raises buyer reach by about 30 percent and product scope by about 9 percent. Halving the product-adoption cost raises product scope by about 12 percent and buyer reach by about 5 percent. The spillovers are positive but smaller than the direct responses, consistent with the decomposition above. Because the link between the margins runs mainly through matching, a larger buyer base raises observed scope primarily by helping existing products find buyers rather than by inducing substantial new development.

These comparative statics carry a policy reading that depends on the decomposition. A policy that lowers buyer–supplier matching costs raises not only the buyer base but also the number of products suppliers sell. In the calibration disciplined by within-link moments, however, the additional products mostly reflect existing products finding new buyers rather than new product development. Had the co-movement been driven by amortization, as calibration A allows, the same policy would stimulate product development. Distinguishing the two channels is therefore not only a question of interpretation, it determines whether easing network frictions is also an innovation policy. This is the quantitative counterpart of the policy argument in the introduction, and it is a channel that a single-margin model, in which product scope and buyer acquisition are chosen independently, cannot deliver. Because the model is partial equilibrium, these responses abstract from adjustments in buyers’ price indices.

We vary only F^P and $\bar{f}^L$ because they govern the two extensive margins directly. The third fixed cost, the per-product activation cost $\bar{f}^A$, plays a different role, it determines how developed products are allocated across buyer relationships. In the calibration disciplined by within-link moments, this cost is large enough that activation binds on most links, which is what makes product–buyer selection, and hence the matching channel, quantitatively important. Its level is pinned down by the number of products per link and the share of single-product links, rather than by the firm-level margins studied here.

Shutting down one extensive margin at a time. The comparative statics before move the two margins jointly; a complementary exercise shuts each margin down entirely and asks which one the sales distribution depends on. In the products-only economy every supplier faces a single representative buyer of average quality, so the buyer margin is switched off and cross-firm sales variation comes only from productivity and product scope. In the buyers-only economy every

Table 7: Comparative statics in the two fixed costs

Cost reduction	Change relative to baseline	
	Product scope $\widehat{G}_s$	Buyer reach N_s
Product-adoption cost F^P : -50%	+12%	+5%
Buyer-relationship cost $\bar{f}^L$: -50%	+9%	+30%

Notes: Each row re-solves the calibrated economy after reducing one fixed cost by 50 percent, holding all other parameters fixed. Entries report percentage changes in average product scope and buyer reach relative to the baseline calibrated economy.

supplier is restricted to a single product, so the product margin is switched off and variation comes only from productivity and the buyer base. Both are computed on the calibrated economy, holding each firm’s realized objects fixed, so productivity remains active in both and neither distribution collapses to a point.

Table 8 reports the result, and it is asymmetric. Removing the buyer margin (products only) cuts the dispersion of log sales by 34%; removing the product margin (buyers only) cuts it by only 12%. Equivalently, the buyer margin on its own reproduces about 88% of the full sales dispersion while the product margin on its own reproduces only about 66%. The buyer margin, not the product margin, is doing most of the work in generating firm-size dispersion and sales concentration. This is the demand-side counterpart of the calibration result that within- ϕ dispersion in scope is carried by the buyer base: firm size in this model is primarily a story about how many buyers a supplier reaches, with product scope amplifying rather than originating the dispersion.

Table 8: Shutting down one extensive margin at a time

	Full model	Products only (representative buyer)	Buyers only (single product)	Equal buyer quality (buyer counts kept)
SD(log T_s)	1.34	0.88 (-34%)	1.17 (-12%)	1.95 (+46%)
Top 1% sales share	0.32	0.23 (-29%)	0.28 (-12%)	0.59 (+85%)

Notes: Accounting exercise holding each supplier’s realized buyer set, product set and link-level product sets fixed, with match quality set to its mean. Full model: $\phi_s^{\sigma-1} \sum_b D_b \sum_{g \in \mathcal{G}_{sb}} h(g)^{1-\sigma}$. Products only: $\phi_s^{\sigma-1} \bar{D} \sum_{g \in \widehat{\mathcal{G}}_s} h(g)^{1-\sigma}$. Let $\widehat{\mathcal{G}}_s \equiv \bigcup_{b \in B_s} \mathcal{G}_{sb}$ denote the supplier’s realized set of products sold to at least one buyer, with $\widehat{G}_s \equiv |\widehat{\mathcal{G}}_s|$. Buyers only: $\phi_s^{\sigma-1} \sum_b D_b$. Equal buyer quality: the full-model expression with $D_b = \bar{D}$. SD(log T_s) of realized sales is 1.39.

7.3 A comparative-static analog of the growth decomposition

The empirical facts in Section 3 decompose realized supplier growth into four buyer×product margins: old-link–old-product (OLOP), new-link–old-product (NLOP), old-link–new-product (OLNP), and new-link–new-product (NLNP). Although the model is static, it admits a natural comparative-static analog. We perturb the calibrated economy, re-solve each supplier’s buyer and product

decisions, and decompose the resulting change in sales using the baseline buyer and product sets as the “old” reference. We report the exercise for a uniform 50% increase in productivity, which raises every ϕ_s by half. A common demand shock produces the same qualitative patterns.

Because the positive productivity shock weakly expands each supplier’s buyer and product sets, baseline buyers and products remain available after the shock, making the decomposition directly comparable to its empirical counterpart. OLOP captures additional sales on baseline buyer–product cells; NLOP captures sales to newly activated buyers of baseline products; OLNP captures sales of newly developed products to baseline buyers; and NLNP captures sales combining a new buyer with a new product. Within each bin of Figure 9, we sum each component across suppliers and divide by the bin’s total sales growth, so that the four components sum to one.

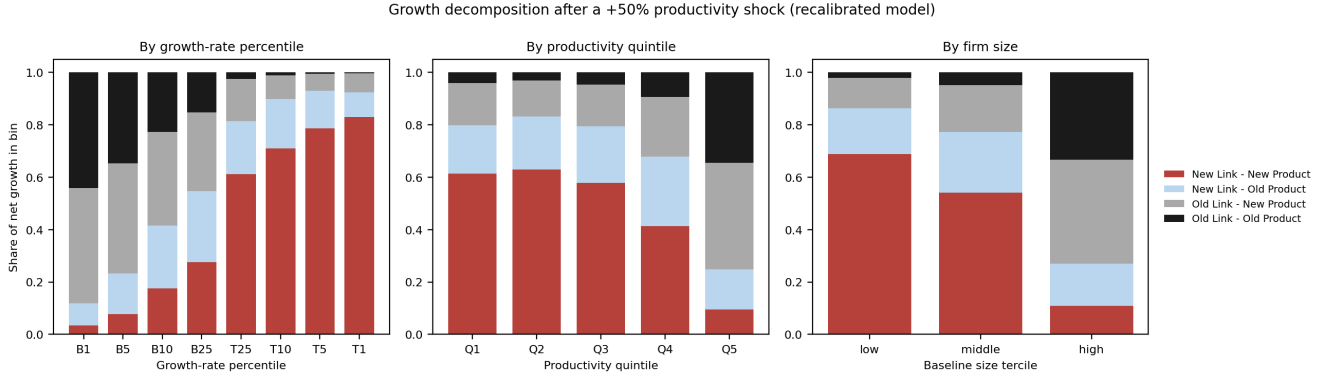
The exercise reproduces the main qualitative gradient in the empirical decomposition without targeting it. Suppliers with the largest simulated growth responses rely relatively more on the extensive margins, particularly sales of existing products to new buyers, which account for 77 percent of growth among the fastest-growing percentile. The same extensive-margin tilt is stronger among initially smaller and less-productive suppliers. By contrast, larger and more-productive suppliers rely relatively more on additional sales through baseline buyer–product relationships. Thus, as in the data, the composition of expansion shifts from greater reliance on new buyers and products among smaller suppliers toward established buyer–product relationships among larger suppliers, which draw about 78 percent of their growth from existing buyer–product cells. The joint new-buyer–new-product margin is smaller in the model than in the data (at most about 20 percent of growth), consistent with the model’s limited development response. We view this pattern as an untargeted validation of the model’s heterogeneous adjustment across the firm distribution rather than as a fitted result.

Two caveats discipline the interpretation. First, the static model has no product or link exit. The lower part of the simulated growth distribution therefore contains suppliers that expand the least, rather than suppliers that contract as in the data. Second, because the productivity shock is common and positive, all suppliers expand; the informative object is therefore the composition of their adjustment rather than the distribution of growth rates itself. Figure 9 should consequently be interpreted as a comparative-static validation of how the composition of expansion varies across suppliers, not as a structural reproduction of the 2017–2023 transition.

8 Conclusion

This paper studies the joint role of buyer relationships and product scope in shaping firm size inside production networks. Using the universe of firm-to-firm transactions in Chile, we show that these are empirically distinct but closely connected margins. Buyer relationships have shorter

Figure 9: Growth decomposition after a +50% productivity shock (calibrated model)



Notes: The figure reports the comparative-static analog of the empirical growth decomposition. Every supplier’s productivity ϕ_s is increased by 50% and the supplier’s buyer and product decisions are re-solved. The resulting change in sales is decomposed into new-link–new-product (NLNP), new-link–old-product (NLOP), old-link–new-product (OLNP), and old-link–old-product (OLOP), where “old” refers to the supplier’s baseline buyer and product sets. Within each bin, components are aggregated across suppliers and normalized by total sales growth, so the four shares sum to one. The left panel groups suppliers by their simulated growth response, the middle panel by baseline productivity, and the right panel by baseline size. Because the model is static and the shock is common and positive, the figure speaks to the composition of expansion rather than to firm dynamics or contraction.

and less persistent life cycles than supplier–product relationships, and large supplier expansions disproportionately combine new buyers with new products. More importantly for the model, buyer breadth and product scope co-move strongly within suppliers. Conditional on supplier fixed effects, a 10 percent increase in the number of buyers is associated with a 5.6 percent increase in product scope, about two thirds of it among products sold to incumbent buyers. The same pattern appears for product entry. Suppliers tend to introduce more products in years when they also serve more buyers. The response is concentrated among peripheral products, consistent with buyer expansion raising the return to products near the margin of adoption.

We develop a static model in which suppliers jointly choose their firm-level product portfolio and the buyers they serve, and in which buyer breadth raises product scope through two channels. Under amortization, a firm-level product-adoption cost paid once generates returns across many buyer relationships. Under matching, buyer-specific product–buyer matches give each developed product more chances to find a buyer as the buyer base grows. The data support the need for buyer-specific matching. A product’s importance in the supplier’s overall portfolio only partly predicts its importance within individual buyer relationships, and different buyers of the same supplier often purchase different, non-nested subsets of products.

The model reproduces the within-supplier relationship between buyer breadth and product scope and allows us to decompose it into its two underlying channels. The decomposition depends on which features of the data discipline the model. With firm-level moments alone, amortization can account for up to about 40 percent of the co-movement. Once the model is also disciplined by how products are allocated across buyer relationships, matching becomes the dominant channel,

while amortization continues to account for a smaller but positive share. The empirical decomposition among incumbent buyers, however, points to a larger role for product development than the static model generates. Because this response occurs within existing buyer relationships, it cannot be explained by the mechanical matching of existing products to new buyers. This gap suggests that buyer expansion also affects firms' product-development decisions through forces that are not fully captured by the static model. These forces may reflect a stronger dynamic amortization mechanism or other forms of adjustment in product development. Distinguishing among them requires a dynamic framework in which buyer relationships and product portfolios evolve jointly over time.

The quantitative exercises also show that the interaction between the two margins matters for the distribution of firm sales. Changes in either the product-adoption cost or the buyer-relationship cost mainly move their own margin, with spillovers to the other. The two margins are not equally important quantitatively. Removing the buyer margin reduces the dispersion of log sales by 34 percent, compared with 12 percent when the product margin is removed. Equivalently, the buyer margin alone reproduces about 88 percent of the sales dispersion generated by the full model, while the product margin alone reproduces about 66 percent. Buyer reach is therefore the main source of firm-size dispersion in the calibrated economy, with product scope amplifying differences generated through the buyer network.

Although some of our empirical facts are measured over time, the benchmark model is static. We therefore interpret its growth exercises as comparative statics rather than as a model of firm dynamics. Nevertheless, when the calibrated economy is perturbed, the model reproduces the qualitative reallocation across old and new buyers and products observed in the data: smaller and less productive suppliers adjust relatively more through extensive margins, whereas larger and more productive suppliers rely more heavily on existing buyer-product relationships. A natural extension is therefore to introduce buyer accumulation and product adjustment dynamically. Such a model would allow us not only to speak directly to the life-cycle patterns documented in the data, but also to explain the incumbent-buyer evidence and determine whether the additional product-development response reflects dynamic amortization or other forces linking the evolution of customer relationships to product adoption.

More broadly, the results show that product scope cannot be studied separately from the network of buyers over which products are sold. Policies that reduce frictions to forming supplier-buyer relationships raise the number of products suppliers sell; whether they also stimulate product development depends on the relative strength of amortization and matching, which within-link and incumbent-buyer evidence can discipline. Conversely, policies that lower the cost of product adoption can increase the value of forming new buyer relationships. Accounting for these interactions is therefore important for understanding how buyer-supplier networks and multiproduct

production jointly shape the distribution of firm activity.

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Appendix

A Buyer Breadth Relative to Product Scope

Figure 10 provides a direct counterpart to the buyer–product density measures reported in Figure 2. For each supplier–year, let B_{st} denote the number of buyers, G_{st} the number of products, and R_{st} the number of active buyer–product pairs.

The average number of buyers per product is R_{st}/G_{st} , while the average number of products per buyer is R_{st}/B_{st} . Their ratio is

$$\frac{R_{st}/G_{st}}{R_{st}/B_{st}} = \frac{B_{st}}{G_{st}}.$$

Thus, B_{st}/G_{st} is not itself the average number of buyers per product. Instead, it measures buyer breadth relative to product scope: how large the supplier’s customer base is compared with the size of its product portfolio.

Figure 10 shows that this ratio increases systematically with supplier sales. Larger suppliers therefore expand their customer base more rapidly than their product portfolio. This provides a direct summary of the asymmetry documented in Figure 2, where the average number of buyers reached by each product rises more strongly with supplier size than the average number of products sold within each buyer relationship.

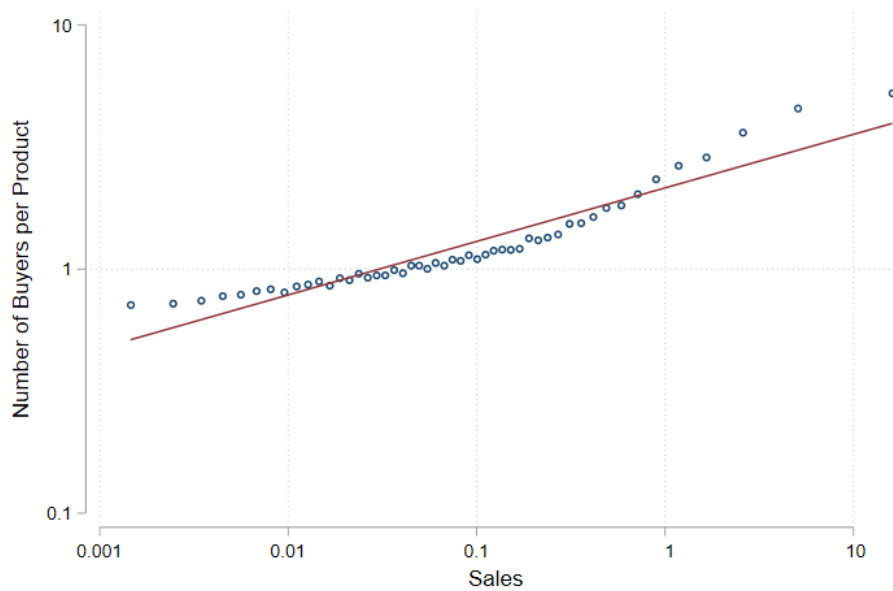


Figure 10: Buyer Breadth Relative to Product Scope

Notes: The figure presents a binned scatterplot relating supplier size to the ratio of the number of buyers to the number of products, B_{st}/G_{st} . Supplier size is measured by sales. Both axes are shown on a log scale. Each dot represents a bin of suppliers based on sales, and the solid line reports the fitted relationship.

A.1 Buyer and Product Margins in Supplier-Size Dispersion

This appendix provides a complementary cross-sectional decomposition of supplier-size dispersion into buyer and product margins. The exercise asks how much of the variation in supplier sales is associated with differences in the number of relationships or products, as opposed to differences in average sales conditional on those extensive margins.

On the buyer side, observed supplier sales can be written as

$$S_s = B_s \times \bar{S}_s^b,$$

where B_s is the number of buyers served by supplier s and $\bar{S}_s^b = S_s/B_s$ is average sales per buyer. In logs,

$$\log S_s = \log B_s + \log \bar{S}_s^b.$$

We use this identity to decompose the variance of log supplier sales into a component associated with the number of buyers and a component associated with average sales per buyer.

Analogously, on the product side,

$$S_s = G_s \times \bar{S}_s^g,$$

where G_s is the number of products sold and $\bar{S}_s^g = S_s/G_s$ is average sales per product. Hence,

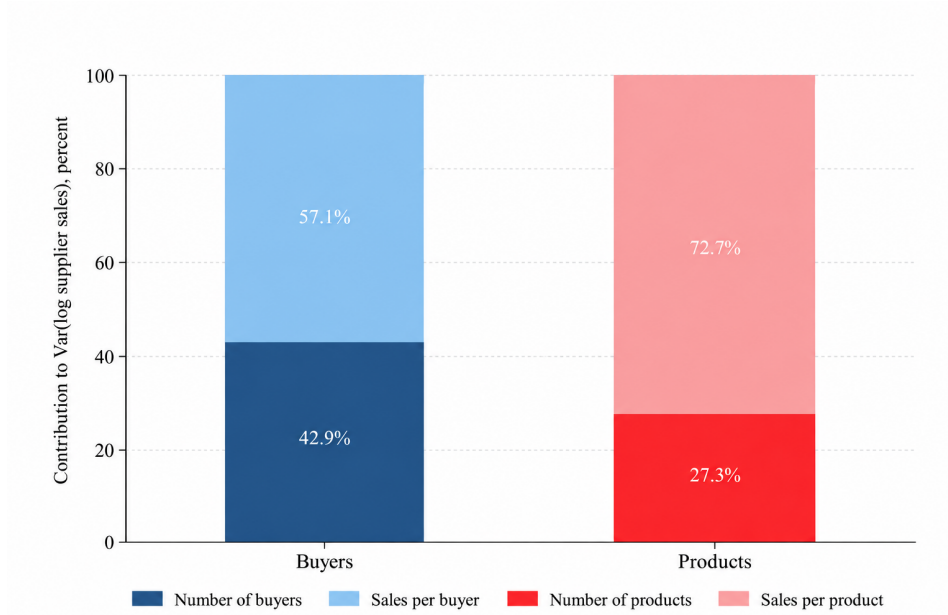
$$\log S_s = \log G_s + \log \bar{S}_s^g,$$

which gives the corresponding decomposition into product scope and average sales per product.

Figure 11 reports both decompositions for the 2017 cross section. The buyer extensive margin accounts for 42.9 percent of the variance of log supplier sales, while average sales per buyer accounts for the remaining 57.1 percent. In the analogous product decomposition, product scope accounts for 27.3 percent of firm-size dispersion and average sales per product for 72.7 percent. Thus, both extensive margins are quantitatively important, but differences in buyer reach account for a substantially larger share of cross-sectional supplier-size dispersion than differences in product scope.

These decompositions should be interpreted separately. The buyer and product shares are alternative ways of decomposing the same variation in supplier sales and therefore should not be added across decompositions. The comparison is instead informative about the relative importance of customer reach and product scope for cross-sectional differences in supplier size.

Figure 11: Buyer and product margins in supplier-size dispersion



Notes: The figure reports two alternative decompositions of the variance of log supplier sales in the 2017 cross section. The buyer decomposition separates the contribution of the number of buyers from average sales per buyer. The product decomposition separates the contribution of the number of products from average sales per product. The number of buyers accounts for 42.9 percent of the variance of log supplier sales and sales per buyer for 57.1 percent. The corresponding product-side contributions are 27.3 percent and 72.7 percent, respectively. Each bar represents a separate decomposition and sums to 100 percent; contributions should not be added across the buyer and product decompositions.

B Dynamics of Buyer Links and Supplier–Product Relationships

This appendix provides additional evidence that supplier–buyer links and supplier–product relationships are empirically distinct objects. Buyer links turn over more quickly and exhibit shorter sales lifecycles than supplier–product relationships. These patterns motivate the model’s separation of buyer reach and product scope. Because the benchmark model is static, however, the lifecycle profiles and survival patterns documented here are not quantitative targets of the model.

B.1 Relationship age and cohort measurement

For the lifecycle analysis, we define the age of a supplier–buyer link as the number of years since that pair is first observed trading:

$$\text{Age}_{sbt}^L = t - \min \{ \tau : S_{sb\tau}^{\text{link}} > 0 \}.$$

Analogously, the age of a supplier–product relationship is

$$\text{Age}_{sgt}^P = t - \min \{ \tau : S_{sg\tau}^{\text{prod}} > 0 \}.$$

Supplier age is defined separately as the number of years since the supplier is first observed with positive sales:

$$\text{Age}_{st}^F = t - \min \{ \tau : S_{s\tau}^{B2B} > 0 \}.$$

Because the transaction panel begins in 2015, relationship entry is observed without left-censoring only for links and supplier–product relationships that begin after the start of the sample. Lifecycle regressions therefore use relationships whose first observed activity occurs within the sample and should be interpreted as describing observed relationship age rather than the complete age of relationships that may have existed before 2015.

Buyer links and supplier–product relationships follow different lifecycle patterns. For supplier–buyer links, we estimate

$$\log S_{sbt}^{\text{link}} = \alpha^L + \sum_{j \geq 1} \beta_j^L \mathbf{1}\{\text{Age}_{sbt}^L = j\} + \mu_{sb} + \gamma_t + \varepsilon_{sbt}^L, \quad (40)$$

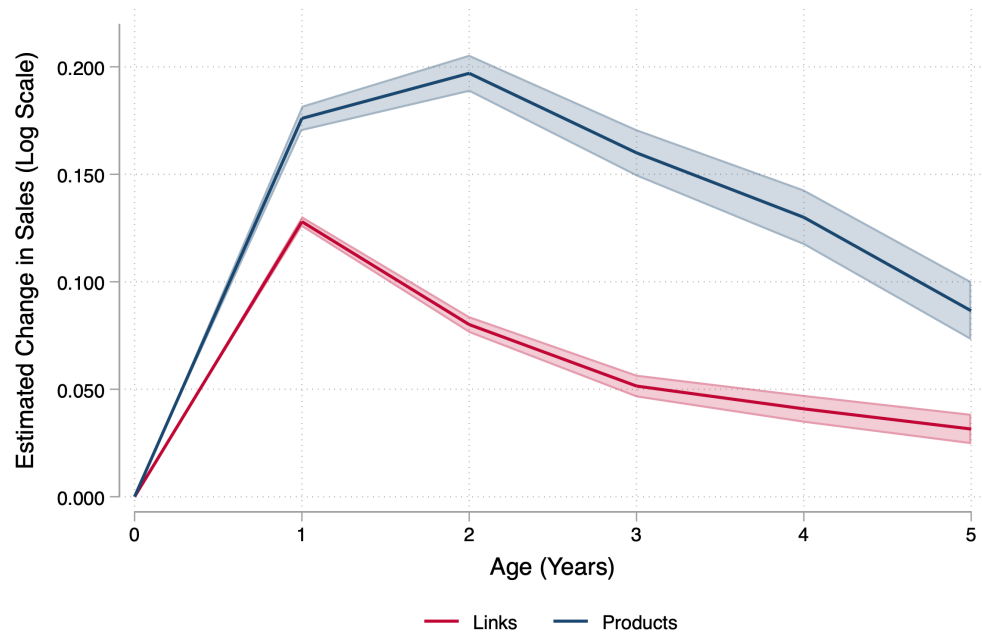
where age zero is the omitted category, μ_{sb} is a supplier–buyer-link fixed effect, and γ_t is a year fixed effect. For supplier–product relationships, we estimate

$$\log S_{sgt}^{\text{prod}} = \alpha^P + \sum_{j \geq 1} \beta_j^P \mathbf{1}\{\text{Age}_{sgt}^P = j\} + \mu_{sg} + \gamma_t + \varepsilon_{sgt}^P, \quad (41)$$

where μ_{sg} is a supplier–product fixed effect. Standard errors are clustered at the relationship level in each specification.

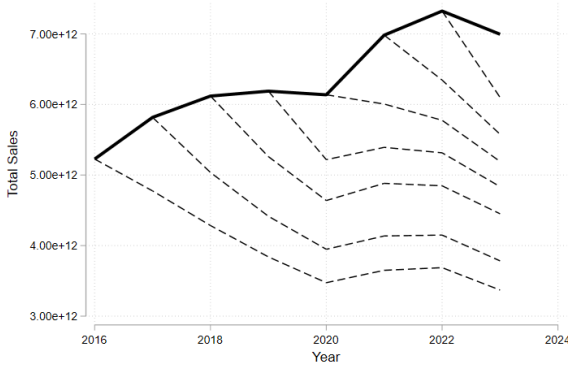
Figure 12 plots the estimated age coefficients. Both relationships display hump-shaped sales profiles, but buyer links peak earlier and decline more quickly. Link sales peak at approximately 0.13 log points above their entry-year level at age one. Supplier–product sales continue rising through age two, reach a peak of approximately 0.20 log points, and remain above the link profile at subsequent ages. By age five, the supplier–product premium remains close to 0.09 log points, whereas the link premium has fallen to approximately 0.03.

Figure 12: Sales over the lifecycle of buyer links and supplier–product relationships

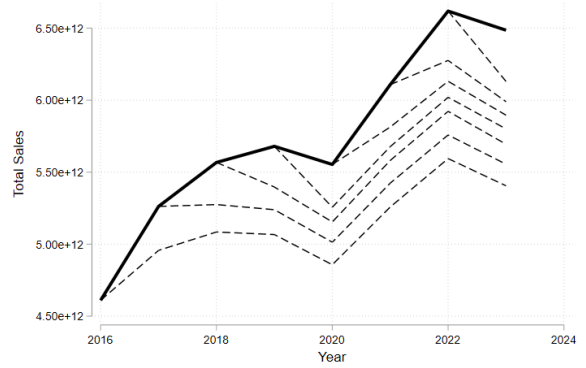


Notes: The figure plots the estimated coefficients from equations (40) and (41). The link specification includes supplier–buyer-link and year fixed effects; the product specification includes supplier–product and year fixed effects. Age zero is the omitted category, so each coefficient measures the average within-relationship change in log sales relative to the first observed year of the relationship. Shaded bands report 95 percent confidence intervals. Standard errors are clustered at the corresponding relationship level.

B.2 Evolution of total sales by cohort and year

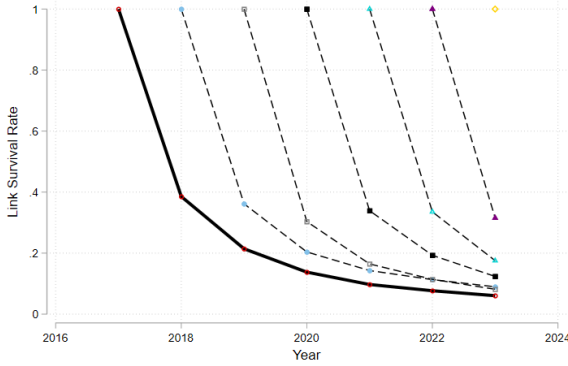


(a) Link total sales per cohort and year

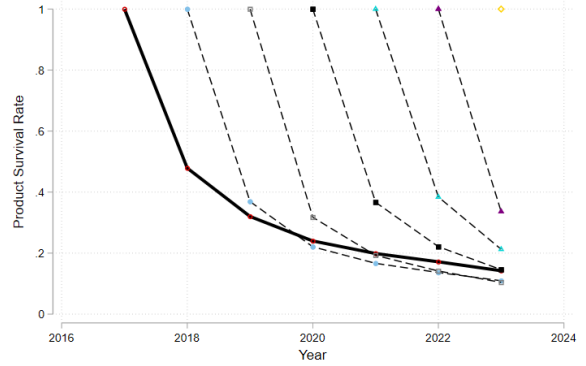


(b) Product total sales per cohort and year

B.3 Survival rate of links and products



(a) Link survival rate



(b) Product survival rate

C Additional Decompositions of Supplier Growth

C.1 Composition of Extensive-Margin Growth

The main decomposition in Section 3.1 includes the OLOP component, which combines changes in sales within the supplier’s baseline buyer–product space with losses from exiting relationships. To focus more directly on adjustment involving new buyers or new products, this appendix reports the three extensive-margin components—NLOP, OLNP, and NLNP—without plotting OLOP.

For each supplier, define the total extensive-margin growth as

$$\Delta S_s^{\text{EXT}} = \Delta S_s^{\text{NLOP}} + \Delta S_s^{\text{OLNP}} + \Delta S_s^{\text{NLNP}}.$$

Then, for suppliers with nonzero extensive-margin adjustment, define the contribution of margin c to total extensive-margin sales growth as

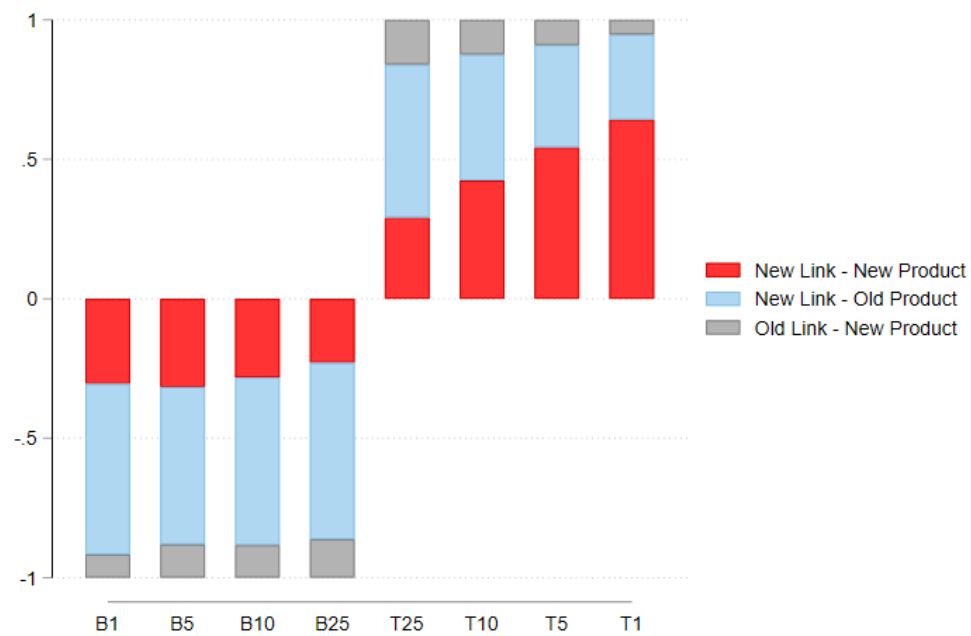
$$\tilde{s}_s^c = \frac{\Delta S_s^c}{\Delta S_s^{\text{EXT}}}, \quad c \in \{\text{NLOP}, \text{OLNP}, \text{NLNP}\}.$$

Because each component is normalized by total extensive-margin adjustment, ΔS_s^{EXT} , the shares need not all be positive. A negative contribution indicates that a particular extensive margin moves in the opposite direction from the supplier's net extensive-margin adjustment.

Figure 15 reports these contributions across the same intervals of the realized-growth distribution used in Figure 3. The composition of extensive-margin adjustment changes markedly across the growth distribution. In the lower tail, some extensive-margin components contribute negatively to the net adjustment. As realized supplier growth increases, however, the composition shifts increasingly toward NLNP. Among the fastest-growing suppliers, a large share of extensive-margin adjustment is associated with transactions involving both buyers and products that were not present in the baseline year. NLOP also remains quantitatively important, whereas OLNP contributes relatively little.

The figure therefore reinforces the main Fact 1 result from a different perspective. Even after focusing only on adjustment involving new buyers or new products, the joint new-buyer-new-product margin becomes especially important among the largest expansions. The fastest-growing suppliers do not simply expand through buyer acquisition or product introduction separately; their extensive-margin growth disproportionately combines both margins.

Figure 15: Composition of Extensive-Margin Growth



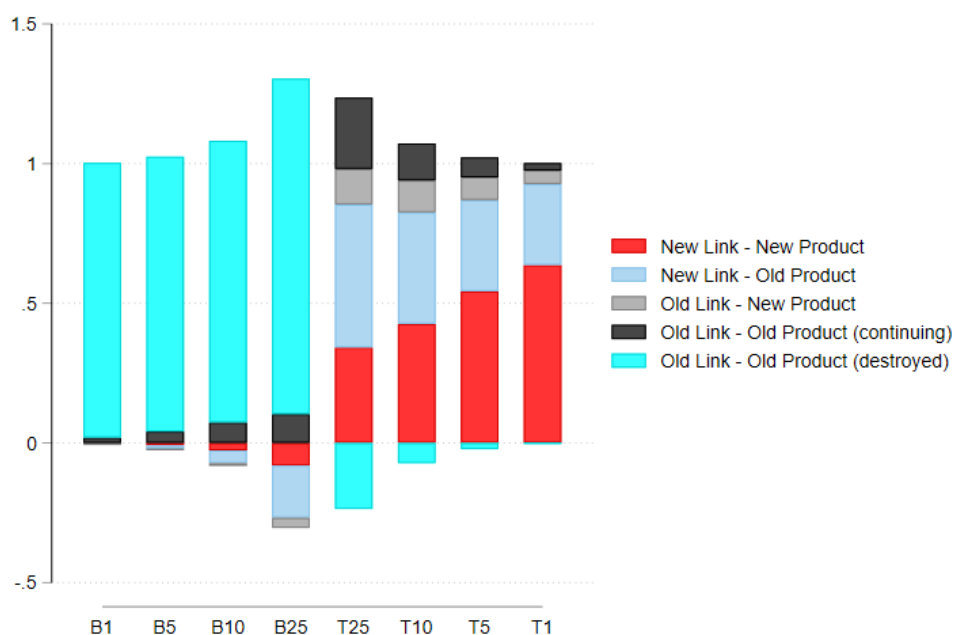
Notes: The figure reports the shares of positive extensive-margin growth accounted for by NLNP, NLOP, and OLNP across intervals of the realized supplier-growth distribution. The groups are defined using adjacent percentile cutoffs, as in Figure 3.

C.2 The Role of Relationship Exit

A further decomposition of OLOP distinguishes changes in sales among old buyer–product relationships that remain active through 2023 from losses associated with relationships that disappear between 2017 and 2023. Figure 16 shows that relationship exit is the dominant source of contraction in the lower tail of the realized-growth distribution. Among the suppliers with the largest declines, nearly all of the fall in sales is accounted for by the disappearance of baseline buyer–product relationships, while changes in sales among continuing OLOP relationships play only a small role. In some of the other declining groups, losses from destroyed relationships exceed the supplier’s net contraction because they are partially offset by sales generated through new buyers or new products.

The pattern changes markedly among growing suppliers. Losses from relationship exit continue to offset some expansion, but become progressively smaller relative to total growth. At the same time, growth increasingly comes from new buyer–product combinations, particularly NLNP, while continuing OLOP relationships also contribute positively. Thus, the asymmetry across the growth distribution reflects not only differences in the importance of extensive-margin expansion, but also substantial differences in relationship destruction: contraction is driven primarily by the disappearance of existing buyer–product relationships, whereas the largest expansions rely increasingly on the formation of new ones.

Figure 16: Decomposition of OLOP Growth into Continuing and Destroyed Relationships



Notes: The figure decomposes supplier sales growth between 2017 and 2023 across intervals of the realized-growth distribution. The old link–old product (OLOP) component is separated into changes in sales among baseline buyer–product relationships that remain active in 2023 (continuing) and losses from baseline buyer–product relationships that are no longer active in 2023 (destroyed). The remaining components are new link–new product (NLNP), new link–old product (NLOP), and old link–new product (OLNP). Each component is normalized by the supplier’s total change in sales. For shrinking suppliers, losses from destroyed relationships therefore enter with a positive share because both the loss and total sales growth are negative, while positive sales generated through new buyers or products appear as negative shares because they offset the overall contraction. B denotes intervals in the bottom half and T intervals in the top half of the realized-growth distribution; categories are defined using adjacent percentile cutoffs as in Figure 3.

C.3 Heterogeneity by Initial Supplier Sales

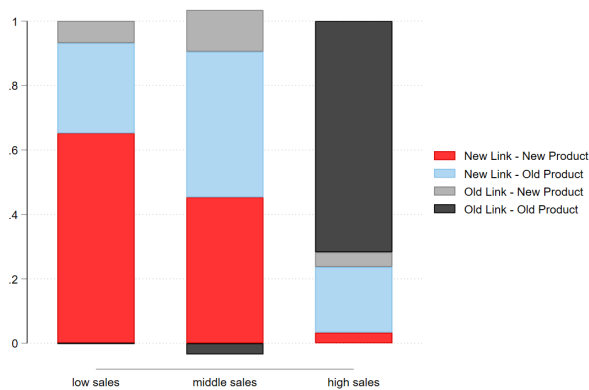
We next examine whether the composition of supplier growth varies with suppliers' initial scale. We divide suppliers into low-, middle-, and high-sales categories based on their sales in 2017 and repeat the growth decomposition across the four buyer-product margins. Figure 17 reports the results. Panel (a) shows the full decomposition, while Panel (b) excludes OLOP and focuses on the composition of growth across the three extensive margins.

The composition of growth varies systematically with initial supplier size. Among initially smaller suppliers, NLNP accounts for the largest share of growth, so expansion disproportionately combines new buyers with new products. As initial sales increase, the importance of NLNP declines, while growth through existing buyer-product relationships becomes more important. Larger suppliers therefore rely relatively less on the joint expansion of buyer and product scope and more on margins involving products already in their portfolios.

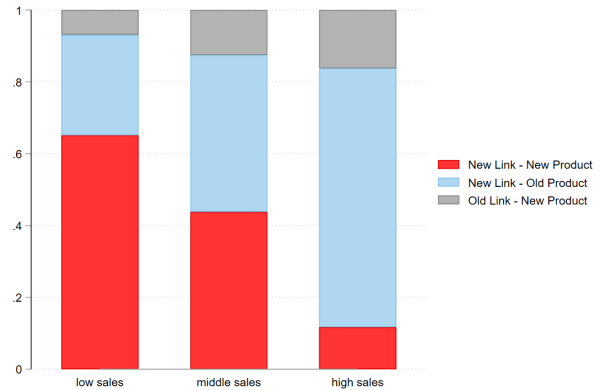
Panel (b) shows that this pattern is not driven solely by the greater importance of OLOP among larger suppliers. Restricting attention to the extensive margins, the contribution of NLNP declines with initial sales, while NLOP becomes increasingly important. Thus, smaller suppliers are relatively more likely to expand buyer and product scope jointly, whereas larger suppliers are more likely to reach new buyers with products they already sell. OLNP remains comparatively small across all three sales categories.

This size gradient provides a useful target for the model's heterogeneous comparative statics. In particular, the model should generate a shift from the joint expansion of buyer and product scope among smaller suppliers toward greater reliance on established products as supplier scale increases.

Figure 17: Growth decomposition by sales category



(a) Growth decomposition by sales category



(b) Growth decomposition by sales without OLOP

C.4 Heterogeneity over the Supplier Life Cycle

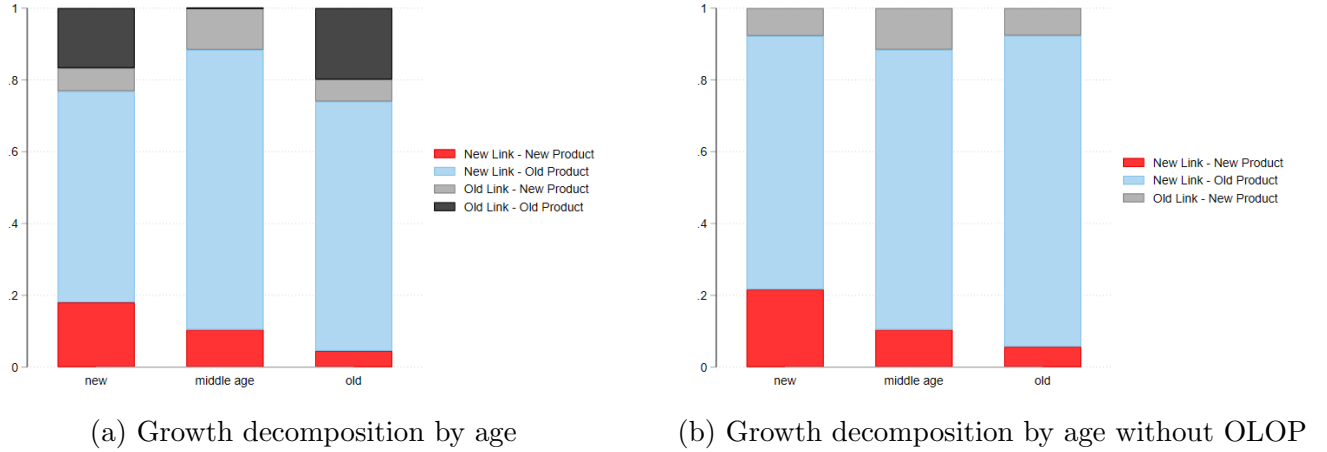
We then ask whether the composition of growth also varies over the supplier life cycle, while conditioning on suppliers' sales in 2017. This exercise separates differences associated with supplier age from those arising simply because older suppliers tend to operate at a different initial scale. Figure 18 reports the decomposition for young, middle-aged, and old suppliers.

The clearest age pattern appears in the composition of extensive-margin growth. Younger suppliers rely relatively more on NLNP, jointly adding new buyers and new products, whereas the importance of NLOP increases with supplier age. Older suppliers are therefore relatively more likely to expand by forming new buyer relationships for products already included in their existing portfolios. OLNP remains small across age groups.

Panel (b) makes this shift particularly clear. After excluding OLOP, the share of extensive-margin growth accounted for by NLNP falls monotonically with supplier age, while the share accounted for by NLOP rises. Conditional on initial sales, younger suppliers therefore expand buyer and product scope more jointly, whereas older suppliers increasingly expand their customer base using products they have already adopted.

These patterns are consistent with a life-cycle interpretation in which younger suppliers are still jointly developing their customer base and product portfolio, while more mature suppliers increasingly extend the market reach of an established product portfolio. Because supplier age is not an explicit state variable in our static model, however, we interpret this exercise as complementary descriptive evidence rather than as a moment that the model is required to match directly.

Figure 18: Growth decomposition by age



Notes: The figure decomposes supplier sales growth between 2017 and 2023 into NLNP, NLOP, OLNP, and OLOP by supplier age group, conditioning on suppliers' sales in 2017. Young suppliers are between one and five years old, middle-aged suppliers are between six and fifteen years old, and old suppliers are more than fifteen years old. Supplier age is measured as of 2017.

Taken together, the initial-size and age decompositions show that the relative importance of buyer and product expansion varies systematically across suppliers. Initially smaller suppliers expand relatively more through the joint addition of new buyers and new products, whereas larger suppliers rely more heavily on established products and existing buyer-product relationships. A related pattern appears over the supplier life cycle: conditional on initial sales, younger suppliers expand relatively more through the joint addition of buyers and products, while older suppliers increasingly extend established products to new customers. The initial-size results provide a direct target for the model's heterogeneous comparative statics, while the age patterns provide complementary descriptive evidence on how these margins vary with supplier maturity.

D Buyer breadth and product scope among incumbent buyers

A potential concern with the relationship documented in Section 3.3 is that the positive association between buyer breadth and product scope could arise mechanically from the formation of new buyer relationships. If newly acquired buyers purchase products that the supplier does not sell to its existing customers, an increase in the number of buyers would mechanically coincide with an increase in measured product scope, even in the absence of broader adjustments across the supplier's existing relationships.

To assess this possibility, we repeat the analysis measuring product scope using only products sold to incumbent buyers. An incumbent buyer is a buyer with an existing relationship with

the supplier prior to year t . The dependent variable is therefore the log number of distinct products sold by supplier s in year t to its incumbent buyers. We estimate specifications analogous to those in Table 2, progressively including sector and supplier fixed effects and controlling for contemporaneous supplier sales.

Table 9 shows that the relationship remains positive and statistically significant when product scope is restricted to incumbent relationships. In the specification with supplier and year fixed effects, the elasticity of product scope to incumbent buyers with respect to buyer breadth is 0.382. Controlling additionally for contemporaneous supplier sales reduces the coefficient to 0.288. Thus, conditional on persistent supplier heterogeneity and current firm size, a 10 percent increase in buyer breadth is associated with approximately a 2.9 percent increase in the number of products sold to incumbent buyers.

The magnitude is smaller than for total product scope, where the corresponding coefficient in the specification that also controls for contemporaneous supplier sales is 0.486 (Table 2, column 6). This difference is consistent with part of the overall co-movement between buyer breadth and product scope occurring through products associated with newly formed buyer relationships. Importantly, however, the positive relationship remains when these newly formed relationships are excluded from the outcome. Suppliers that serve more buyers also tend to operate a broader product portfolio within their pre-existing customer relationships.

This exercise therefore suggests that the Fact 3 relationship is not driven solely by the mechanical addition of products accompanying new buyers. Instead, changes in buyer breadth are associated with broader product scope even among established relationships. As in the baseline analysis, we interpret this evidence as a conditional association rather than a causal effect of buyer acquisition. The result reinforces the motivation for a model in which buyer and product scope are jointly determined and adjustments along one margin can affect incentives along the other.

E Buyer Breadth and Product Additions

In the main text, Fact 3 documents a strong positive relationship between a supplier's buyer breadth and its overall product scope. This appendix shows that the same pattern is also present at the margin of product additions. We define a product as new to a supplier in year t if the supplier sells that CPC in year t but did not sell it in year $t - 1$. Thus, this measure captures additions to the supplier's product portfolio relative to the previous year rather than the product's first-ever appearance in the data.

Table 10 relates the log number of product additions to the supplier's contemporaneous buyer breadth. The specifications mirror those in Table 2 in the main text. Columns (1)–(2) include year fixed effects, columns (3)–(4) additionally control for supplier-sector fixed effects, and columns (5)–(6) instead include supplier fixed effects. Even-numbered columns additionally control for

Table 9: Buyer Breadth and Product Scope Among Incumbent Buyers

<i>Dependent variable: Log Number of Products Sold to Incumbent Buyers</i>						
Log Number of Buyers	0.447*** (0.00190)	0.402*** (0.00225)	0.463*** (0.00179)	0.387*** (0.00219)	0.382*** (0.00459)	0.288*** (0.00487)
Log Sales		0.0692*** (0.00169)		0.116*** (0.00179)		0.185*** (0.00388)
Observations	106,300	106,143	106,299	106,142	104,166	103,983
Firms	21,627	21,622	21,626	21,621	19,493	19,462
R^2	0.330	0.340	0.452	0.474	0.862	0.867
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Sector FE	No	No	Yes	Yes	No	No
Supplier FE	No	No	No	No	Yes	Yes

Notes: The dependent variable is the log number of distinct products (CPCs) sold by supplier s in year t to incumbent buyers. The main explanatory variable is the log number of distinct buyers served by the supplier in the same year. Columns (1)–(2) include year fixed effects; columns (3)–(4) additionally include supplier-sector fixed effects; and columns (5)–(6) include supplier fixed effects instead. Even-numbered columns additionally control for contemporaneous log supplier sales. Heteroskedasticity-robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

contemporaneous supplier sales.

The relationship between buyer breadth and product additions is positive and precisely estimated across all specifications. Most importantly, it remains strong after introducing supplier fixed effects, which absorb persistent differences across firms. In the specification with year and supplier fixed effects, the coefficient on log buyers is 0.392. Controlling additionally for contemporaneous sales reduces the coefficient to 0.340, but it remains highly statistically significant. Therefore, within the same supplier and conditional on current firm size, a 10 percent increase in buyer breadth is associated with approximately a 3.4 percent increase in the number of product additions.

These results complement the evidence in the main text. The positive relationship between buyer breadth and product scope is not driven solely by differences in firms' existing portfolios: suppliers with broader customer bases also add more products. As in the main analysis, we interpret these estimates as conditional associations rather than causal effects of buyer acquisition on product adoption. They provide additional evidence for the model-facing fact that buyer breadth and product scope systematically co-move within suppliers.

Table 10: Buyer Breadth and Product Additions

<i>Dependent variable: Log Number of New Products</i>						
Log Buyers	0.261*** (0.00149)	0.246*** (0.00171)	0.289*** (0.00136)	0.255*** (0.00163)	0.392*** (0.00331)	0.340*** (0.00362)
Log Sales		0.0217*** (0.00129)		0.0491*** (0.00128)		0.0948*** (0.00271)
Observations	138,784	138,534	138,784	138,534	138,219	137,966
Firms	23,382	23,380	23,382	23,380	22,817	22,812
R^2	0.225	0.226	0.420	0.426	0.735	0.739
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Sector FE	No	No	Yes	Yes	No	No
Supplier FE	No	No	No	No	Yes	Yes

Notes: The dependent variable is the log number of products sold by supplier s in year t that were not sold by the supplier in year $t - 1$. The main explanatory variable is the log number of distinct buyers served by the supplier in year t . Columns (1)–(2) include year fixed effects; columns (3)–(4) additionally include supplier-sector fixed effects; and columns (5)–(6) include supplier fixed effects instead. Even-numbered columns additionally control for contemporaneous log supplier sales. Heteroskedasticity-robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

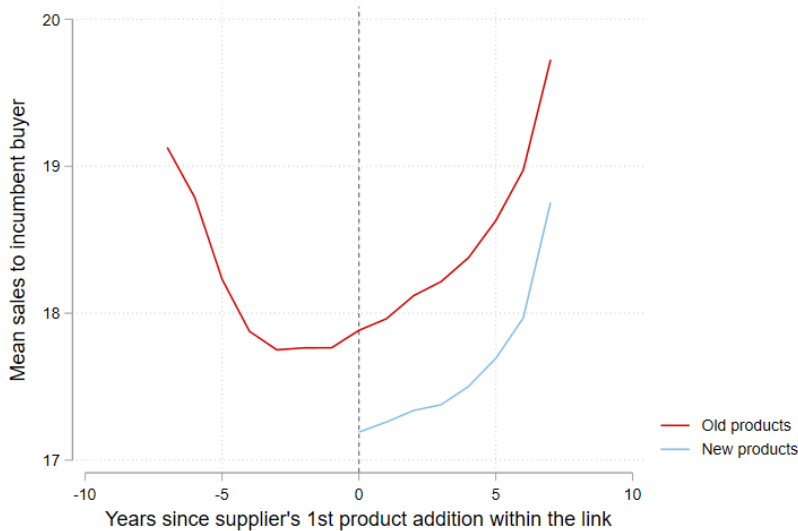
F Ruling Out Cannibalization

The two dimensions of firm scope studied in the paper raise two distinct cannibalization concerns. First, when a supplier introduces an additional product to an existing buyer, sales of the new product may come at the expense of products that the supplier was already selling to that buyer. Second, when a supplier acquires a new buyer, sales to the new relationship may come at the expense of its incumbent buyer relationships. This appendix examines these two possibilities separately.

F.1 Product Introductions within Existing Buyer Relationships

We first ask whether introducing an additional product within an existing supplier–buyer relationship displaces sales of the supplier’s incumbent products. Figure 19 provides a descriptive view of the event. We define $t = 0$ as the first year in which the supplier adds a new product within an existing buyer relationship and follow sales of the products already sold in that relationship and of the newly introduced products. Restricting attention to supplier–buyer links that survive throughout the event window, incumbent-product sales do not decline following the introduction. Instead, sales of old and new products expand together.

Figure 19: Sales around a product introduction within an existing link



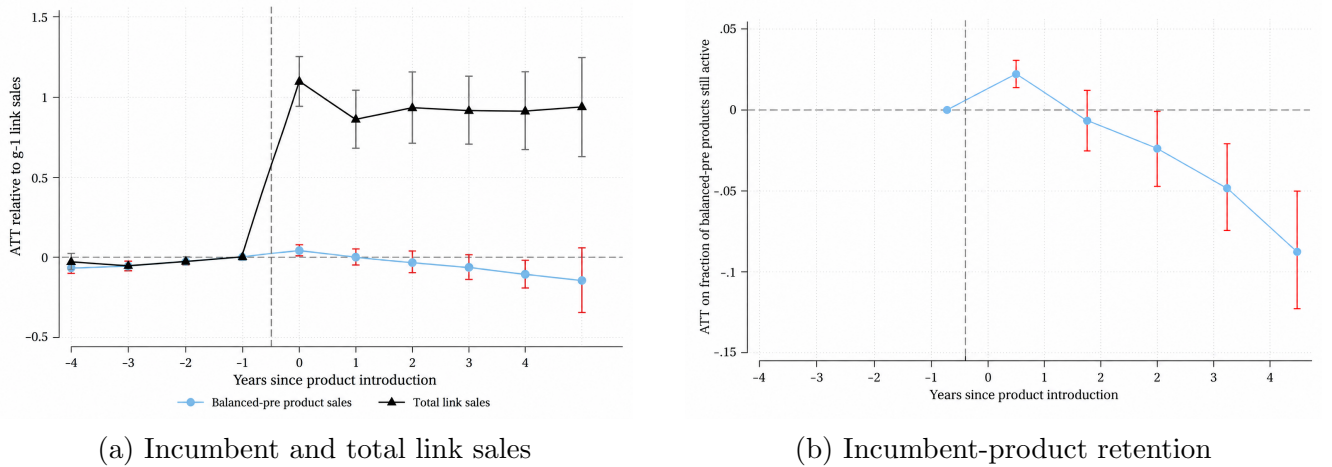
Notes: The figure plots mean log sales to the incumbent buyer for old products and new products in years surrounding the first addition of a new product within an existing supplier–buyer relationship. Event time $t = 0$ denotes the product introduction. Old products are those sold within the relationship before the new product is introduced; new products are those first added at $t = 0$. The sample contains supplier–buyer relationships that remain active throughout the event window. The figure is descriptive and plots raw sales paths rather than estimates relative to a comparison group.

Figure 20 complements these raw sales paths with a balanced-pre comparison. For each product-introduction cohort, we define incumbent products as those with positive sales in every pre-event year from $t = -4$ through $t = -1$. Treated relationships are compared with never-treated and not-yet-treated relationships, using $t = -1$ as the common baseline. To make sales changes comparable across relationships, product and total-link sales are normalized by total sales in the supplier–buyer relationship at $t = -1$. We focus here on relationships that remain active through the end of the sample, so that changes in incumbent products are not mechanically generated by the disappearance of the supplier–buyer relationship itself.

Panel a shows a sharp expansion in total supplier–buyer sales following the introduction of a new product. In contrast, sales of the balanced-pre incumbent products remain broadly stable. Thus, the additional product primarily expands the size of the relationship rather than being offset by a reduction in revenues from products already sold to the buyer.

Panel b shows an important qualification. Although incumbent-product revenues do not systematically decline, the fraction of incumbent products that remains active falls gradually at longer horizons. Product introduction therefore appears to generate some longer-run portfolio reorganization. The evidence points to gradual product replacement on the extensive margin rather than immediate cannibalization of incumbent-product revenues.

Figure 20: Product introduction and incumbent products: balanced-pre comparisons



Notes: The figure studies the first addition of a new product within an existing supplier–buyer relationship. Incumbent products are products with positive sales in each of the four years preceding the event, $t = -4, \dots, -1$. The sample shown here is restricted to supplier–buyer relationships that remain active through the end of the sample. For each treatment cohort and event horizon, treated relationships are compared with never-treated and not-yet-treated relationships, with $t = -1$ as the common baseline. Panel (a) plots estimated effects on sales of balanced-pre incumbent products and on total supplier–buyer sales, with both outcomes normalized by total link sales at $t = -1$. Panel (b) reports the estimated effect on the fraction of balanced-pre incumbent products that continue to have positive sales. Vertical bars report 95 percent confidence intervals; standard errors are clustered at the supplier level.

Taken together, Figures 19 and 20 provide little evidence of economically important within-link sales cannibalization. New-product sales are associated with a substantial expansion of the overall supplier–buyer relationship, while revenues from incumbent products remain broadly stable. At longer horizons, some incumbent products disappear, indicating gradual portfolio turnover even in the absence of systematic displacement of incumbent-product sales. This pattern is consistent with the separability assumption maintained in the benchmark and, in the nested-CES extension of Appendix N, with the case in which the within-supplier and across-supplier elasticities are similar, $\varepsilon \approx \nu$.

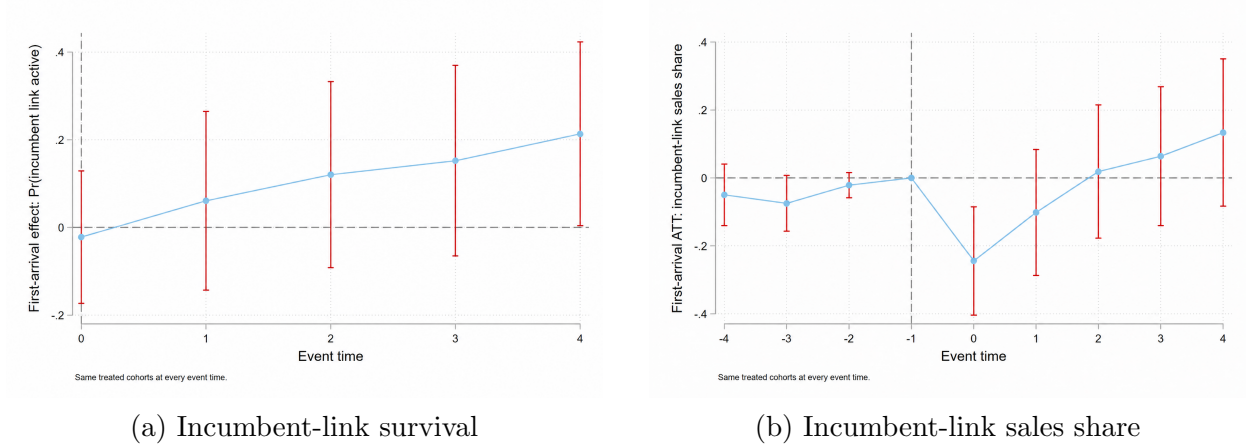
F.2 New-Buyer Introductions and Incumbent Buyer Relationships

The second exercise studies cannibalization along the buyer margin. We ask whether acquiring a new buyer diverts activity away from the supplier’s existing buyer relationships. Here, the event is different from the exercise above: $t = 0$ denotes the arrival of a new buyer rather than the introduction of a new product within an existing relationship. Incumbent links are supplier–buyer relationships that were already active before the new buyer arrives.

Figure 21 examines two potential forms of displacement. Panel a asks whether the arrival of a new buyer makes incumbent relationships more likely to disappear. We find no systematic decline in incumbent-link survival. If anything, the estimates are positive after the event, although they are imprecisely estimated.

Panel b instead considers the intensive allocation of the supplier’s sales across buyers. The sales share accounted for by incumbent buyers falls on impact when the new buyer arrives, as one would mechanically expect when an additional buyer enters the supplier’s customer portfolio. The decline is temporary: the incumbent share subsequently recovers and is close to its pre-event level within roughly two years. Thus, the addition of a new buyer generates some short-run reallocation of the supplier’s sales across customers, but we find no evidence of persistent displacement of incumbent buyer relationships.

Figure 21: New-buyer introduction and incumbent buyer relationships



Notes: The figure examines incumbent buyer relationships around the arrival of a new buyer. Event time $t = 0$ denotes the year in which the new buyer enters the supplier's customer portfolio. Incumbent links are buyer relationships that were active before the event. Panel (a) reports the estimated effect on the probability that an incumbent relationship remains active in each post-event year. Panel (b) reports the evolution of incumbent buyers' share of supplier sales around the event, with $t = -1$ as the omitted reference period. The sample is restricted to cohorts observed at every event time shown in each panel. Vertical bars report 95 percent confidence intervals; standard errors are clustered at the supplier level.

The two exercises address different forms of potential cannibalization. Following a product introduction within an existing relationship, we find little evidence that the new product displaces revenues from products already sold to that buyer. Total relationship sales expand substantially, while incumbent-product sales remain broadly stable, although some incumbent products are gradually dropped at longer horizons. Following a new-buyer introduction, we similarly find no persistent deterioration in the supplier's incumbent buyer relationships. Incumbent buyers temporarily account for a smaller share of supplier sales when the new relationship is formed, but their share subsequently recovers and their relationships remain active.

Taken together, these results suggest that cannibalization is not a first-order force along either dimension of firm scope in our setting. Product expansion is predominantly associated with expansion of existing buyer relationships rather than redistribution of a fixed amount of sales across products, while buyer expansion does not generate persistent displacement of existing customers.

G Rank structure

This matching structure yields a sharp, testable implication for the relationship between firm-level and link-level product hierarchies. A rigid common ladder implies that a product’s rank in the supplier’s overall portfolio should predict its rank within any individual link almost perfectly: every buyer descends the same ordered list. A matching structure instead predicts a positive but partial association where core products tend to be more important within links because they match more often and sell more when matched, but idiosyncratic matches make the within-link hierarchy differ from the firm-level one buyer by buyer. Table 11 shows that the data display exactly this partial alignment. A higher firm-level rank score is associated with only a 0.27–0.29 higher link-level score, and products in the supplier’s top three are 0.08–0.10 more likely to be among a link’s top three; these associations are positive and significant but far from the near-perfect alignment a rigid ladder would imply. Most tellingly, being the supplier’s single top product does not significantly predict being the top product within a given link (column 3). This is difficult to reconcile with a common ladder, under which the core product should be the within-link leader almost everywhere, but is precisely what the matching model predicts: even the firm’s best product fails to match some buyers, and other products lead within those links. We therefore adopt matching as the baseline and treat the correlation in Table 11 as an identifying moment for the dispersion of the match shock (the parameter θ_ζ introduced in Section 4.1): stronger alignment maps to less dispersed, more deterministic matches, and conversely.

Appendix H shows that the buyer–product complementarity documented in Table 2 is not uniform across the supplier’s product portfolio. We study product-level sales around the addition of a large new buyer and allow the response to vary with the product’s pre-event position in the supplier’s portfolio. Products are classified as core, middle, or peripheral according to their pre-event sales rank, with core products serving as the omitted category. The response is strongly concentrated toward the periphery of the portfolio. Before the arrival of the new buyer, sales of middle and peripheral products evolve similarly to those of core products. Beginning in the event year, however, peripheral products experience a sharp and persistent increase in sales relative to core products. The response of middle products is positive but substantially smaller. The effect therefore becomes progressively stronger as we move from core toward peripheral products.

This heterogeneity is consistent with the amortization mechanism in Section 4.2. Because the firm-level product adoption cost is amortized across buyer relationships, adding a buyer raises the return to maintaining and expanding products closer to the margin of the supplier’s portfolio. The evidence therefore complements the product-introduction results in Table 10: buyer acquisition is associated disproportionately with activity in peripheral products rather than with further expansion of the products that already account for the largest share of the supplier’s sales.

Table 11: Within-link product-rank correlations

	(1)	(2)	(3)	(4)	(5)	(6)
	Link score	Link rank	Link top-1 dummy	Link top-1 score	Link top-3 dummy	Link top-3 score
<i>Panel A: Link FE + Year FE</i>						
Firm rank score	0.266*** (0.0223)			0.142*** (0.0383)		0.270*** (0.0379)
Firm rank		0.0599*** (0.00482)				
Firm top-1			0.0167 (0.0550)			
Firm top-3					0.105*** (0.0216)	
R-squared	0.012	0.583	0.112	0.115	0.274	0.272
<i>Panel B: Link×Year FE</i>						
Firm score	0.287*** (0.0270)			0.120*** (0.0457)		0.225*** (0.0410)
Firm rank		0.0465*** (0.00625)				
Firm top-1			0.000456 (0.0640)			
Firm top-3					0.0775*** (0.0245)	
R-squared	0.013	0.628	0.131	0.133	0.319	0.320
Observations	56,565,645	56,565,645	56,565,645	56,565,645	48,832,461	48,832,461
Firms	52,334	52,334	52,334	52,334	45,023	45,023
Cluster	Supplier					

Notes: The unit of observation is a supplier-buyer-product-year. Dependent variables capture the importance of product g within link (s, b) at year t : Link score is the product's sales share within the link, Link rank is its rank position by sales within the link, Link top-1 dummy equals one if the product is the top seller within the link, Link top-1 score is its sales share if it is the top seller zero otherwise, and analogous definitions apply for the top-3 outcomes. The independent variables are the corresponding supplier-level analogs, computed using the supplier's total observed B2B sales of each product across all buyers. Panel A uses additive link and year fixed effects; Panel B replaces them with link-by-year fixed effects, which absorb any link-specific shocks that vary over time. Sample sizes and firm counts are identical across panels for a given column. Standard errors clustered at the supplier level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

H Buyer–Product Complementarity across the Product Ladder

The results in Table 2 show that buyer acquisition and product expansion move together. In this appendix, we ask whether the effect of adding a buyer is concentrated in particular parts of the supplier’s existing product portfolio. This provides a more direct test of the model’s prediction that the value of an additional buyer should be particularly important for products closer to the margin of the firm’s product scope.

We implement an event-study around the first year in which a supplier adds a large new buyer. Buyer size is measured using the buyer’s total purchases, and a large buyer is defined as one in the top decile of the buyer-size distribution. We classify the supplier’s products according to their pre-event position in its sales distribution. Core products are those in the top tercile, peripheral products are those in the bottom tercile, and the remaining products form the middle category. Because the classification is determined before the event, subsequent changes in product sales do not mechanically alter a product’s position in the ladder.

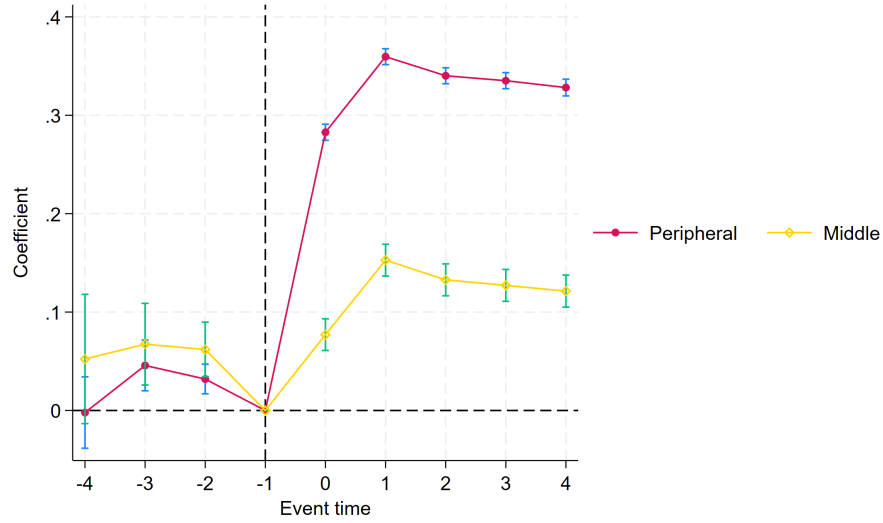
Figure 22 reports the event-study estimates, with core products serving as the omitted category. Event time $t = -1$, one year before the addition of the large buyer, is the reference period. The coefficients therefore trace how the sales of middle and peripheral products evolve around buyer acquisition relative to the corresponding change for core products. There is little evidence of differential pre-trends before buyer acquisition. In the years preceding the event, the coefficients for both middle and peripheral products are small relative to their post-event values and close to the core-product benchmark. The pattern changes sharply when the new buyer arrives. Peripheral products experience an immediate increase in sales relative to core products, and this differential rises further in the following year before remaining persistently large. Middle products also experience a positive relative increase, but its magnitude is substantially smaller.

The response therefore displays a clear gradient across the product ladder: the effect of buyer acquisition becomes stronger as products move from the core toward the periphery of the supplier’s portfolio. This pattern is consistent with the product-adoption mechanism in Section 4.3. A product-level opportunity that generates relatively little revenue with the supplier’s existing customer base can become more valuable when an additional buyer is added. Because the fixed cost of product adoption is borne at the supplier level and its return can be realized across buyer relationships, buyer acquisition disproportionately raises the value of products closer to the margin of the firm’s portfolio.

Importantly, the event study should not be interpreted as directly measuring new product introduction: products are assigned to the ladder using their pre-event sales and are therefore already present before the new buyer arrives. Rather, the exercise shows that the increase in

product-level sales associated with buyer acquisition is concentrated among products that were initially peripheral. Combined with the evidence in Table 10 that years with broader buyer bases are also associated with greater product introduction, this provides complementary evidence that buyer expansion is disproportionately associated with activity in products near the margin of the supplier's portfolio.

Figure 22: Buyer–Product Complementarity across the Product Ladder



Notes: The figure reports event-study estimates of product-level log sales around the first year in which a supplier adds a large new buyer. A large buyer is defined as a buyer in the top decile of the buyer-size distribution, where buyer size is measured by total purchases. Products are classified according to their pre-event position in the supplier's product-sales distribution: core products are in the top tercile, peripheral products are in the bottom tercile, and middle products are in the remaining tercile. Core products are the omitted category. Event time $t = -1$, one year before the new buyer is added, is the reference period. The plotted coefficients therefore measure the evolution of middle and peripheral product sales relative to core product sales and relative to $t = -1$. Vertical bars report 95% confidence intervals. Standard errors are clustered at the supplier level.

I CES Buyer Demand and Revenue Derivation

This appendix derives the buyer demand system underlying the product-level revenue equation used in the benchmark model. The main text states the resulting revenue equation directly; here we derive the associated CES price index, quantity demand, and expenditure shares.

Buyer b has total expenditure $E_b > 0$ on the relevant intermediate-input bundle. Let $\mathcal{S}_b \subseteq \mathcal{S}$ denote the endogenous set of suppliers active on buyer b . Supplier s develops a firm-level portfolio of G_s products, but only a buyer-specific subset $\mathcal{G}_{sb} \subseteq \{1, \dots, G_s\}$ is available on link (s, b) . When allocating expenditure, buyer b takes the realized sets $\mathcal{S}_b$ and $\{\mathcal{G}_{sb}\}_{s \in \mathcal{S}_b}$ as given. Products that are not active on a link do not enter the buyer's choice set.

I.1 CES aggregator and expenditure minimization

For each active supplier-product variety (s, g) , let q_{sbg} denote the quantity purchased by buyer b and p_{sbg} its price. Buyer-product match quality is denoted by $\zeta_{sbg} > 0$. Buyer b combines the available supplier-product varieties according to

$$M_b = \left[\sum_{s \in \mathcal{S}_b} \sum_{g \in \mathcal{G}_{sb}} \zeta_{sbg}^{1/\sigma} q_{sbg}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \quad \sigma > 1. \quad (42)$$

The normalization $\zeta_{sbg}^{1/\sigma}$ implies that the corresponding quantity demand and revenue equations are linear in the product-buyer match shifter ζ_{sbg} .

To derive the associated price index, consider the expenditure-minimization problem for a target level $\bar{M}_b$:

$$\min_{\{q_{sbg}\}} \sum_{s \in \mathcal{S}_b} \sum_{g \in \mathcal{G}_{sb}} p_{sbg} q_{sbg} \quad (43)$$

subject to

$$\left[\sum_{s \in \mathcal{S}_b} \sum_{g \in \mathcal{G}_{sb}} \zeta_{sbg}^{1/\sigma} q_{sbg}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} \geq \bar{M}_b. \quad (44)$$

Let λ_b denote the multiplier on the CES constraint. The Lagrangian is

$$\begin{aligned} \mathcal{L}_b = & \sum_{s \in \mathcal{S}_b} \sum_{g \in \mathcal{G}_{sb}} p_{sbg} q_{sbg} \\ & + \lambda_b \left\{ \bar{M}_b - \left[\sum_{s \in \mathcal{S}_b} \sum_{g \in \mathcal{G}_{sb}} \zeta_{sbg}^{1/\sigma} q_{sbg}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} \right\}. \end{aligned} \quad (45)$$

For any active variety (s, g) , the first-order condition is

$$p_{sbg} = \lambda_b \bar{M}_b^{1/\sigma} \zeta_{sbg}^{1/\sigma} q_{sbg}^{-1/\sigma}. \quad (46)$$

Rearranging,

$$q_{sbg} = \bar{M}_b \zeta_{sbg} \left(\frac{\lambda_b}{p_{sbg}} \right)^\sigma. \quad (47)$$

Substituting (47) into (44) and imposing equality at the optimum gives

$$1 = \lambda_b^{\sigma-1} \sum_{s \in \mathcal{S}_b} \sum_{g \in \mathcal{G}_{sb}} \zeta_{sbg} p_{sbg}^{1-\sigma}. \quad (48)$$

Hence

$$\lambda_b = \left[\sum_{s \in \mathcal{S}_b} \sum_{g \in \mathcal{G}_{sb}} \zeta_{sbg} p_{sbg}^{1-\sigma} \right]^{\frac{1}{1-\sigma}}. \quad (49)$$

The multiplier is therefore the buyer-level CES price index,

$$P_b = \left[\sum_{s \in \mathcal{S}_b} \sum_{g \in \mathcal{G}_{sb}} \zeta_{sbg} p_{sbg}^{1-\sigma} \right]^{\frac{1}{1-\sigma}}. \quad (50)$$

Equivalently,

$$P_b^{1-\sigma} = \sum_{s \in \mathcal{S}_b} \sum_{g \in \mathcal{G}_{sb}} \zeta_{sbg} p_{sbg}^{1-\sigma}. \quad (51)$$

Using $\lambda_b = P_b$ in (47), Hicksian demand for an active supplier–product variety is

$$q_{sbg}^H = \bar{M}_b \zeta_{sbg} \left(\frac{p_{sbg}}{P_b} \right)^{-\sigma}. \quad (52)$$

The associated expenditure function is

$$\sum_{s \in \mathcal{S}_b} \sum_{g \in \mathcal{G}_{sb}} p_{sbg} q_{sbg}^H = P_b \bar{M}_b. \quad (53)$$

Thus P_b is the unit-expenditure function for buyer b 's intermediate-input bundle.

I.2 Quantity demand and product-level revenue

In the benchmark partial-equilibrium environment, buyer b has total expenditure E_b . Expenditure exhaustion implies

$$E_b = P_b M_b, \quad \text{or equivalently} \quad M_b = \frac{E_b}{P_b}. \quad (54)$$

Substituting (54) into (52) gives quantity demand:

$$\begin{aligned} q_{sbg} &= \frac{E_b}{P_b} \zeta_{sbg} \left(\frac{p_{sbg}}{P_b} \right)^{-\sigma} \\ &= E_b \zeta_{sbg} p_{sbg}^{-\sigma} P_b^{\sigma-1}. \end{aligned} \quad (55)$$

Multiplying quantity demand by the product price gives expenditure on supplier–product variety (s, g) , which is also the supplier’s product-level revenue from buyer b :

$$\begin{aligned} r_{sbg} &\equiv p_{sbg} q_{sbg} = E_b \zeta_{sbg} p_{sbg}^{1-\sigma} P_b^{\sigma-1} \\ &= E_b \zeta_{sbg} \left(\frac{p_{sbg}}{P_b} \right)^{1-\sigma}. \end{aligned} \quad (56)$$

Equation (56) is the product-level revenue equation used in the main text. The corresponding share of buyer b ’s expenditure allocated to product (s, g) is

$$\frac{r_{sbg}}{E_b} = \zeta_{sbg} \left(\frac{p_{sbg}}{P_b} \right)^{1-\sigma}. \quad (57)$$

Summing expenditure shares across all varieties available to buyer b gives

$$\begin{aligned} \sum_{s \in \mathcal{S}_b} \sum_{g \in \mathcal{G}_{sb}} \frac{r_{sbg}}{E_b} &= P_b^{\sigma-1} \sum_{s \in \mathcal{S}_b} \sum_{g \in \mathcal{G}_{sb}} \zeta_{sbg} p_{sbg}^{1-\sigma} \\ &= P_b^{\sigma-1} P_b^{1-\sigma} \\ &= 1, \end{aligned} \quad (58)$$

where the second equality uses (51). Thus total expenditure is exhausted across the active supplier–product varieties.

I.3 Pricing and the supplier-side revenue equation

For completeness, this appendix also shows how (56) maps into the reduced-form revenue equation used in the supplier’s problem.

Product g of supplier s has productivity

$$\phi_{s,g} = \frac{\phi_s}{h(g)}, \quad (59)$$

so its marginal cost is

$$c_{sbg} = \frac{w}{\phi_{s,g}} = \frac{w h(g)}{\phi_s}. \quad (60)$$

Under monopolistic competition with elasticity $\sigma > 1$, optimal pricing is a constant markup over marginal cost:

$$p_{sbg} = \tilde{\sigma} c_{sbg}, \quad \tilde{\sigma} \equiv \frac{\sigma}{\sigma - 1}. \quad (61)$$

Substituting (60) and (61) into the revenue equation gives

$$\begin{aligned} r_{sbg} &= E_b \zeta_{sbg} \left(\frac{\tilde{\sigma} w h(g)}{\phi_s P_b} \right)^{1-\sigma} \\ &= E_b \left(\frac{P_b}{\tilde{\sigma} w} \right)^{\sigma-1} \phi_s^{\sigma-1} h(g)^{-(\sigma-1)} \zeta_{sbg}. \end{aligned} \quad (62)$$

Defining

$$D_b \equiv E_b \left(\frac{P_b}{\tilde{\sigma} w} \right)^{\sigma-1}, \quad (63)$$

we obtain

$$r_{sbg} = D_b \phi_s^{\sigma-1} h(g)^{-(\sigma-1)} \zeta_{sbg}. \quad (64)$$

Since variable operating profits under CES pricing are a constant share of revenue,

$$\pi_{sbg} = \frac{1}{\sigma} r_{sbg} = \frac{D_b}{\sigma} \phi_s^{\sigma-1} h(g)^{-(\sigma-1)} \zeta_{sbg}. \quad (65)$$

These are the revenue and operating-profit equations used in the supplier's product-activation and link-formation decisions.

I.4 Active sets and the partial-equilibrium interpretation

The active sets in the preceding derivation are endogenous from the supplier's perspective but are taken as given when buyer b allocates expenditure. The supplier first chooses firm-level product scope G_s and evaluates buyer relationships. Conditional on an active relationship, product–buyer match realizations determine the set $\mathcal{G}_{sb} \subseteq \{1, \dots, G_s\}$ of products that are activated for buyer b . Products outside this set have $q_{sbg} = r_{sbg} = 0$ and do not enter the buyer's CES price index.

The same match shifter ζ_{sbg} therefore affects both the profitability of activating a product for a particular buyer and, conditional on activation, the product's realized expenditure share. This is why the benchmark matching model uses the same source of product–buyer heterogeneity for selection into trade and for sales conditional on trade.

From the perspective of an individual supplier, the buyer-side demand environment summarized by D_b is taken as given. In the underlying atomless-supplier environment, the realization of supplier s 's own product–buyer match ζ_{sbg} has a negligible effect on buyer b 's price index P_b , and therefore does not feed back into D_b . The supplier can consequently evaluate expected product-level profitability conditional on (ϕ_s, D_b) while holding the buyer demand shifter fixed.

With the normalization $\mathbb{E}[\zeta_{sbg}] = 1$, expected operating profits prior to conditioning on product activation are therefore

$$\mathbb{E}[\pi_{sbg}(\phi_s, \zeta_{sbg}) \mid \phi_s, D_b] = \frac{D_b}{\sigma} \phi_s^{\sigma-1} h(g)^{-(\sigma-1)}. \quad (66)$$

This expectation is used when evaluating the profitability of products and buyer relationships in the supplier's problem.

The theoretical demand system satisfies buyer-level expenditure exhaustion by construction. In the quantitative implementation, however, the buyer-side profitability shifter D_b is treated as a reduced-form primitive and is not recomputed from the realized finite set of simulated suppliers. The finite supplier population is therefore interpreted as an approximation to an underlying atomless supplier environment. Accordingly, the quantitative simulation need not impose the buyer-level accounting identity

$$\sum_{s \in \mathcal{S}_b} \sum_{g \in \mathcal{G}_{sb}} r_{sbg} = E_b \quad (67)$$

exactly for every simulated buyer. An individual supplier instead takes D_b and the buyer-side demand environment as given when solving its product-scope, product-activation, and link-formation problems.

I.5 Regularity conditions

Supplier productivity follows a Pareto distribution,

$$\phi_s \sim \text{Pareto}(\theta_\phi, \phi_{\min}), \quad \phi_s \geq \phi_{\min} > 0. \quad (68)$$

For a Pareto random variable,

$$\mathbb{E}[\phi_s^k] < \infty \quad \Longleftrightarrow \quad \theta_\phi > k. \quad (69)$$

In the matching model, the product–buyer match shock is also Pareto distributed,

$$\zeta_{sbq} \sim \text{Pareto}(\theta_\zeta, \zeta_{\min}), \quad \theta_\zeta > 1, \quad (70)$$

with $\zeta_{\min}$ chosen so that $\mathbb{E}[\zeta_{sbq}] = 1$.

As shown in the characterization of expected product surplus, the relevant interior-Pareto term scales with supplier productivity as

$$\phi_s^{\theta_\zeta(\sigma-1)}. \quad (71)$$

Existence of the corresponding population moment therefore requires

$$\theta_\phi > \theta_\zeta(\sigma - 1). \quad (72)$$

This condition is stronger than the standard requirement $\theta_\phi > \sigma - 1$ that applies in the absence of the additional selection induced by product–buyer matching.

Condition (72) is sufficient but not necessary when developed scope is bounded. For any finite $G_{\max}$, sufficiently productive suppliers have every product in the region where activation is certain, where expected surplus is linear in $\phi_s^{\sigma-1}$. The relevant population moments then exist whenever $\theta_\phi > \sigma - 1$, which the calibrated $\theta_\phi = 4.5$ satisfies. Although finite simulations mechanically generate finite sample moments when this restriction is violated, the corresponding population moments would not exist, and simulated outcomes could become increasingly sensitive to extreme productivity realizations as the simulated supplier population grows.

I.6 Summary of model primitives

Table 12: Model primitives

Class	Object	Role
Supplier	ϕ_s	Supplier productivity
Supplier	c_s	Supplier-level capability
Buyer	D_b	Buyer profitability shifter
Buyer	$\bar{f}_b^L$	Average link-formation difficulty
Buyer	$\bar{f}_b^A$	Average product-activation difficulty
Supplier–buyer	η_{sb}	Pair-specific cost heterogeneity
Product–buyer	ζ_{sbq}	Match quality and realized appeal

Notes: The matching model uses ζ_{sbq} both to determine product–buyer selection and to generate realized sales dispersion conditional on trade. The deterministic common-ladder benchmark, as in [Arkolakis et al. \(2021\)](#), instead imposes a common product ranking across buyers and therefore has no product–buyer match shock. We allow a separate ex-post appeal shock ξ_{sbq} , normalized so that $\mathbb{E}[\xi_{sbq}] = 1$, to generate residual product-level sales dispersion in this benchmark.

The primitive sources of heterogeneity satisfy

$$\phi_s \perp D_b \perp \eta_{sb} \perp c_s \perp \zeta_{sb}, \quad (73)$$

subject to the buyer-level heterogeneity encoded in $(D_b, \bar{f}_b^L, \bar{f}_b^A)$.

Although the underlying shocks in (73) are mutually independent, the realized link-formation and product-activation costs are not independent because both contain the common components η_{sb} and c_s . This shared structure allows supplier–buyer pairs that are easier to establish to also be easier to operate along the product margin.

J Product–Buyer Matching and Expected Surplus

Fix supplier scope G_s and define

$$A_{sb} \equiv \frac{D_b}{\sigma} \phi_s^{\sigma-1}.$$

Product g is activated for buyer b whenever

$$\zeta_{sb} \geq z_{sb}(g) \equiv \frac{f_{sb,0}^A}{A_{sb}} g^\kappa, \quad \kappa \equiv \mu + \alpha(\sigma - 1).$$

Given the Pareto distribution of ζ , the activation probability is

$$p_{sb}(g) \equiv \Pr(\zeta \geq z_{sb}(g)) = \min \left\{ 1, \left(\frac{\zeta_{\min}}{z_{sb}(g)} \right)^{\theta_\zeta} \right\}. \quad (74)$$

Expected net surplus from product g is

$$\Psi_{sb}(g) = \mathbb{E} \left[(A_{sb} g^{-\alpha(\sigma-1)} \zeta - f_{sb,0}^A g^\mu)^+ \right]. \quad (75)$$

Evaluating the Pareto integral gives

$$\Psi_{sb}(g) = \begin{cases} A_{sb} g^{-\alpha(\sigma-1)} - f_{sb,0}^A g^\mu, & z_{sb}(g) \leq \zeta_{\min}, \\ \frac{f_{sb,0}^A g^\mu p_{sb}(g)}{\theta_\zeta - 1}, & z_{sb}(g) > \zeta_{\min}. \end{cases} \quad (76)$$

For products in the interior Pareto region, $z_{sb}(g) > \zeta_{\min}$, this becomes

$$\Psi_{sb}(g) = \psi_{sb} g^{\mu - \kappa \theta_\zeta}, \quad (77)$$

where

$$\psi_{sb} = \frac{\zeta_{\min}^{\theta_\zeta}}{\theta_\zeta - 1} A_{sb}^{\theta_\zeta} (f_{sb,0}^A)^{1-\theta_\zeta}. \quad (78)$$

K Numerical Solution

The quantitative model is solved using the exact discrete supplier problem. For each supplier s and candidate scope $G \in \{0, 1, \dots, |\Omega|\}$, we first compute

$$C_{sb}(G) = \sum_{g=1}^G \Psi_{sb}(g)$$

for every potential buyer. Expected net value from buyer b is

$$V_{sb}(G) = [C_{sb}(G) - f_{sb}^L]^+.$$

Supplier profits at candidate scope G are therefore

$$\Pi_s(G) = \sum_{b \in \mathcal{B}} V_{sb}(G) - F^P(G). \quad (79)$$

Optimal product scope is

$$G_s^* \in \arg \max_G \Pi_s(G). \quad (80)$$

Given G_s^* , the link-formation rule determines $\mathcal{B}_s^*$. Conditional on an active link, product–buyer match shocks are drawn and determine the realized set $\mathcal{G}_{sb}^*$ and corresponding sales.

Existence follows from the finite product-scope choice set. The closed-form expression in Section 4.5 is used for economic interpretation; all quantitative results use the exact discrete solution.

K.1 Distributions and simulation design

The quantitative model draws all primitives from parametric distributions. Supplier productivity is Pareto,

$$\phi_s = \phi_{\min} (1 - u_s)^{-1/\theta_\phi}, \quad u_s \sim U(0, 1).$$

Supplier relationship-cost shifters, buyer demand shifters, and buyer-level cost scales are lognormal, each normalized to have the stated mean:

$$\ln c_s \sim \mathcal{N}\left(-\frac{\omega_c^2}{2}, \omega_c^2\right), \quad \ln D_b \sim \mathcal{N}\left(-\frac{\omega_B^2}{2}, \omega_B^2\right),$$

$$\bar{f}_b^L = \bar{f}^L \epsilon_b^L, \quad \bar{f}_b^A = \bar{f}_0^A \epsilon_b^A, \quad \ln \epsilon_b^j \sim \mathcal{N}\left(-\frac{\omega_{fj}^2}{2}, \omega_{fj}^2\right), \quad j \in \{L, A\},$$

so that $\mathbb{E}[c_s] = \mathbb{E}[D_b] = \mathbb{E}[\epsilon_b^L] = \mathbb{E}[\epsilon_b^A] = 1$ and $\bar{f}^L$ and $\bar{f}_0^A$ are the mean link and activation cost scales. The pair-specific cost shock is lognormal,

$$\ln \eta_{sb} \sim \mathcal{N}\left(-\frac{\omega_\eta^2}{2}, \omega_\eta^2\right),$$

drawn independently for every supplier–buyer pair. The same draw enters both costs of the pair, $f_{sb}^L = \bar{f}_b^L \eta_{sb} c_s$ and $f_{sb,0}^A = \bar{f}_b^A \eta_{sb} c_s$. Product–buyer match qualities are Pareto,

$$\zeta_{sbg} = \zeta_{\min} (1 - v_{sbg})^{-1/\theta_\zeta}, \quad v_{sbg} \sim U(0, 1), \quad \zeta_{\min} = \frac{\theta_\zeta - 1}{\theta_\zeta},$$

drawn independently for each active link and each developed product after the supplier’s scope and link decisions. They therefore affect realized selection and sales but not the ex-ante decisions, which depend only on the expected surplus in equation (22). All draws are mutually independent.

Buyer pool and simulation. The simulated economy consists of S suppliers and a common pool of B potential buyers. Buyer characteristics $(D_b, \bar{f}_b^L, \bar{f}_b^A)$ are drawn once and are the same for all suppliers, so buyers differ in how attractive they are to every supplier, while η_{sb} and ζ_{sbg} generate pair- and product-specific heterogeneity. Realized match qualities are then drawn on active links, and all moments are computed from realized trade. We use $S = 2,000$, $B = 8,000$, $G_{\max} = 300$, and a fixed random seed, so that the simulated moments are smooth in the parameters and the SMM objective is reproducible.

L Additional diagnostics for buyer–product co-movement

This appendix provides additional diagnostics for the relationship between buyer breadth and product scope in the calibrated matching model. The main text shows that product scope and buyer breadth remain positively related after conditioning on supplier productivity. Here we report two complementary exercises.

First, Table 13 decomposes the variation in log product scope into the portion associated with supplier productivity and the additional portion associated with buyer breadth. Supplier productivity alone explains only a fifth of the cross-sectional variation in product scope (0.202). Conditional on productivity, buyer breadth accounts for most of the remaining variation: adding $\log n_s$ raises the explained share of $\text{Var}(\log \hat{G}_s)$ from 0.202 to 0.839, about 80 percent of the residual variation. Supplier relationship-cost heterogeneity plays a similar role: adding $\log c_s$ instead raises the explained share to 0.822.

Second, Table 14 reports the correlation between product scope and buyer breadth separately

Table 13: Variance decomposition of $\log \widehat{G}_s$ (calibrated model)

	Share of $\text{Var}(\log \widehat{G}_s)$
Explained by $\log \phi_s$ alone	0.202
Explained by $\log \phi_s + \log n_s$	0.839
Explained by $\log \phi_s + \log c_s$	0.822
Within- ϕ residual share	0.798
Additional share explained by $\log n_s$ as % of within- ϕ residual	0.637 79.8%

within productivity quintiles. The relationship remains strongly positive within every quintile, both in levels and in logs. This confirms that the positive co-movement documented in the main text is not driven solely by sorting of more productive suppliers into both larger product portfolios and larger buyer bases.

Table 14: Within- ϕ -quintile correlations between $\widehat{G}_s$ and n_s (calibrated model)

Quintile	$\rho(\widehat{G}_s, n_s)$	$\rho(\log \widehat{G}_s, \log n_s)$
Q1	0.810	0.867
Q2	0.832	0.897
Q3	0.867	0.877
Q4	0.846	0.899
Q5	0.870	0.934
Pooled	0.877	0.915

L.1 How broadly is the return to product adoption spread across buyers?

The match-adjusted buyer-base aggregator in equation (34) places heterogeneous weights on active buyers. Let

$$w_{sb} \equiv \frac{D_b^{\theta_\zeta} (f_{sb,0}^A)^{1-\theta_\zeta}}{\Lambda_s^M}, \quad b \in \mathcal{B}_s,$$

denote buyer b 's share of the aggregator, so that $\sum_{b \in \mathcal{B}_s} w_{sb} = 1$. To summarize how concentrated these contributions are, we use the effective number of buyers, the inverse Herfindahl index of these weights:

$$N_s^{\text{eff}} \equiv \left(\sum_{b \in \mathcal{B}_s} w_{sb}^2 \right)^{-1}. \quad (81)$$

The effective number equals N_s when all active buyers contribute equally and approaches one when a single buyer dominates.

Table 15 shows that the return to product adoption is concentrated in a few high-value relationships. The effective number of buyers rises only from about 2 to about 4 as the active buyer count rises from a handful to several hundred, and the largest buyer accounts for about 44 per cent of the aggregator even among suppliers with more than 256 buyers. Because an additional marginal buyer adds little to the return to developing a product, this concentration is consistent with the small amortization share in calibration B.

Table 15: Effective number of buyers in the product-adoption aggregator

Active buyers N_s	Median N_s	Median N_s^{eff}	N_s^{eff}/N_s	Top-1 share
1	1	1.00	1.000	1.000
2–3	2	1.73	0.704	0.703
4–7	5	2.30	0.445	0.571
8–15	10	2.87	0.271	0.506
16–31	23	3.28	0.149	0.460
32–63	43	3.66	0.089	0.429
64–127	90	4.12	0.044	0.430
128–255	174	3.89	0.023	0.440
256+	339	3.94	0.009	0.449
All firms	10	2.60	0.276	0.537
Link-weighted	115	3.75	0.034	0.441

Notes: Active (ex-ante) links. Weights $w_{sb} \propto D_b^{\theta_\zeta} (f_{sb,0}^A)^{1-\theta_\zeta}$, normalized to sum to one within supplier; $N_s^{\text{eff}} = (\sum_b w_{sb}^2)^{-1}$. Columns report medians within each group. The link-weighted row weights suppliers by N_s .

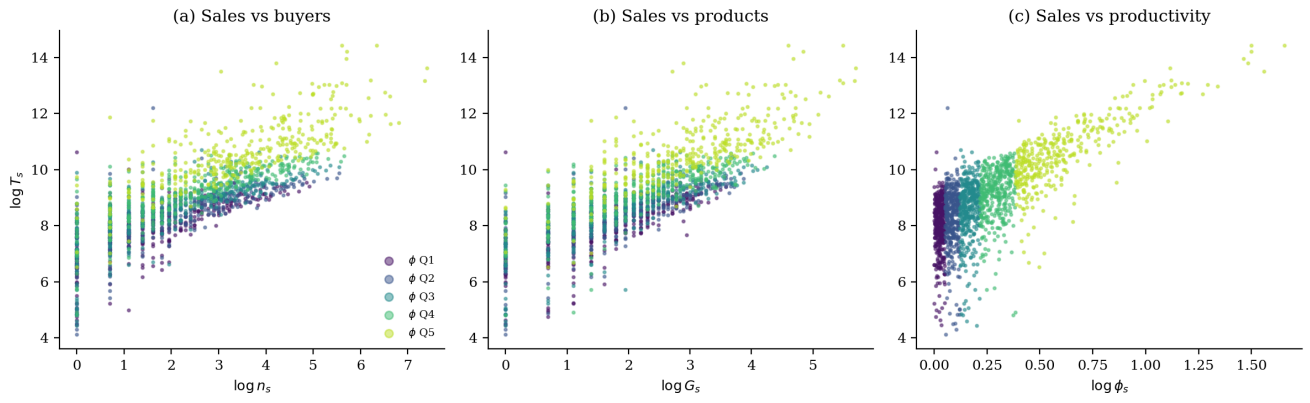
L.2 Sales across margins

Figure 23 relates simulated supplier sales to buyer breadth, product scope, and supplier productivity. Sales increase with all three objects, but the relationships differ in tightness because each captures a different component of firm heterogeneity.

Product scope is particularly strongly associated with sales because it is itself an endogenous outcome of supplier productivity, buyer profitability, and the supplier’s buyer network. Buyer count also co-moves strongly with sales, while substantial dispersion remains within buyer-count cells because suppliers differ in productivity and buyer composition. The sales–productivity relationship is less tight because firms with similar productivity can operate with substantially different buyer bases and product scopes.

These patterns provide an additional internal-consistency check on the calibrated model and mirror the empirical fact that larger suppliers operate with broader buyer and product scopes.

Figure 23: Total sales across margins (calibrated model)



Notes: Log total sales $\log T_s$ against buyer count $\log N_s$ (left), product scope $\log \hat{G}_s$ (middle), and productivity $\log \phi_s$ (right), colored by productivity quintile.

M Deterministic-Ladder Benchmark (M1)

This appendix presents the deterministic-ladder benchmark, M1, used in the quantitative comparisons in Section 5. M1 preserves the same basic supplier productivity, buyer heterogeneity, fixed-cost structure, and firm-level product-adoption decision as the matching model in the main text. The key difference is the mapping from the supplier’s developed portfolio into products sold to individual buyers.

In M1, products follow a common deterministic profitability ranking across buyers. Conditional on serving buyer b , supplier s sells a prefix of its firm-level product ladder,

$$\{1, \dots, G_{sb}\}, \quad G_{sb} \leq G_s.$$

Thus, buyers can differ in how far they extend down the supplier’s product ranking, but they cannot purchase non-nested subsets of products. There is no product–buyer match shock ζ_{sb} . Instead, realized product-level sales contain an ex-post appeal shock ξ_{sb} with $\mathbb{E}[\xi_{sb}] = 1$. Because ξ_{sb} is realized after the scope and link decisions, it generates residual sales dispersion but does not affect which products are activated or which links are formed.

M.1 Stage 2: Deterministic link-level product scope

Fix supplier s ’s firm-level product scope G_s . As in the baseline model, define the buyer–supplier profitability shifter

$$A_{sb}(\phi_s) \equiv \frac{D_b}{\sigma} \phi_s^{\sigma-1}. \quad (82)$$

Products are ranked according to

$$h(g) = g^\alpha,$$

and the buyer-specific activation cost is

$$f_{sb}^A(g) = f_{sb,0}^A g^\mu, \quad f_{sb,0}^A = \bar{f}_b^A \eta_{sb} c_s.$$

Because the ex-post appeal shock has mean one and does not affect decisions, the expected operating profit generated by rank g before paying its buyer-specific activation cost is

$$A_{sb}(\phi_s) g^{-\alpha(\sigma-1)}. \quad (83)$$

The incremental gross payoff from activating rank g on link (s, b) is therefore

$$\Delta u_{sb}(g; \phi_s) = A_{sb}(\phi_s) g^{-\alpha(\sigma-1)} - f_{sb,0}^A g^\mu. \quad (84)$$

Define, as in the main text,

$$\kappa \equiv \mu + \alpha(\sigma - 1). \quad (85)$$

When $\kappa > 0$, the ratio of activation cost to operating profit rises with rank, so the incremental payoff in (84) crosses zero at most once. The unconstrained buyer-specific product cutoff is therefore

$$\begin{aligned} \tilde{G}_{sb}(\phi_s) &= \max \{g : A_{sb}(\phi_s)g^{-\alpha(\sigma-1)} \geq f_{sb,0}^A g^\mu\} \\ &= \left\lfloor \left(\frac{A_{sb}(\phi_s)}{f_{sb,0}^A} \right)^{1/\kappa} \right\rfloor \\ &= \left\lfloor \left(\frac{D_b \phi_s^{\sigma-1}}{\sigma \bar{f}_b^A \eta_{sb} c_s} \right)^{1/\kappa} \right\rfloor. \end{aligned} \quad (86)$$

Conditional on the firm's developed scope G_s , the desired product depth of buyer b is

$$G_{sb}^0(G_s; \phi_s) = \min \{G_s, \tilde{G}_{sb}(\phi_s)\}. \quad (87)$$

Hence every active buyer receives a deterministic prefix of the supplier's product hierarchy.

Before paying the link fixed cost, the gross value of the relationship is

$$U_{sb}^{\text{M1}}(G_s; \phi_s) = \sum_{g=1}^{G_{sb}^0(G_s; \phi_s)} [A_{sb}(\phi_s)g^{-\alpha(\sigma-1)} - f_{sb,0}^A g^\mu]. \quad (88)$$

Define net link surplus as

$$S_{sb}^{\text{M1}}(G_s; \phi_s) \equiv U_{sb}^{\text{M1}}(G_s; \phi_s) - f_{sb}^L. \quad (89)$$

The supplier activates the relationship whenever

$$I_{sb}^{\text{M1}}(G_s; \phi_s) = \mathbf{1} \{S_{sb}^{\text{M1}}(G_s; \phi_s) \geq 0\}. \quad (90)$$

Thus the realized link-level scope is

$$G_{sb} = I_{sb}^{\text{M1}}(G_s; \phi_s) G_{sb}^0(G_s; \phi_s). \quad (91)$$

Conditional on trade, realized product-level sales are

$$r_{sbg}^{\text{M1}} = D_b \phi_s^{\sigma-1} g^{-\alpha(\sigma-1)} \xi_{sbg}, \quad g \leq G_{sb}, \quad (92)$$

where $\mathbb{E}[\xi_{sbg}] = 1$. Unlike ζ_{sbg} in M2, ξ_{sbg} affects realized sales only and plays no role in selection into product-buyer trade.

M.2 Stage 1: Firm-level product scope

Anticipating the Stage 2 policy, supplier s chooses its firm-level scope G_s to maximize

$$\Pi_s^{\text{M1}}(G_s; \phi_s) = \sum_{b \in \mathcal{B}} [S_{sb}^{\text{M1}}(G_s; \phi_s)]^+ - F^{\text{P}}(G_s), \quad (93)$$

where

$$F^{\text{P}}(G_s) = \sum_{g=1}^{G_s} f^{\text{P}}(g), \quad f^{\text{P}}(g) = f_0^{\text{P}} g^\delta.$$

The exact optimal scope is therefore

$$G_s^{\star, \text{M1}} \in \arg \max_{G_s \in \{0, 1, \dots, |\Omega|\}} \Pi_s^{\text{M1}}(G_s; \phi_s). \quad (94)$$

The marginal value of increasing firm scope from $g-1$ to g is

$$\Delta \Pi_s^{\text{M1}}(g) = \sum_{b \in \mathcal{B}} \left([S_{sb}^{\text{M1}}(g)]^+ - [S_{sb}^{\text{M1}}(g-1)]^+ \right) - f^{\text{P}}(g). \quad (95)$$

This marginal value can again be separated into intensive and extensive components. Define

$$\mathcal{B}_s^{\text{bind}}(g) \equiv \left\{ b \in \mathcal{B}_s(g-1) : \tilde{G}_{sb} \geq g \right\}, \quad (96)$$

the previously active buyers for which rank g remains below the buyer-specific cutoff, and

$$\mathcal{B}_s^{\text{new}}(g) \equiv \mathcal{B}_s(g) \setminus \mathcal{B}_s(g-1), \quad (97)$$

the relationships activated when the supplier develops product g . Then

$$\begin{aligned} \Delta \Pi_s^{\text{M1}}(g) = & \underbrace{\sum_{b \in \mathcal{B}_s^{\text{bind}}(g)} [A_{sb} g^{-\alpha(\sigma-1)} - f_{sb}^A(g)]}_{\text{intensive: deepen existing prefixes}} \\ & + \underbrace{\sum_{b \in \mathcal{B}_s^{\text{new}}(g)} S_{sb}^{\text{M1}}(g)}_{\text{extensive: newly activated links}} - f^{\text{P}}(g). \end{aligned} \quad (98)$$

The important difference from M2 is the first term. In M1, buyer b contributes to the marginal return from product g only if $\tilde{G}_{sb} \geq g$. Thus, as the supplier moves farther down the product ladder, buyers progressively leave the amortization base.

M.3 A scope ceiling in the deterministic ladder

The buyer-specific cutoffs imply an upper bound on firm-level product scope that is absent from the matching model.

[Scope ceiling in the deterministic-ladder model] Suppose $f^P(g) > 0$ for products above the current portfolio. Then optimal firm-level scope in M1 satisfies

$$G_s^{*,M1} \leq \max_{b \in \mathcal{B}} \tilde{G}_{sb} = \max_{b \in \mathcal{B}} \left[\left(\frac{D_b \phi_s^{\sigma-1}}{\sigma \bar{f}_b^A \eta_{sb} c_s} \right)^{1/\kappa} \right]. \quad (99)$$

Proof. Consider any $G_s > \max_b \tilde{G}_{sb}$. For every buyer, equation (87) implies

$$G_{sb}^0(G_s) = \tilde{G}_{sb}.$$

Increasing G_s further therefore changes neither the product set nor the gross value of any potential relationship. The first term of (93) is unchanged, while the firm continues to pay additional firm-level adoption costs. Hence such a value of G_s cannot be strictly optimal. $\square$

This result clarifies an important limitation of the deterministic ladder. Amortization can raise product scope only while the marginal product is below the cutoff of at least one buyer. Once G_s reaches the largest buyer-specific cutoff, additional buyers' payoffs cannot be summed to support a product beyond that ceiling. The ceiling is therefore determined by the buyer with the deepest feasible product range rather than by the sum of surplus contributions across buyers.

At the same time, this feature generates a mechanical relationship between buyer breadth and product scope. A supplier with more buyers draws more buyer-specific cutoffs, increasing the probability that at least one buyer has a high $\tilde{G}_{sb}$. Firm scope can therefore increase with buyer count through the order statistic

$$\max_{b \in \mathcal{B}_s} \tilde{G}_{sb},$$

even in the absence of a firm-level cost to amortize. This is why the buyer–product relationship in M1 survives when F^P is set to zero. The matching model has an analogous non-amortization channel: with buyer-specific matches, observed scope is the union of products selected by at least one buyer (equation (36)), which also grows with the number of buyers. The relocated-cost experiment of Section 7.1 measures the size of this channel directly.

M.4 Amortization in M1 and the activation-cost correction

M1 nevertheless contains a genuine amortization force below the scope ceiling. Holding the active and binding buyer sets locally fixed, the marginal condition for product g can be written as

$$\frac{\phi_s^{\sigma-1}}{\sigma} g^{-\alpha(\sigma-1)} \sum_{b \in \mathcal{B}_s^{\text{bind}}(g)} D_b \geq f_0^P g^\delta + c_s g^\mu \sum_{b \in \mathcal{B}_s^{\text{bind}}(g)} \bar{f}_b^A \eta_{sb}. \quad (100)$$

The first term on the right-hand side is the firm-level adoption cost, which is paid once. The second is different: activation costs are paid separately for every buyer to which the marginal product is supplied. Consequently, a larger buyer base raises both the return to adopting the product and the total activation cost associated with deploying it.

To make this correction explicit, define the activation-cost index at the marginal rank by

$$\Xi_s \equiv g^{\mu-\delta} \sum_{b \in \mathcal{B}_s^{\text{bind}}(g)} \bar{f}_b^A \eta_{sb}. \quad (101)$$

Then the local marginal condition can be expressed as

$$\frac{\phi_s^{\sigma-1}}{\sigma} \Lambda_s^{\text{bind}} g^{-\alpha(\sigma-1)} \geq (f_0^P + c_s \Xi_s) g^\delta, \quad \Lambda_s^{\text{bind}} \equiv \sum_{b \in \mathcal{B}_s^{\text{bind}}(g)} D_b. \quad (102)$$

Under the local proportional-buyer approximation—holding the composition of the relevant buyer set fixed while varying its size—the elasticity of scope with respect to buyer breadth is therefore attenuated relative to the formula that ignores buyer-specific activation costs:

$$\varepsilon_{G,N}^{\text{M1}} \simeq \underbrace{\frac{1}{\delta + \alpha(\sigma - 1)}}_{\text{clean amortization term}} \underbrace{\frac{f_0^P}{f_0^P + c_s \Xi_s}}_{\text{activation-cost correction}}. \quad (103)$$

The correction factor is strictly below one whenever buyer-specific activation costs are positive. The reason is economic: an additional buyer helps amortize the firm-level cost f_0^P , but it does not amortize the activation costs that must be paid again on each buyer relationship. Consequently, the expression $1/[\delta + \alpha(\sigma - 1)]$ by itself overstates the strength of the amortization channel in M1.

Equation (103) should be interpreted as a local comparative-static approximation, not as a global characterization of M1. The exact solution is given by (94) and is additionally constrained by the buyer-specific scope ceiling in Proposition M.3.

M.5 Marginal buyers and monotonicity

For any fixed firm-level scope G_s , buyers must be ordered by their net link surplus rather than by gross link value. Let

$$S_{sb}^{\text{M1}}(G_s; \phi_s) = U_{sb}^{\text{M1}}(G_s; \phi_s) - f_{sb}^L,$$

as in equation (89), and order potential buyers from highest to lowest according to this object. Let $b(n; G_s)$ denote the buyer in position n . The number of active buyers is then

$$N_s(G_s; \phi_s) = \max \{n : S_{s,b(n; G_s)}^{\text{M1}}(G_s; \phi_s) \geq 0\}. \quad (104)$$

When an interior marginal buyer exists,

$$\begin{aligned} U_{s,b(N_s; G_s)}^{\text{M1}}(G_s; \phi_s) &\geq f_{s,b(N_s; G_s)}^L, \\ U_{s,b(N_s+1; G_s)}^{\text{M1}}(G_s; \phi_s) &< f_{s,b(N_s+1; G_s)}^L. \end{aligned} \quad (105)$$

Hence the relevant ordering for link formation is by net surplus, not by U_{sb}^{M1} alone.

The model generates the “more products $\Rightarrow$ more buyers” direction weakly. Because all ranks up to $\tilde{G}_{sb}$ have nonnegative incremental operating surplus, $U_{sb}^{\text{M1}}(G_s)$ is weakly increasing in G_s . It follows that

$$\mathcal{B}_s^{\text{M1}}(G_s) \subseteq \mathcal{B}_s^{\text{M1}}(G_s + 1). \quad (106)$$

The reverse statement requires more care. For a product rank g , an additional buyer increases the gross marginal return to firm-level adoption only when that buyer itself supports rank g , that is, when

$$\tilde{G}_{sb} \geq g \iff A_{sb} g^{-\alpha(\sigma-1)} - f_{sb}^A(g) \geq 0. \quad (107)$$

Under this gross-payoff condition, adding the buyer weakly raises the left-hand side of the marginal adoption condition and therefore weakly raises the incentive to adopt product g . The condition cannot instead be imposed on arbitrary net link payoffs, because a potential buyer can have negative net surplus after the relationship fixed cost f_{sb}^L is paid and therefore never enter the amortization base.

This qualification is specific to the deterministic ladder. In M2, every active buyer contributes strictly positive expected surplus $\Psi_{sb}(g) > 0$ for every finite product threshold, so the corresponding buyer contribution to the amortization sum does not disappear at a buyer-specific product ceiling.

M.6 Comparison of the scope elasticities in M1 and M2

Ignoring the scope ceiling and buyer-specific activation-cost correction, the clean power-law approximation in M1 has scope exponent

$$\frac{1}{\delta + \alpha(\sigma - 1)}. \quad (108)$$

By contrast, the interior-Pareto approximation in M2 has exponent

$$\frac{1}{\delta + \kappa\theta_\zeta - \mu}. \quad (109)$$

Using $\kappa = \mu + \alpha(\sigma - 1)$, the denominator in M2 can be written as

$$\delta + \kappa\theta_\zeta - \mu = \delta + \alpha(\sigma - 1) + \kappa(\theta_\zeta - 1). \quad (110)$$

Since $\theta_\zeta > 1$ and $\kappa > 0$, the exponent on the buyer-base aggregator in the M2 closed form is smaller than the corresponding clean M1 exponent when the remaining primitives are held fixed. This does not imply that amortization is weaker in M2. The two expressions are elasticities with respect to different objects. M1 depends on the raw buyer-profitability sum for buyers whose cutoff still reaches the marginal rank, while M2 depends on the match-adjusted expected-surplus aggregator. Moreover, the clean M1 expression abstracts from both the activation-cost correction in equation (103) and the scope ceiling in Proposition M.3. The two exponents therefore cannot be interpreted as a direct structural ranking of the strength of complementarity across the two models.

M.7 Cost accounting and why M1 and M2 are calibrated separately

There is one additional accounting difference between the two specifications. In M1, if buyer b receives a prefix of length G_{sb} , the supplier pays activation costs on every product in that prefix:

$$F_{sb}^{A,M1}(G_{sb}) = \sum_{g=1}^{G_{sb}} f_{sb}^A(g). \quad (111)$$

In M2, by contrast, activation costs are paid only on products whose realized product–buyer match is sufficiently strong to make them active:

$$F_{sb}^{A,M2}(\mathcal{G}_{sb}) = \sum_{g \in \mathcal{G}_{sb}} f_{sb}^A(g). \quad (112)$$

At common finite parameter values, M2 therefore avoids activation costs for developed products that fail to match a particular buyer, while M1 pays the cost of every product in the deterministic prefix.

For this reason, parameter values should not be compared mechanically across M1 and M2. The two models are calibrated separately to their respective moment sets, and their quantitative implications are evaluated at their own calibrated parameter values. The purpose of M1 is not to provide a nested same-parameter counterfactual to M2, but to show what the common deterministic product ladder implies when it is given its own best quantitative fit.

Taken together, these differences explain why M1 is useful as a benchmark but not as the preferred model. The deterministic ladder preserves the economic force that firm-level product costs can be amortized across buyers, but it also imposes common prefix product sets, introduces a buyer-specific scope ceiling, and generates a mechanical deepest-buyer channel through which buyer breadth and product scope can co-move even when the firm-level adoption cost is removed. M2 replaces the deepest-buyer channel with buyer-specific matching, which the data require (coverage below one, non-nested product sets) and whose contribution relative to amortization can be measured.

N Extension 1: Nested CES and within-link cannibalization

The baseline matching model in Section 4 abstracts from within-link cannibalization: conditional on realized product–buyer matches, adding a product on link (s, b) does not reduce the profitability of the supplier’s other products sold to the same buyer. This separability follows from the single CES demand system, under which a supplier’s own products and the products of other suppliers are equally substitutable.

In this section, we relax that assumption by introducing a nested CES demand system, with an outer nest across suppliers, governed by elasticity ν , and an inner nest across products within each supplier, governed by elasticity ε . To isolate the cannibalization mechanism tractably, we write the within-link scope problem in the deterministic rank-based form of Appendix M: conditional on serving buyer b , supplier s chooses how far to extend down its product ranking on that relationship. The extension should therefore be interpreted as a robustness exercise for the separability assumption rather than as a full nested-CES version of the buyer-specific matching structure in Section 4. The firm-level adoption decision, buyer-link decision, and distinction between firm-level and link-level scope are preserved. The new ingredient is that products sold by the same supplier to the same buyer interact through the inner CES nest, generating within-link cannibalization whenever $\varepsilon \neq \nu$.

N.1 Demand: nested CES across suppliers and products

Let $\mathcal{S}_b \subseteq \mathcal{S}$ denote the endogenous set of suppliers active on buyer b , and let $\mathcal{G}_{sb} \subseteq 1, \dots, G_s$ denote the set of products that supplier $s \in \mathcal{S}_b$ activates on link (s, b) , with cardinality $G_{sb} = |\mathcal{G}_{sb}|$. In the rank-based formulation used in this extension, $\mathcal{G} * sb = 1, \dots, G_{sb}$. These objects are determined by the supplier's two-stage problem below and are taken as given when buyer b allocates expenditure. Products in the supplier's adopted portfolio that are not activated on the link, as well as suppliers that are not active on buyer b , do not enter the buyer's choice set.

Outer nest: across suppliers. Let X_{sb} denote the composite input that buyer b purchases from supplier $s \in \mathcal{S}_b$, the supplier's "variety" inside buyer b . The buyer's overall intermediate-input bundle is

$$M_b = \left[\sum_{s \in \mathcal{S}_b} X_{sb}^{\frac{\nu-1}{\nu}} \right]^{\frac{\nu}{\nu-1}}, \quad \nu > 1, \quad (113)$$

where ν governs substitution across suppliers: a higher ν means that buyer b can more easily substitute expenditure across suppliers.

Inner nest: across products within a supplier. For each active supplier $s \in \mathcal{S} * b$, the within-supplier product-line composite is

$$X_{sb} = \left[\sum_{g \in \mathcal{G}_{sb}} x_{sbg}^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{\varepsilon}{\varepsilon-1}}, \quad \varepsilon > 1, \quad (114)$$

where x_{sbg} is the quantity of product g that supplier s sells to buyer b , and ε governs substitution across products within a supplier's product line. When $\varepsilon = \nu$, the two nests collapse to the single-CES structure used in the separable benchmark, and the within-link cannibalization force described below disappears.

Price indices. Let p_{sbg} denote the delivered price of product g sold by supplier s to buyer b . The within-supplier, or inner-nest, price index is

$$P_{sb} = \left[\sum_{g \in \mathcal{G} * sb} p_{sbg}^{1-\varepsilon} \right]^{\frac{1}{1-\varepsilon}}, \quad (115)$$

which is dual to (114) and gives the minimum unit cost of one unit of the supplier composite X_{sb} given prices p_{sbg} for $g \in \mathcal{G}_{sb}$. The buyer-level, outer-nest price index across suppliers is

$$P_b = \left[\sum_{s \in \mathcal{S}_b} P_{sb}^{1-\nu} \right]^{\frac{1}{1-\nu}}, \quad (116)$$

which is dual to (113). Appendix I derives the corresponding expenditure-minimization problem and revenue equations for the single-nest CES system used in the baseline model.

Two-stage budgeting. Given total expenditure E_b , buyer b first allocates spending across suppliers and then across products within each supplier:

$$R_{sb} \equiv P_{sb} X_{sb} = E_b \left(\frac{P_{sb}}{P_b} \right)^{1-\nu}, \quad (\text{outer}) \quad (117)$$

$$r_{sbg} \equiv p_{sbg} x_{sbg} = R_{sb} \left(\frac{p_{sbg}}{P_{sb}} \right)^{1-\varepsilon}, \quad g \in \mathcal{G}_{sb}, \quad (\text{inner}) \quad (118)$$

Equations (117)–(118) make the two effects of adding a product explicit. Adding a product to $\mathcal{G}_{sb}$ lowers the inner-nest price index P_{sb} , which raises supplier s 's share of buyer b 's total expenditure through the outer nest. At the same time, expenditure within the supplier is reallocated across a larger set of products. This second force generates within-link cannibalization. Its importance is governed by the relative substitutability of products within the supplier and suppliers across the buyer's input bundle. Cannibalization is strongest when products within a supplier are closer substitutes than different suppliers' product lines, that is, when $\varepsilon > \nu$.

N.2 Technology, pricing, and link-level revenue

The marginal-cost schedule is the same as in the benchmark:

$$c_{sbg} = \frac{w h(g)}{\phi_s}, \quad (119)$$

where w is the input price, ϕ_s is supplier productivity, and $h(g)$ is the rank schedule, increasing in g . Optimal pricing under monopolistic competition with outer-nest elasticity ν gives a constant

markup over marginal cost:¹¹

$$p_{sbg} = \tilde{\nu} c_{sbg} = \tilde{\nu} \frac{w h(g)}{\phi_s}, \quad \tilde{\nu} \equiv \frac{\nu}{\nu - 1} > 1. \quad (120)$$

If supplier s activates exactly G products on link (s, b) , so that $G_{sb} = G$ and the active products have ranks $g = 1, \dots, G$, the within-supplier price index becomes

$$P_{sb}(G) = \tilde{\nu} \frac{w}{\phi_s} \left[\sum_{g=1}^G h(g)^{1-\varepsilon} \right]^{\frac{1}{1-\varepsilon}}. \quad (121)$$

Substituting (121) into (117) gives total link-level expenditure as a function of G :

$$R_{sb}(G) = E_b \left(\frac{P_b \phi_s}{\tilde{\nu} w} \right)^{\nu-1} \left[\sum_{g=1}^G h(g)^{1-\varepsilon} \right]^{\frac{\nu-1}{\varepsilon-1}}. \quad (122)$$

Product-level revenue within the link, using the inner-nest allocation in (118), is

$$r_{sbg}(g; G) = R_{sb}(G) \frac{h(g)^{1-\varepsilon}}{\sum_{j=1}^G h(j)^{1-\varepsilon}}, \quad g = 1, \dots, G. \quad (123)$$

Equation (123) makes product-level cannibalization explicit. When the supplier adds a product, the denominator $\sum_{j=1}^G h(j)^{1-\varepsilon}$ changes, reallocating revenue across all infra-marginal products on the link. Since $\nu > 1$ and $\varepsilon > 1$, the exponent $(\nu - 1)/(\varepsilon - 1)$ in (122) is positive, so expanding link-level scope raises total link revenue even though the revenue of individual infra-marginal products may fall.

Under the constant-markup rule, total variable profits are a constant fraction of link revenue:

$$\sum_{g=1}^G \pi_{sb}^V(g; G) = \frac{1}{\nu} \sum_{g=1}^G r_{sbg}(g; G) = \frac{1}{\nu} R_{sb}(G). \quad (124)$$

This is the key analytical simplification. Although product-level revenues are jointly determined through within-link cannibalization, total variable profits on the relationship depend on G only through the scalar $R_{sb}(G)$. The buyer-level problem therefore remains a one-dimensional choice over link-level scope.

¹¹We maintain the constant-markup rule throughout. This is an analytical convenience. Under nested CES with $\varepsilon \neq \nu$, fully strategic multiproduct pricing would generally generate markups that depend on the inner-nest elasticity; see the discussion of multiproduct pricing in [Arkolakis et al. \(2021\)](#) and [Hottman et al. \(2016\)](#). The constant-markup approximation keeps variable profits proportional to revenues and allows us to isolate the scope-cannibalization mechanism on the demand side. It can be interpreted as a benchmark in which suppliers price product by product while taking the supplier-level price index as given.

N.3 The supplier's two-stage problem under nested CES

We preserve the two-stage logic of the model. In Stage 1, supplier s chooses firm-level adopted scope G_s and pays the firm-level product-development cost $F^P(G_s)$. In Stage 2, for each potential buyer b , the supplier decides whether to activate the relationship, paying f_{sb}^L , and how many of its developed products to activate on that link, paying the corresponding product-activation costs. Relative to the deterministic-ladder problem in Appendix M, the key change is that the within-link revenue function $R_{sb}(G)$ now incorporates cannibalization through the inner CES nest.

Stage 2: link-level value function. Given firm-level scope G_s , the maximum operating value that supplier s can obtain from buyer b is

$$V_{sb}(G_s; \phi_s) = \max_{G \in \{0, 1, \dots, G_s\}} \left\{ \frac{1}{\nu} R_{sb}(G) - \mathbf{1}\{G > 0\} f_{sb}^L - F_{sb}^A(G) \right\}, \quad (125)$$

or, substituting (122),

$$V_{sb}(G_s; \phi_s) = \max_{G \in \{0, 1, \dots, G_s\}} \left\{ \frac{E_b}{\nu} \left(\frac{P_b \phi_s}{\tilde{\nu} w} \right)^{\nu-1} \left[\sum_{g=1}^G h(g)^{1-\varepsilon} \right]^{\frac{\nu-1}{\varepsilon-1}} - \mathbf{1}\{G > 0\} f_{sb}^L - F_{sb}^A(G) \right\}. \quad (126)$$

The first term is total variable profit from activating G products on the link, the second is the link fixed cost, paid only when the relationship is active, and the third is the cumulative product-activation cost. Optimal link-level scope and link activation are therefore characterized jointly by

$$G_{sb}^*(G_s; \phi_s) \in \arg \max_{G \in \{0, 1, \dots, G_s\}} \left\{ \frac{1}{\nu} R_{sb}(G) - \mathbf{1}\{G > 0\} f_{sb}^L - F_{sb}^A(G) \right\}, \quad (127)$$

with the relationship activated if and only if $G_{sb}^*(G_s; \phi_s) \geq 1$. The corresponding active buyer set at firm-level scope G_s is

$$\mathcal{B}_s(G_s) = \{b \in \mathcal{B} : G_{sb}^*(G_s; \phi_s) \geq 1\}. \quad (128)$$

Two features distinguish (127) from the deterministic-ladder Stage-2 problem in Appendix M.1. First, the gain from adding another product to the link is no longer additive rank by rank, because the inner-nest term

$$\left[\sum_j h(j)^{1-\varepsilon} \right]^{(\nu-1)/(\varepsilon-1)}$$

depends on the entire set of active products. Second, adding a product changes not only the revenue generated by that product but also the revenues of infra-marginal products through the cannibalization channel. The closed-form rank cutoff $\tilde{G}_{sb}(\phi_s)$ of the deterministic-ladder benchmark therefore no longer applies. Instead, the link-level problem must be solved as a discrete

choice over $G \in 0, 1, \dots, G_s$. Different buyers can optimally choose different scopes because E_b , f_{sb}^L , and $F_{sb}^A(\cdot)$ differ across relationships.

Stage 1: firm-level scope. Substituting the Stage-2 value functions into the supplier's outer problem gives

$$\Pi_s(G_s; \phi_s) = \sum_{b \in \mathcal{B}} V_{sb}(G_s; \phi_s) - F^P(G_s), \quad (129)$$

and optimal firm-level scope satisfies

$$G_s^*(\phi_s) \in \arg \max_{G_s \in 0, 1, \dots, |\Omega|} \Pi_s(G_s; \phi_s). \quad (130)$$

The outer logic is the same as in the baseline model of Section 4 and the deterministic-ladder benchmark of Appendix M.2. For each candidate G_s , the supplier aggregates the value of all potential buyer relationships and subtracts the firm-level product-development cost $F^P(G_s)$. The amortization mechanism is therefore preserved: the firm-level development cost is paid once, whereas the return to a developed product can accrue across multiple buyer relationships. A larger and more valuable buyer base consequently raises the return to expanding firm-level scope.

Joint determination and solution. Firm-level product scope and the active buyer set remain jointly determined. For any candidate G_s , equation (127) determines the optimal link-level decision for every potential buyer and therefore the active buyer set $\mathcal{B}_s(G_s)$. The supplier then evaluates the resulting value in (129) and chooses the maximizing G_s . Because the product-scope choice set is finite, the problem can be solved directly by evaluating Stage 2 for each candidate firm-level scope, in the same spirit as the exact discrete solution described in Section 4.4. The nested-CES extension therefore changes the within-link revenue calculation but does not create an additional equilibrium-existence problem.

Comparison with the deterministic-ladder benchmark. Conditional on G_s , each buyer-level problem in (126) remains a scalar discrete choice. What is lost relative to Appendix M is the simple closed-form rank-cutoff characterization of link-level scope. When $\varepsilon = \nu$, the inner aggregator in (122) reduces to

$$\sum_{g=1}^G h(g)^{-(\nu-1)},$$

and the additive deterministic-ladder structure is recovered. The nested-CES specification therefore generalizes the rank-based benchmark by introducing within-link cannibalization whenever $\varepsilon \neq \nu$, with $\varepsilon > \nu$ corresponding to the empirically relevant case in which a supplier's own products are closer substitutes for one another than different suppliers' product lines are.

N.4 Complementarity in both directions: buyers $\leftrightarrow$ products

The two-way interaction between buyer breadth and firm-level product scope established in Section 4.5 continues to operate under nested CES, although the within-link demand channel is modified by cannibalization.

More buyers $\Rightarrow$ more products. Holding supplier productivity ϕ_s fixed, expanding the active buyer set raises the return to firm-level product development because the cost $F^P(G_s)$ is paid once and can be amortized across more relationships. Formally, the marginal gain from raising firm-level scope from $g - 1$ to g is

$$\Delta\Pi_s(g; \phi_s) = \sum_{b \in \mathcal{B}} [V_{sb}(g; \phi_s) - V_{sb}(g - 1; \phi_s)] - f^P(g). \quad (131)$$

Each additional buyer can contribute a non-negative increment to the gross return from the marginal product, while $f^P(g)$ remains fixed at the firm level. The same amortization logic emphasized in Section 4.5 therefore implies that, holding productivity fixed, optimal firm-level scope is weakly increasing in the size and value of the buyer base.

More products $\Rightarrow$ more buyers. Holding ϕ_s fixed, increasing G_s weakly expands the feasible set in the link-level problem (127). The maximized relationship value $V_{sb}(G_s; \phi_s)$ is therefore weakly increasing in G_s . A larger firm-level portfolio allows a wider link-level product range on relationships for which the firm-level cap was binding. Through (117), a broader active product line lowers the inner-nest price index and can raise the supplier's share of the buyer's expenditure. Previously unprofitable buyer relationships can consequently become profitable as G_s rises. The active-set inclusion

$$\mathcal{B}_s(G_s) \subseteq \mathcal{B}_s(G_s + 1)$$

therefore continues to hold, paralleling the corresponding result in Section 4.5.

Joint determination. The two directions reinforce one another. More buyers raise the return to firm-level product development through amortization, while a broader product portfolio raises the value of potential buyer relationships. Nested CES changes how additional products affect revenues within each active link, but it does not eliminate the qualitative interaction between buyer breadth and product scope that motivates the model and the empirical co-movement documented in Fact 3.

O Extension 2: Dynamic buyer accumulation over the life cycle

The baseline model in Section 4 and the nested-CES robustness exercise in Appendix N are static within-period environments. They characterize how a given buyer base affects firm-level product scope and how product scope, in turn, affects the profitability of buyer relationships, but they do not explain how the buyer base itself is accumulated over time.

The life-cycle evidence in Appendix C.4 provides a natural motivation for adding this margin. Conditional on initial sales, younger suppliers rely relatively more on the joint addition of new buyers and new products, whereas the importance of selling established products to new buyers rises with supplier age. Because age is not a state variable in the benchmark model, these patterns are treated as complementary descriptive evidence rather than quantitative targets. Here we show how a simple buyer-accumulation block can generate a dynamic interpretation of those patterns.

O.1 Buyer-base state variable

The static matching model's approximate scope rule in Section 4.5 depends on the match-adjusted buyer-base aggregator

$$\Lambda_s^M \equiv \sum_{b \in \mathcal{B}_s} D_b^{\theta_\zeta} (f_{sb,0}^A)^{1-\theta_\zeta}.$$

To introduce gradual buyer accumulation, let

$$\Lambda_{s,a}^M \geq 0 \tag{132}$$

denote the corresponding buyer-base state of supplier s at age $a \in 0, 1, 2, \dots$. Define the contribution of buyer b to this aggregator as

$$\lambda_{sb} \equiv D_b^{\theta_\zeta} (f_{sb,0}^A)^{1-\theta_\zeta}.$$

Then

$$\Lambda_{s,a}^M = \sum_{b \in \mathcal{B}_{s,a}} \lambda_{sb} = N_{s,a} \bar{\lambda}_{s,a},$$

where $N_{s,a} = |\mathcal{B}_{s,a}|$ is the number of active buyers and

$$\bar{\lambda}_{s,a} \equiv \frac{1}{N_{s,a}} \sum_{b \in \mathcal{B}_{s,a}} \lambda_{sb}$$

is their average contribution to the adoption aggregator. The state therefore summarizes both buyer breadth and the average value of those relationships for product-development incentives.

In each period, supplier s chooses buyer-acquisition effort

$$n_{s,a}^B \geq 0, \quad (133)$$

which determines the expected number of new buyer relationships reached at age a . Let $m^B > 0$ denote acquisition or matching efficiency, let $\bar{\lambda}^{\text{new}} > 0$ denote the average contribution of a newly acquired relationship to the buyer-base aggregator, and let $\delta^B \in (0, 1)$ denote the per-period survival probability of an existing buyer relationship. The expected law of motion for the buyer-base state is

$$\mathbb{E} [\Lambda_{s,a+1}^M \mid \Lambda_{s,a}^M, n_{s,a}^B] = \delta^B \Lambda_{s,a}^M + m^B \bar{\lambda}^{\text{new}} n_{s,a}^B. \quad (134)$$

The buyer base next period therefore combines surviving current relationships with newly acquired buyers. We keep δ^B age-invariant in this stylized extension so that life-cycle differences in buyer-base growth arise through acquisition rather than mechanically through an age-varying separation rate. This is consistent with the role of the survival evidence in Appendix B.3, while keeping the dynamic block deliberately parsimonious.

O.2 Cost of buyer accumulation

Building the buyer base requires a convex networking or acquisition cost,

$$C_a^B(n_{s,a}^B) \equiv \xi_a^B \frac{f^B}{\gamma_B} (n_{s,a}^B)^{\gamma_B}, \quad \gamma_B > 1, \quad f^B > 0, \quad (135)$$

where $\xi_a^B \geq 1$ is an optional age-specific networking wedge. The convexity parameter $\gamma_B > 1$ makes rapid buyer acquisition increasingly costly and therefore gives firms an incentive to smooth the accumulation of customer relationships over time. If ξ_a^B is higher at young ages, young firms face an additional impediment to building their buyer base quickly, strengthening the gradual-accumulation mechanism.

The cost C_a^B can be interpreted as networking, sales, promotion, search, or prospecting effort aimed at forming new firm-to-firm relationships. The substantive assumption is simply that buyer acquisition is costly and that the marginal cost of rapidly expanding the customer base rises with acquisition effort.

O.3 Dynamic supplier problem

The dynamic block is layered on top of the static matching problem in Section 4. Conditional on current firm-level scope and the current buyer-side state, product–buyer activation and link for-

mation remain within-period decisions. In particular, the supplier's link-formation decision follows the expected-surplus rule in equation (23), while realized products within an active relationship are selected according to equation (27).

To keep the dynamic extension tractable, we summarize the buyer-side state relevant for product-development incentives by $\Lambda_{s,a}^M$. Let

$$\Pi_s^{\text{flow}}(G_s; \phi_s, \Lambda_{s,a}^M) \quad (136)$$

denote the current-period expected profit block inherited from the static matching model, with the buyer-base state summarized by $\Lambda_{s,a}^M$. The supplier's dynamic problem is

$$W_s(\phi_s, a, \Lambda_{s,a}^M) = \max_{G_s; n_{s,a}^B} \Pi_s^{\text{flow}}(G_s; \phi_s, \Lambda_{s,a}^M) - C_a^B(n_{s,a}^B) + \beta \mathbb{E} [W_s(\phi_s, a+1, \Lambda_{s,a+1}^M)] , \quad (137)$$

subject to the law of motion (134), where $\beta \in (0, 1)$ is the discount factor. Each period, the firm therefore chooses both its current product scope G_s , inherited from the Stage-1 decision of the static model, and buyer-acquisition effort $n_{s,a}^B$. The firm internalizes that acquisition today raises the future buyer-base state and therefore changes the future return to product development.

Static versus dynamic decisions. Conditional on $(G_s, \Lambda_{s,a}^M)$, the product-buyer and link-level decisions remain within-period. The supplier evaluates potential relationships as in Sections 4.2–4.3: expected product surplus determines whether a link is formed, and realized product-buyer matches determine which developed products are activated within an active relationship. What the dynamic block adds is a decision above this static problem. The firm chooses $n_{s,a}^B$ today to shape the future evolution of $\Lambda_{s,a+1}^M$, taking into account that a larger future buyer base raises the return to firm-level product scope through the amortization mechanism characterized in Section 4.5.

Euler-style condition for buyer acquisition. The first-order condition with respect to $n_{s,a}^B$ takes the standard intertemporal form. Differentiating (137) and using (134),

$$\underbrace{\xi_a^B f^B(n_{s,a}^B)^{\gamma_B-1}}_{\text{marginal acquisition cost}} = \beta m^B \bar{\lambda}^{\text{new}} \cdot \underbrace{\mathbb{E} \left[\frac{\partial W_s(\phi_s, a+1, \Lambda_{s,a+1}^M)}{\partial \Lambda_{s,a+1}^M} \right]}_{\text{marginal value of the buyer base}}. \quad (138)$$

The right-hand side links the dynamic acquisition decision to the static scope mechanism. An additional unit of the buyer-base state raises future profits directly and, through amortization, increases the future return to developing a broader product portfolio.

O.4 Connection to the benchmark scope rule

Section 4.5 provides an approximate characterization of optimal firm-level scope in the interior Pareto region. Using the match-adjusted buyer-base aggregator, the static relationship is

$$G_s^* \propto \left[\frac{\phi_s^{\theta_\zeta(\sigma-1)} \Lambda_s^M}{f^P(1)} \right]^{\frac{1}{\delta + \kappa\theta_\zeta - \mu}}, \quad \kappa = \mu + \alpha(\sigma - 1).$$

Embedding the same logic in the dynamic environment gives the life-cycle scope rule

$$G_s^*(\phi_s, \Lambda_{s,a}^M) \approx \text{integer} \left[\frac{\phi_s^{\theta_\zeta(\sigma-1)} \Lambda_{s,a}^M}{f^P(1)} \right]^{\frac{1}{\delta + \kappa\theta_\zeta - \mu}}, \quad (139)$$

where the expression is rounded to the feasible integer product-scope choice. Holding productivity fixed, gradual accumulation of $\Lambda_{s,a}^M$ therefore generates gradual expansion of optimal firm-level scope whenever the denominator

$$\delta + \kappa\theta_\zeta - \mu$$

is positive, as required by the finite-scope condition underlying the static approximation.

The same qualitative logic carries over to the nested-CES robustness exercise of Appendix N. A larger buyer base raises the payoff to paying the firm-level development cost $F^P(G_s)$ and can also make additional buyer relationships profitable. Within-link cannibalization prevents the same closed-form characterization, but it does not overturn the basic monotonicity: a larger and more valuable buyer base raises the return to firm-level product development.

O.5 Equivalent truncated-history representation

For comparison with dynamic production-network and customer-accumulation models such as [Einav et al. \(2021\)](#) and [Aekka and Khanna \(2024\)](#), the buyer-base state can equivalently be represented as a truncated history of past acquisition decisions. Iterating (134) forward gives

$$\Lambda_{s,a}^M \approx m^B \bar{\lambda}^{\text{new}} \sum_{\tau=0}^{A_B} (\delta^B)^\tau n_{s,a-1-\tau}^B, \quad (140)$$

where A_B is a truncation horizon chosen so that contributions from sufficiently old acquisition cohorts are negligible. The approximation is accurate when buyer relationships do not survive indefinitely, because the contribution of past cohorts decays geometrically at rate δ^B . The dynamic buyer side can therefore be summarized either by the scalar state $\Lambda_{s,a}^M$ in (134) or, equivalently, by a finite-dimensional history of recent buyer-acquisition decisions.

O.6 Interpretation and mapping to the empirical facts

The static and dynamic blocks provide a unified interpretation of the empirical patterns. The baseline matching model in Section 4 explains why buyer breadth and product scope co-move through amortization and buyer-specific matching. The nested-CES extension in Appendix N shows how this interaction is modified when products within a supplier compete for expenditure inside a buyer relationship. The present extension adds a dynamic margin by allowing the buyer base itself to be accumulated gradually over time.

This dynamic mechanism provides a natural interpretation of the life-cycle evidence documented in Fact 2, Section 3.2, and Appendix C.4. Figure 18 shows that, conditional on initial sales, younger suppliers rely relatively more on the joint new-buyer–new-product margin (NLNP), whereas the importance of selling existing products to new buyers (NLOP) rises with supplier age. OLNP remains comparatively small across age groups.

In the dynamic model, young firms begin with a relatively small buyer-base state $\Lambda_{s,a}^M$. Because expanding that state is costly, buyer accumulation occurs gradually. Since optimal firm-level scope is increasing in $\Lambda_{s,a}^M$ through (139), product scope expands alongside the buyer base. Early in the life cycle, increases in the buyer base therefore tend to coincide with product expansion, providing a natural counterpart to the relatively high NLNP share among younger suppliers. As $\Lambda_{s,a}^M$ grows, the firm operates with a broader established product portfolio, so subsequent buyer acquisition is increasingly accommodated by extending existing products to new customers, consistent with the rising importance of NLOP among older suppliers in Figure 18.

The extension should be interpreted as a qualitative dynamic rationalization rather than as a calibrated model of the life cycle. A natural next step would be to estimate the acquisition-cost curvature γ_B , the age profile of ξ_a^B , and the persistence parameter δ^B using the observed dynamics of buyer relationships, and then quantify the relative contributions of slow buyer accumulation and gradual product development to firm growth over the life cycle.